Ray Optics

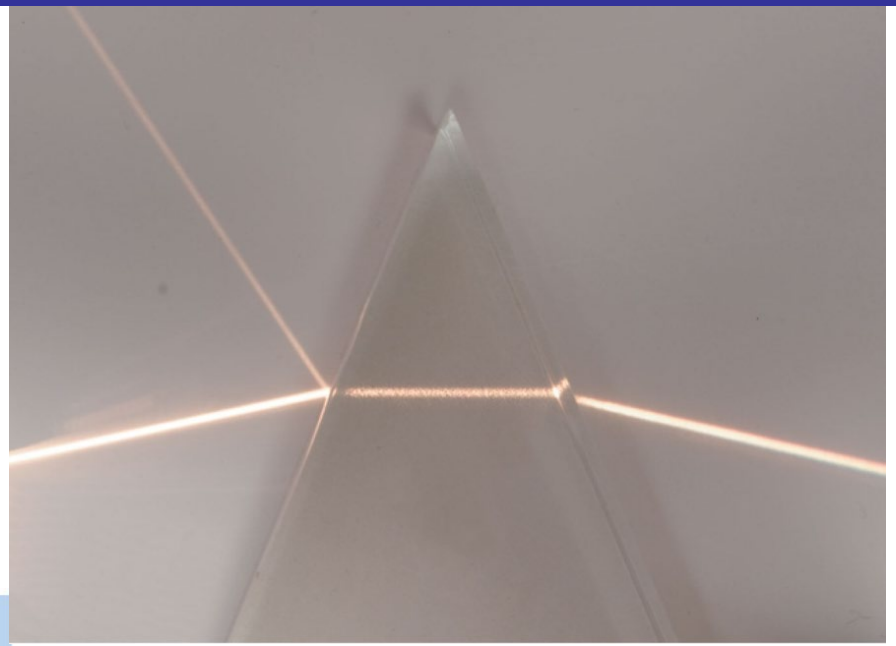
Antisolar Rays

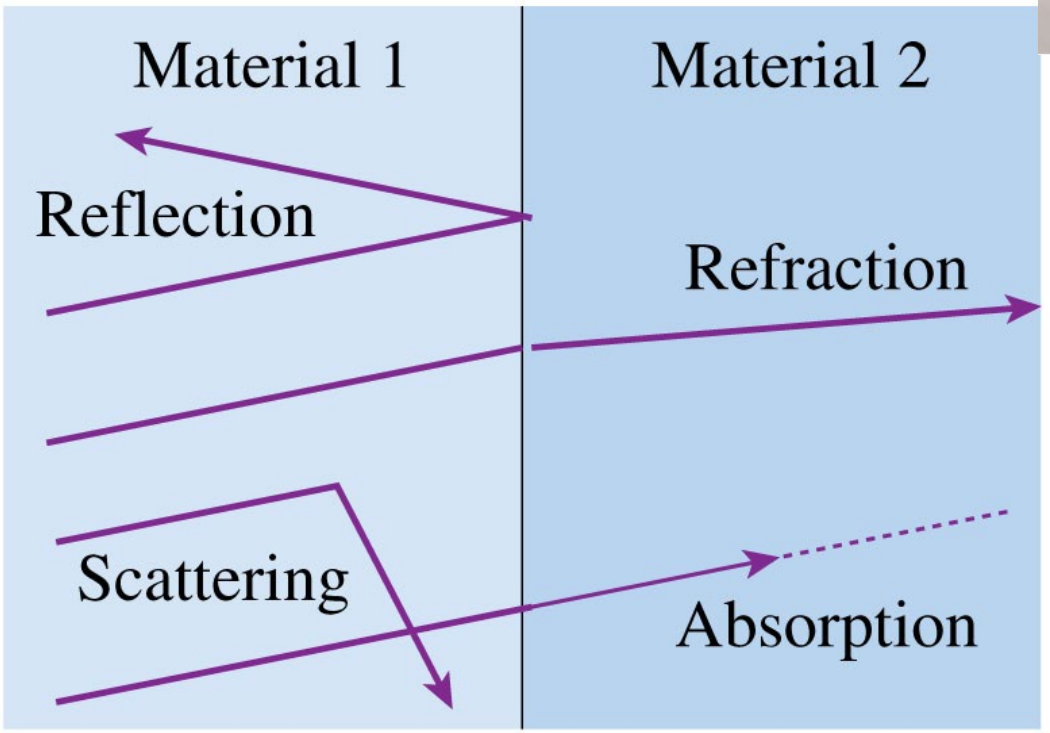
The Ray Model of Light

Light interacts with matter:

At an **interface** between two materials, light can be **reflected** and/or **refracted**.

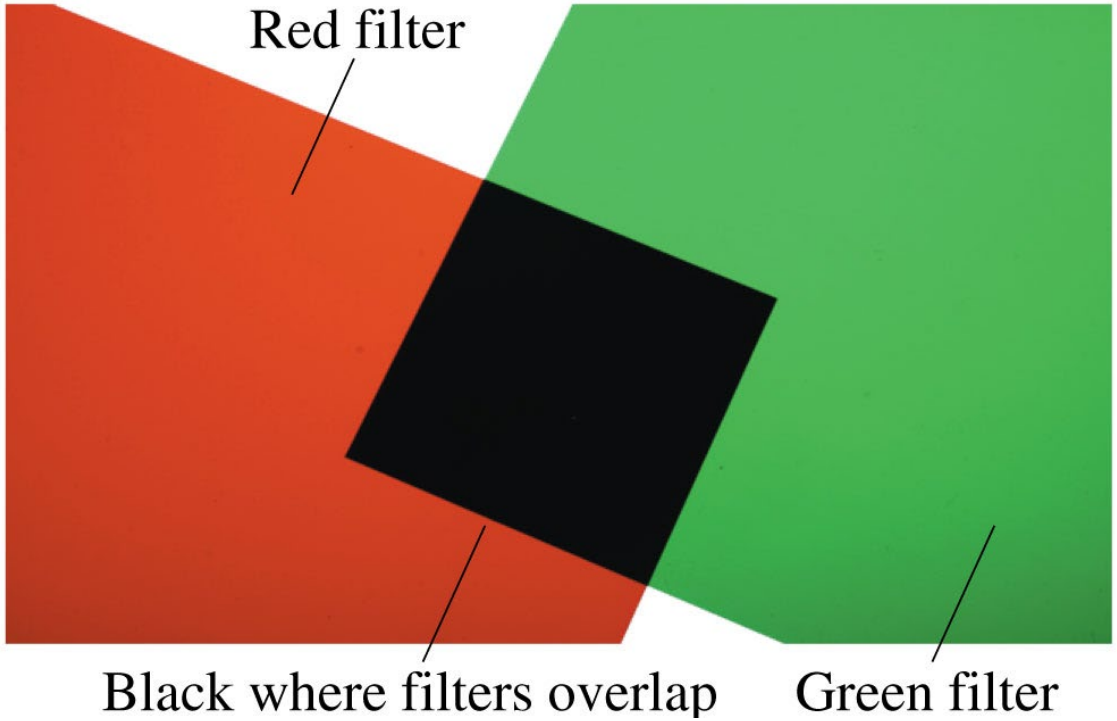
Within a material, light can be **scattered** and/or **absorbed**.





Colored Filters and Colored Objects

- Green glass is green because it **absorbs** any light that is “not green.”
- If a green filter and a red filter are overlapped, no light gets through.
- The green filter **transmits** only green light, which is then absorbed by the red filter because it is “not red.”

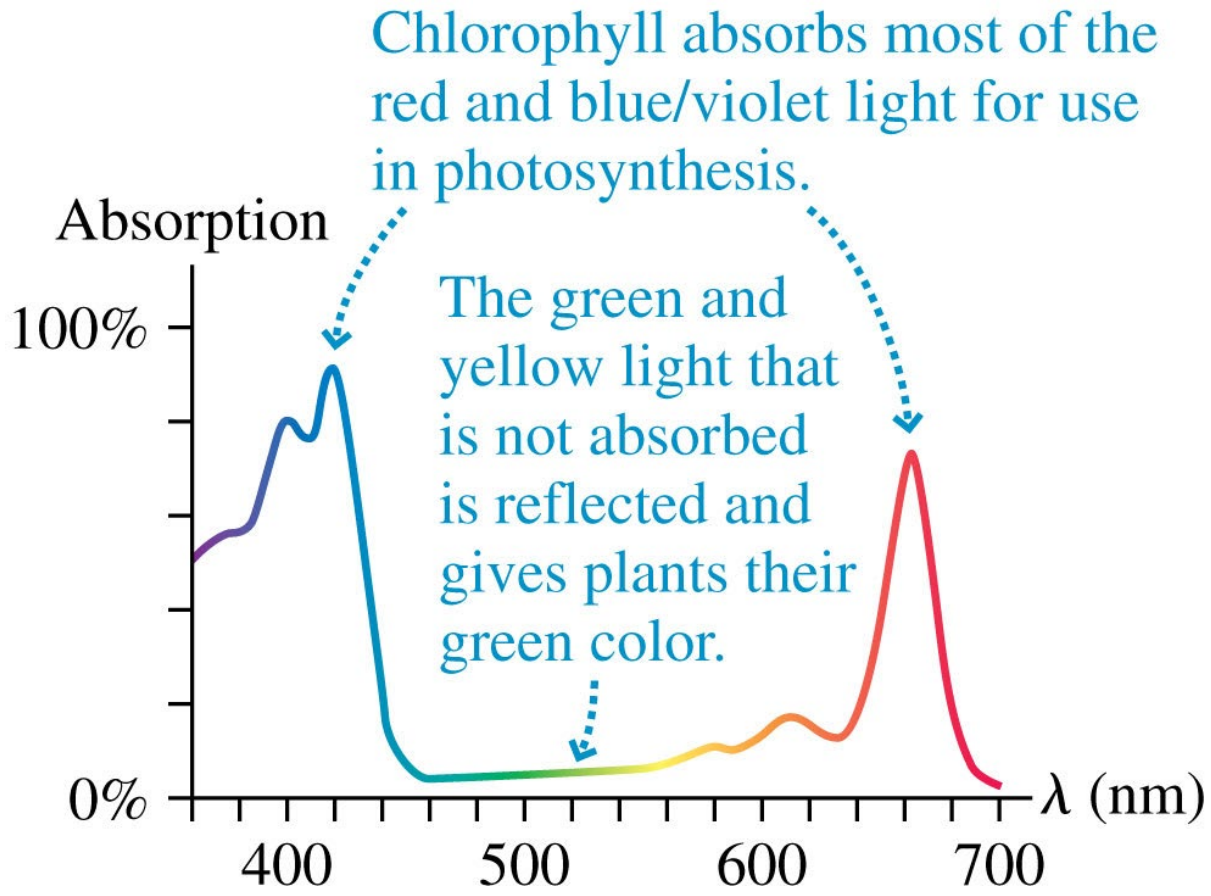


Colored Filters and Colored Objects

- The figure below shows the absorption curve of *chlorophyll*, which is essential for **photosynthesis** in green plants.

- The chemical reactions of photosynthesis absorb red light and blue/violet light from sunlight and puts it to use.

- When you look at the green leaves on a tree, you're seeing the light that was reflected because it *wasn't needed* for photosynthesis.

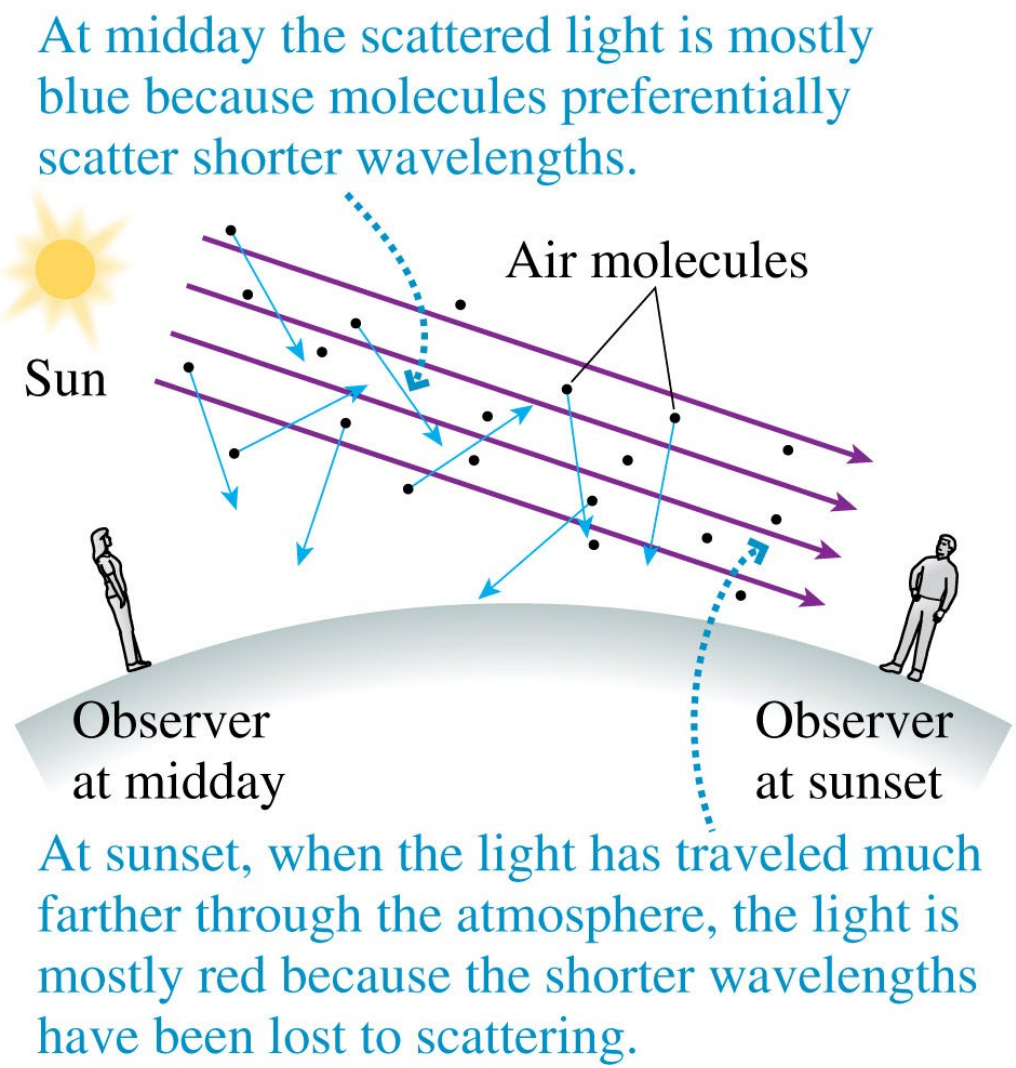


Light Scattering: Blue skies & Red sunsets

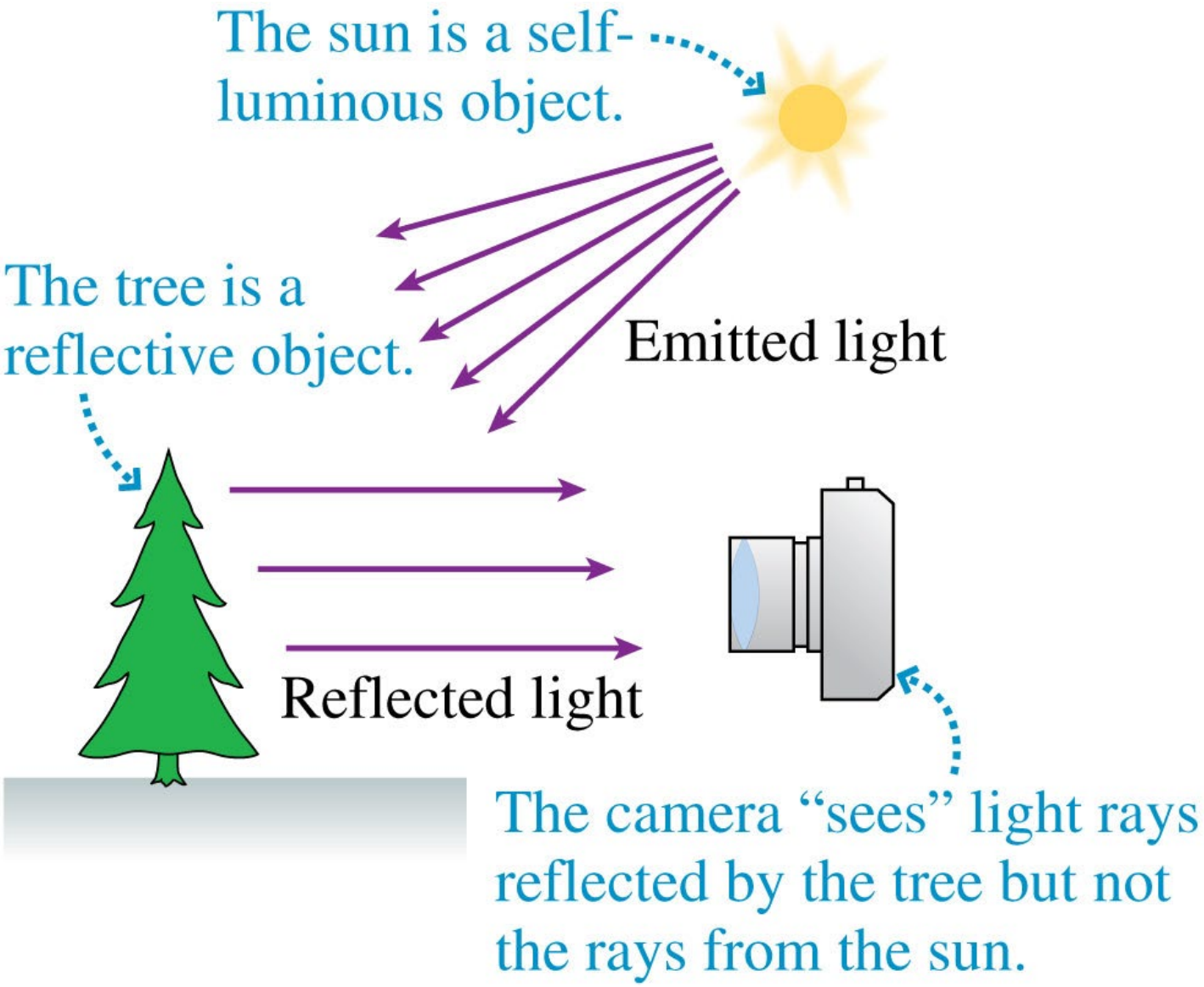
Light Scattering: Blue skies & Red sunsets

- Light can scatter from small particles that are suspended in a medium.
- **Rayleigh scattering** from atoms and molecules depends inversely on the fourth power of the wavelength:

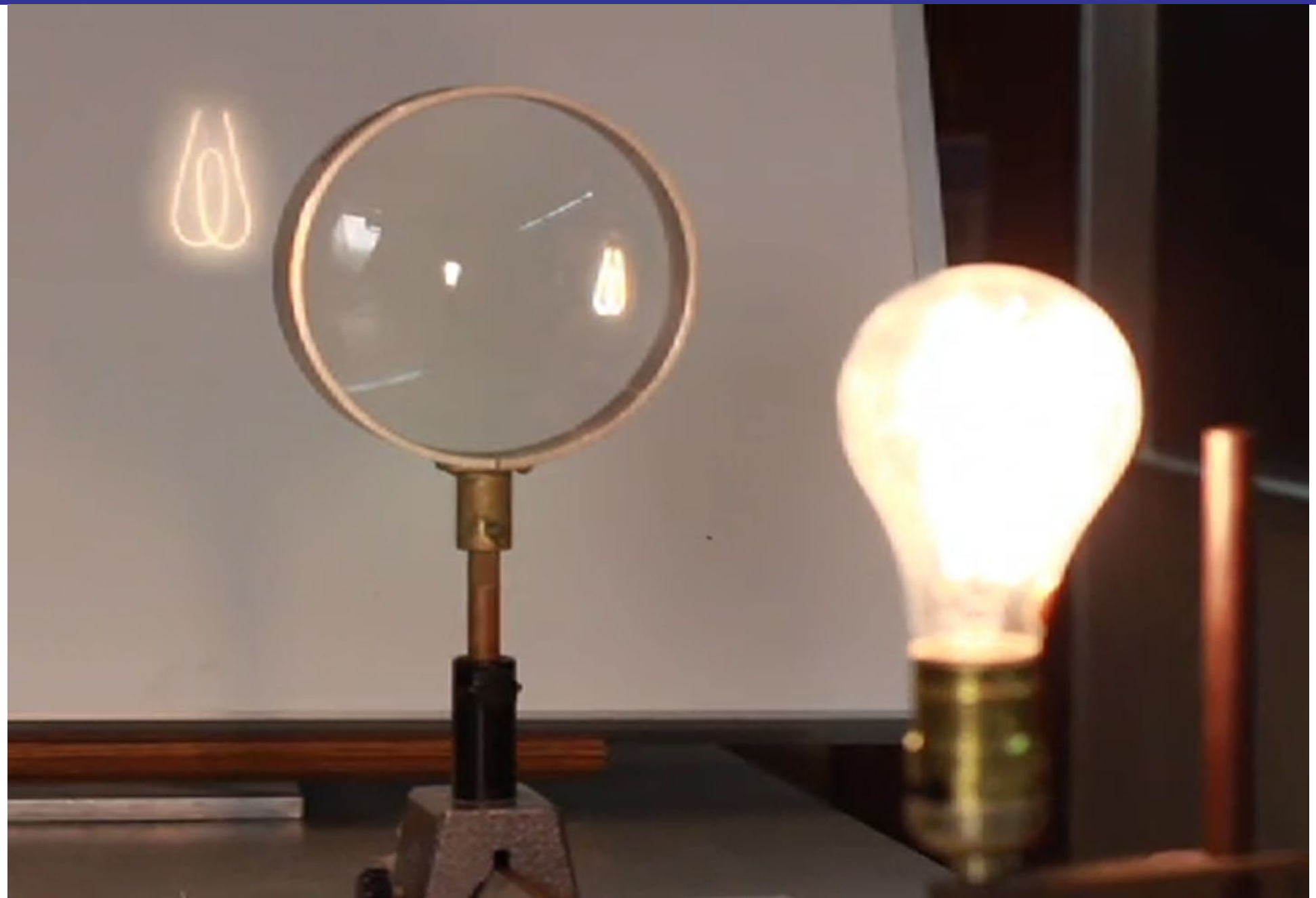
$$I_{\text{scattered}} \propto \lambda^{-4}$$



Objects

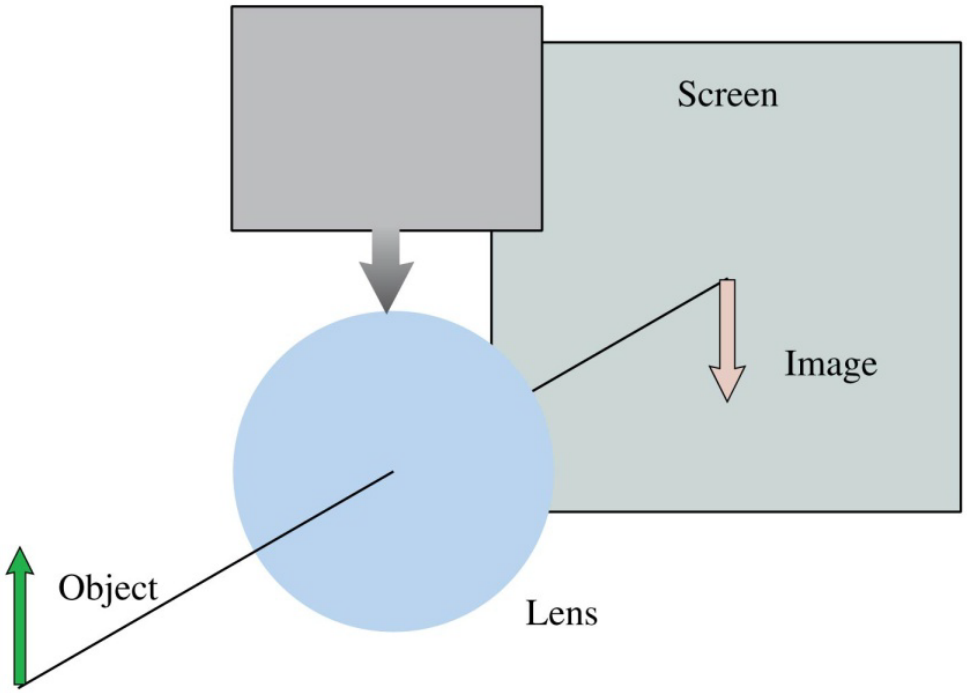

The sun is a self-luminous object.
The tree is a reflective object.
Emitted light
Reflected light
The camera "sees" light rays reflected by the tree but not the rays from the sun.

Demo: Half-covered lens



Q&A 23.10

A lens produces a sharply focused, inverted image on a screen. What will you see on the screen if a piece of dark paper is lowered to cover the top half of the lens?



A. An inverted but blurry image.

✔ B. An image that is dimmer but otherwise unchanged.

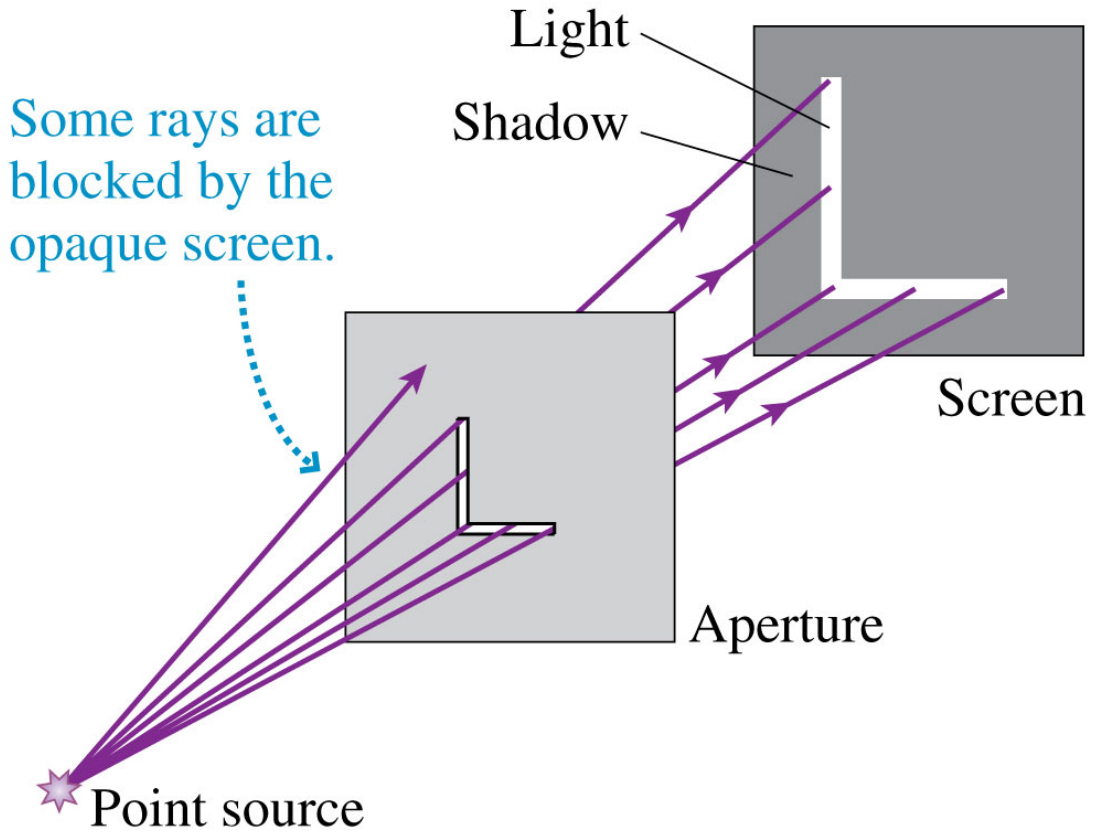
C. Only the top half of the image.

D. Only the bottom half of the image.

E. No image at all.

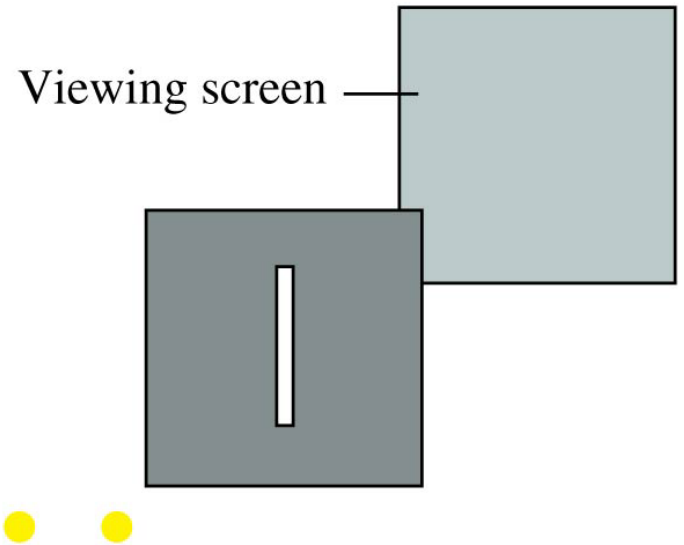
Apertures

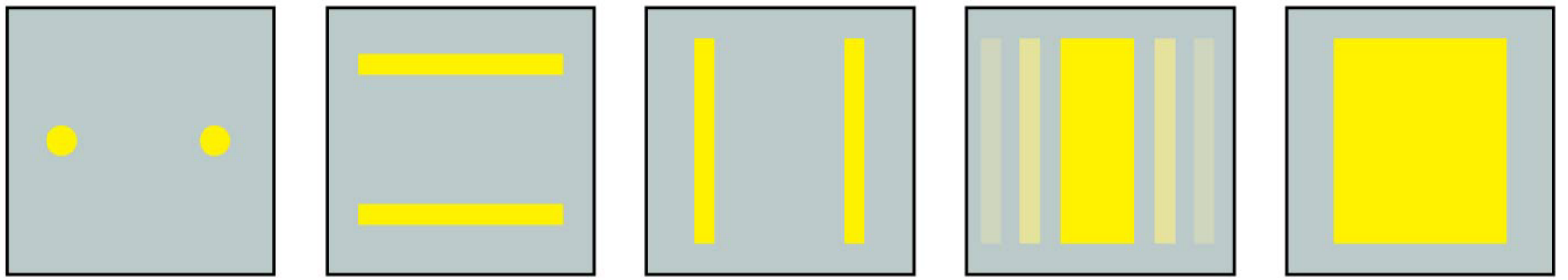
We can apply the ray model to more complex apertures, such as the L-shaped aperture below.



Q 23.2

Two point sources of light illuminate a narrow (≈ 2 mm) vertical aperture in a dark screen. What do you see on the viewing screen?





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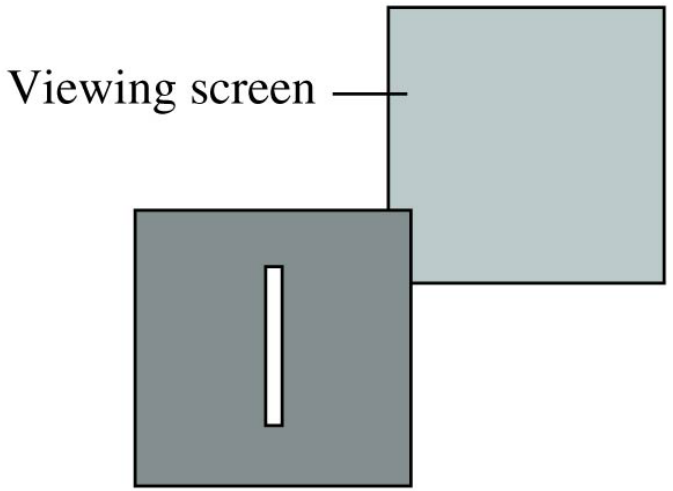
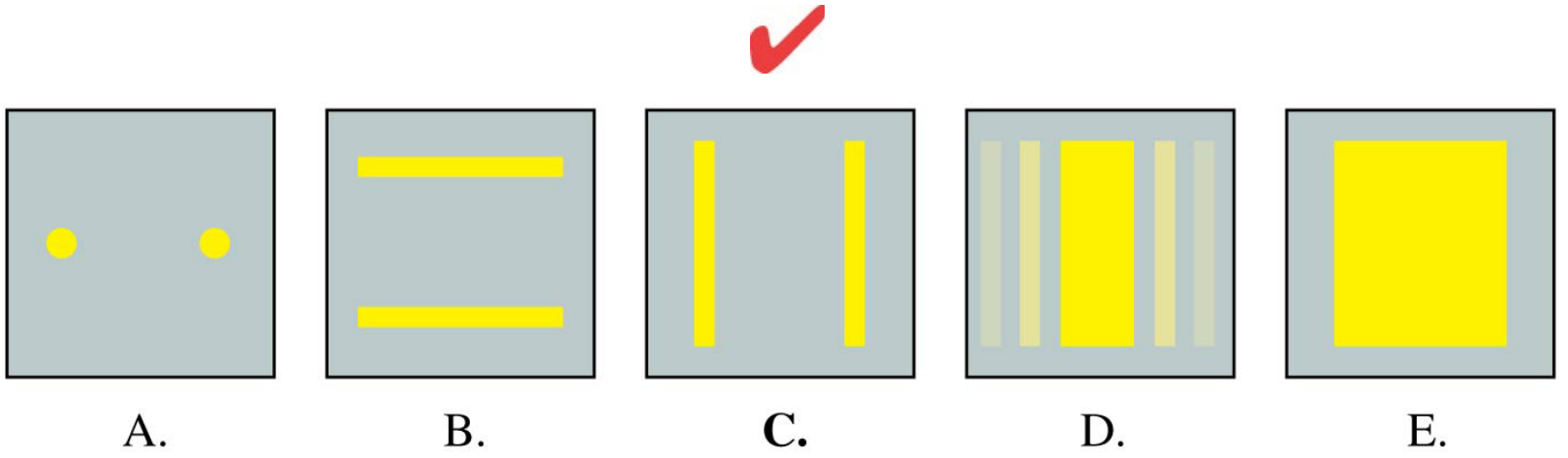
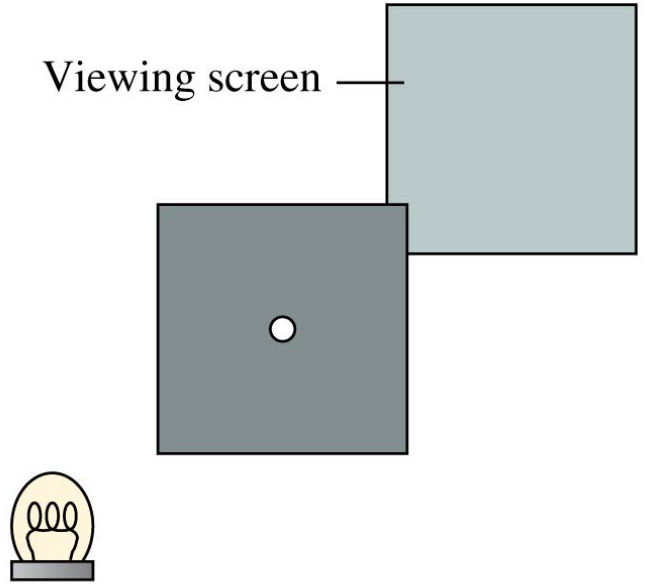
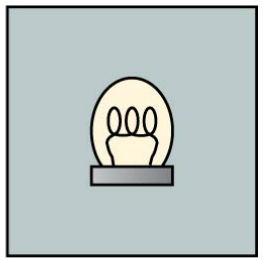




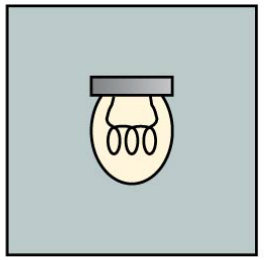
Q 23.1

The dark screen has a small hole, ≈ 2 mm in diameter. The lightbulb is the only source of light. What do you see on the viewing screen?

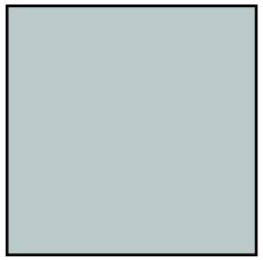




A.



B.



C.

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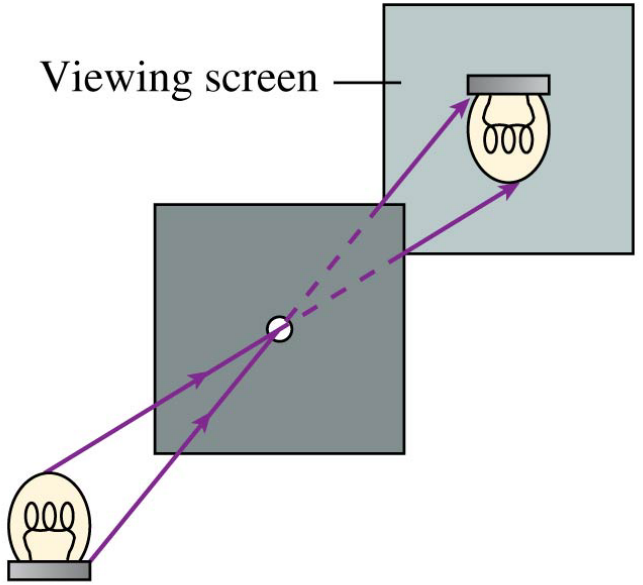
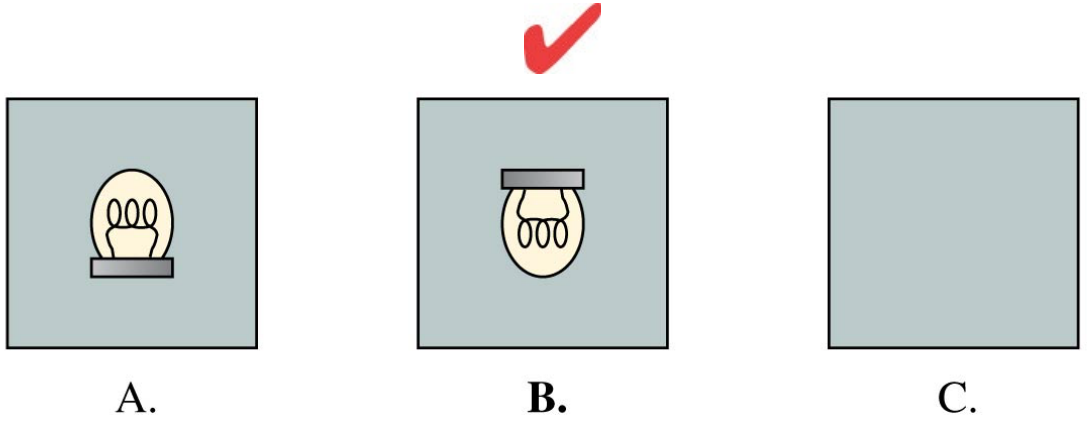
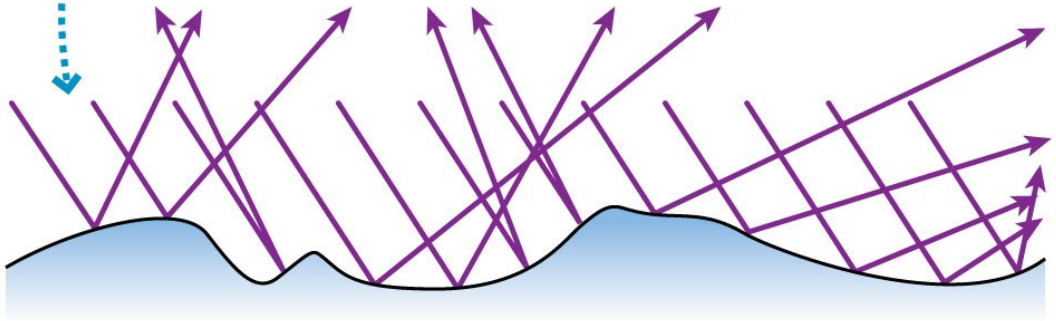




A. B. C.

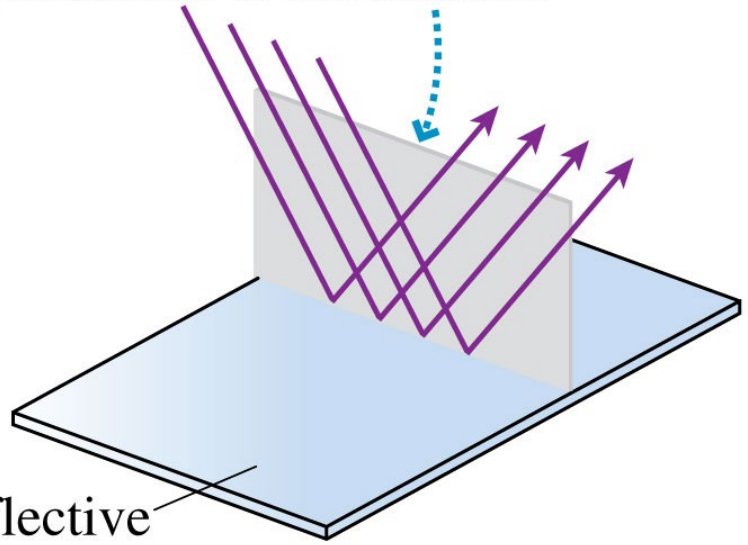
Diffuse vs. Specular Reflection of Light

Each ray obeys the law of reflection at that point, but the irregular surface causes the reflected rays to leave in many random directions.



Magnified view of surface

The incident and reflected rays lie in the plane of incidence, a plane perpendicular to the surface.



Reflective surface

Virtual Image Formed by Convex Mirror

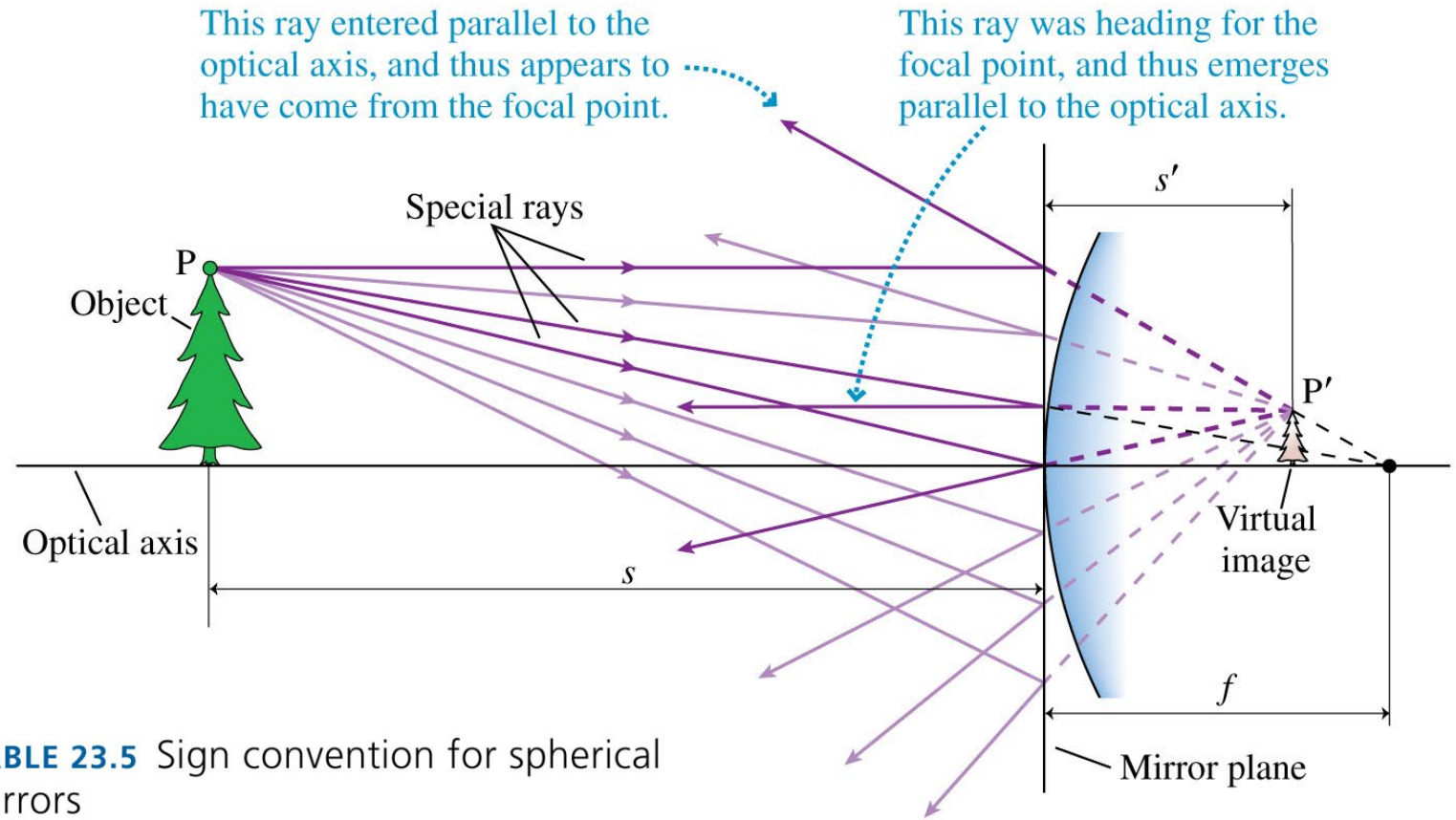


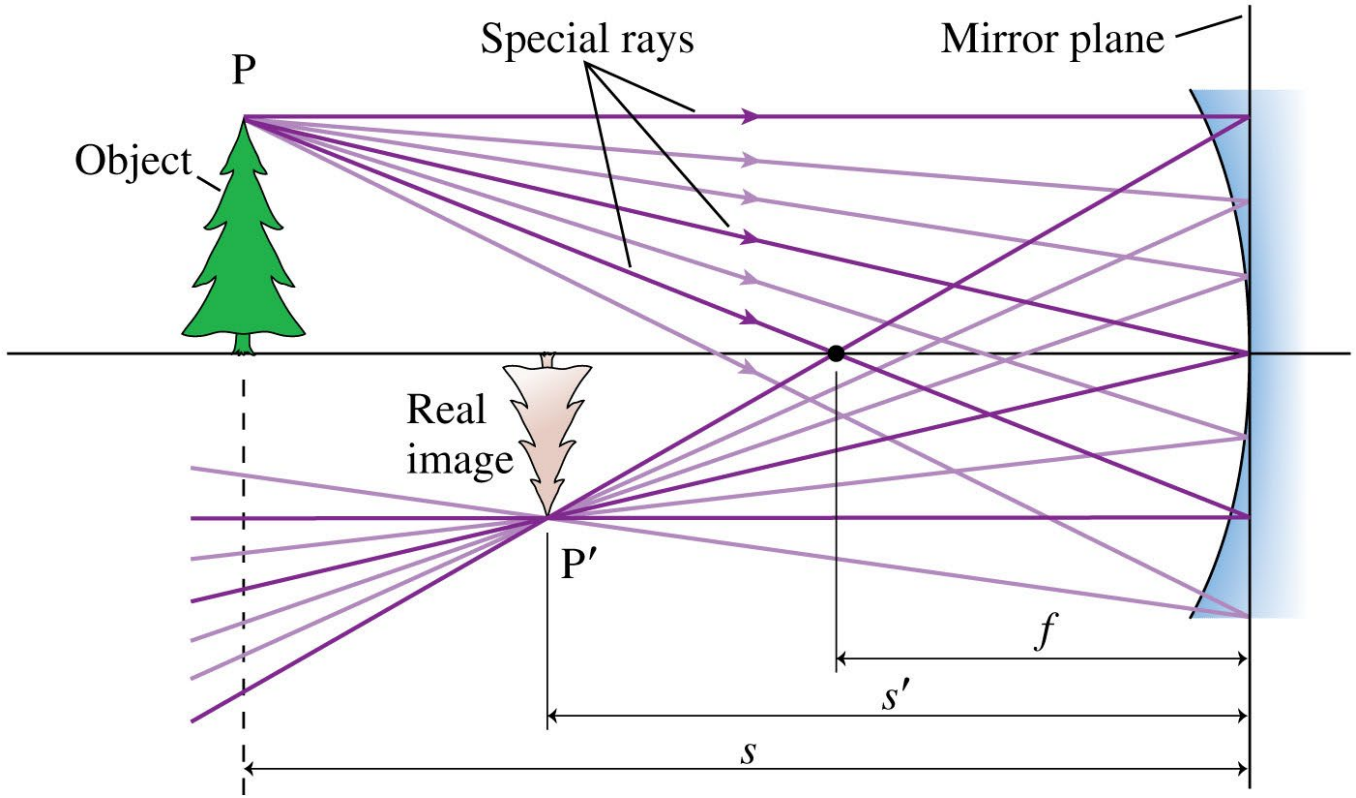
TABLE 23.5 Sign convention for spherical mirrors

	Positive	Negative
R, f	Concave toward the object	Convex toward the object
s'	Real image, same side as object	Virtual image, opposite side from object

$$\frac{1}{s} + \frac{1}{s'} = \frac{1}{f} \qquad \text{(mirror equation)}$$

$$f = \frac{R}{2}$$

Real Image Formed by Concave Mirror



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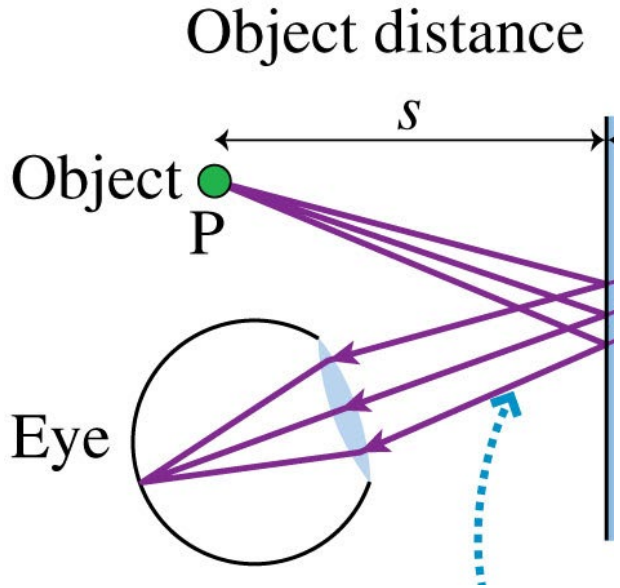
$$\frac{1}{s}+\frac{1}{s'}=\frac{1}{f} \qquad \text{(mirror equation)}$$

$$f=\frac{R}{2}$$

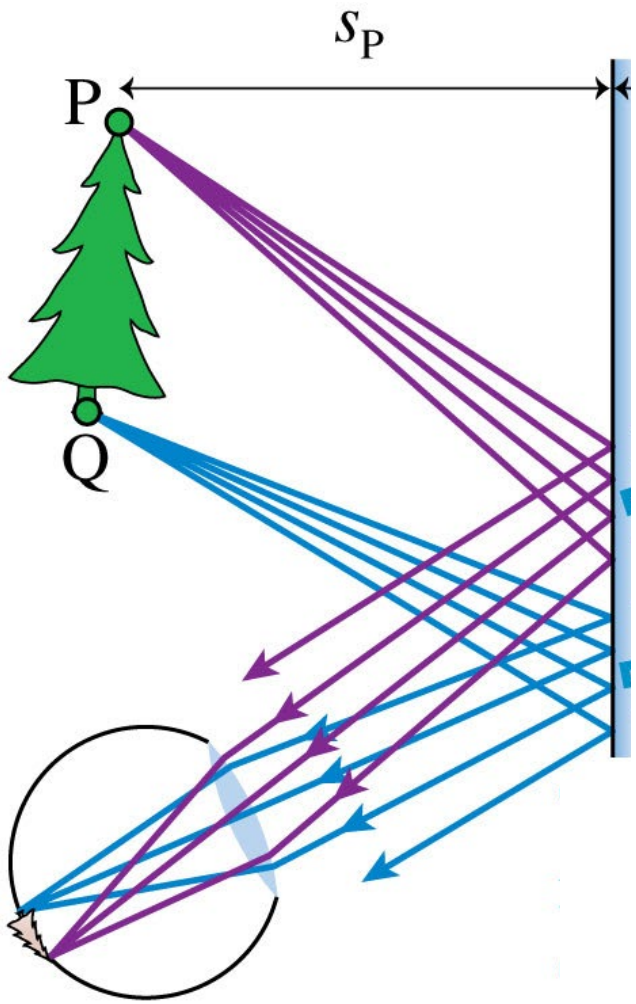
FADEL SENNA/AFP/AFP/GETTY IMAGES
FADEL SENNA/AFP/AFP/GETTY IMAGES

The Plane Mirror

- Consider P, a source of rays which reflect from a mirror.
- The reflected rays appear to emanate from P', the same distance behind the mirror as P is in front of the mirror.
- That is, $s' = s$.



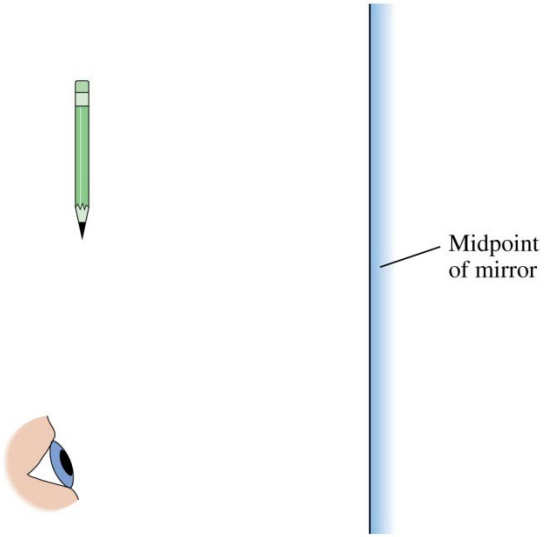
The Plane Mirror



Q 23.3

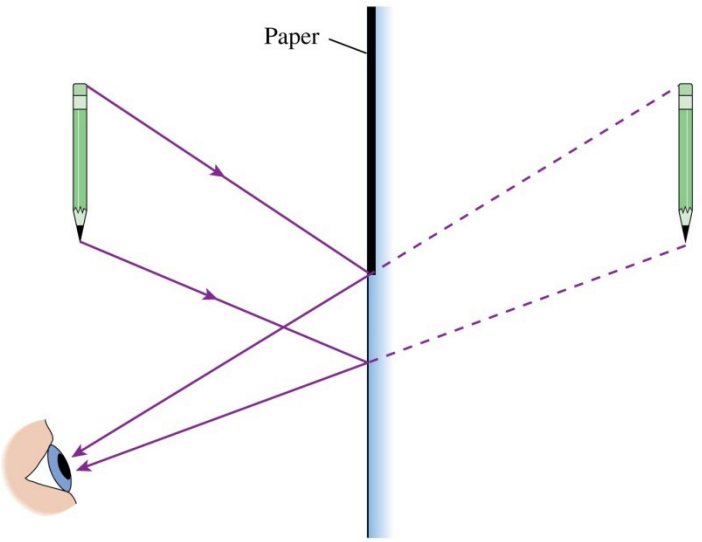
You are looking at the image of a pencil in a mirror. What do you see in the mirror if the top half of the mirror is covered with a piece of dark paper?

A. The full image of the pencil.
B. The top half only of the pencil.
C. The bottom half only of the pencil.
D. No pencil, only the paper.



Q&A 23.3

You are looking at the image of a pencil in a mirror. What do you see in the mirror if the *top half* of the mirror is covered with a piece of dark paper?



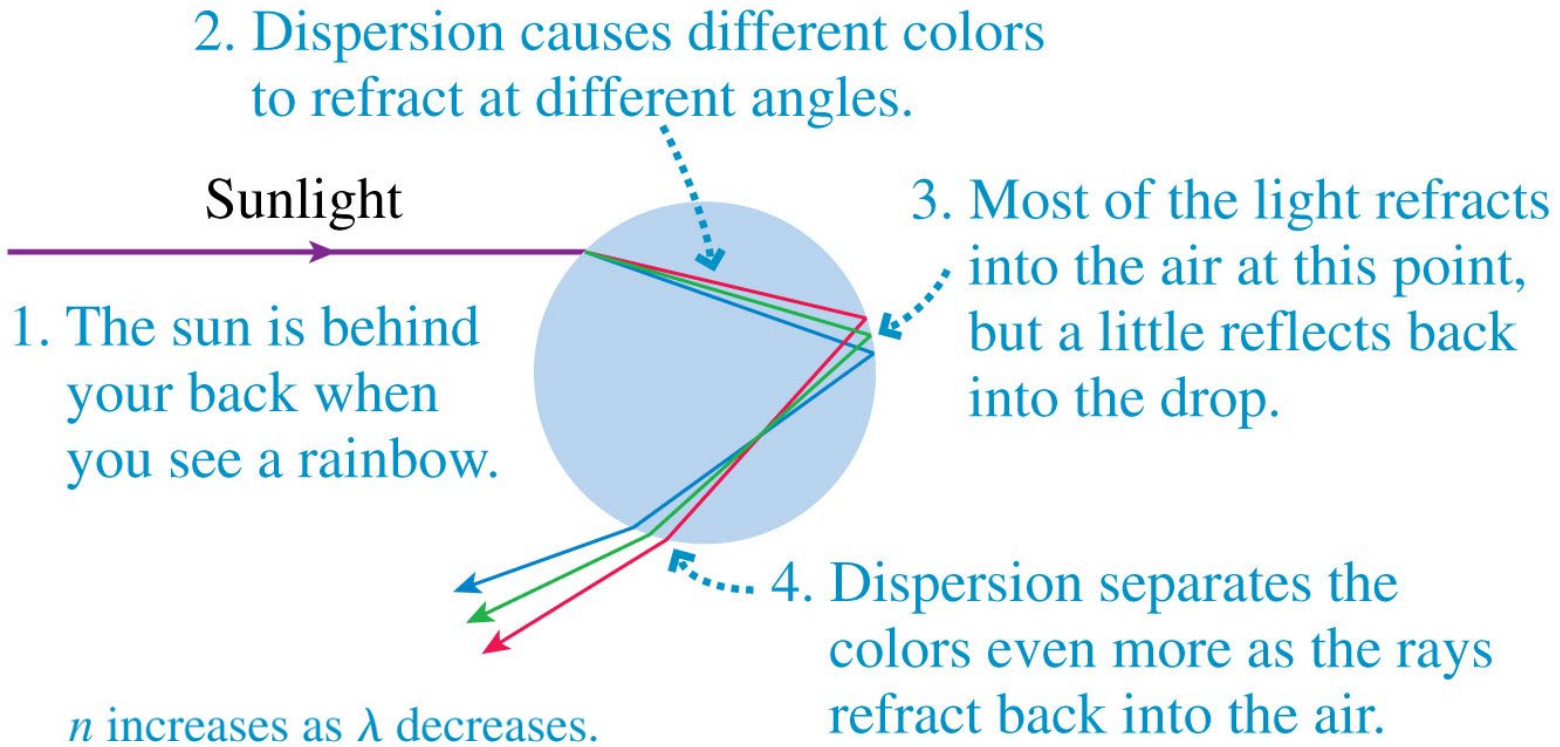
✔ **A. The full image of the pencil.**

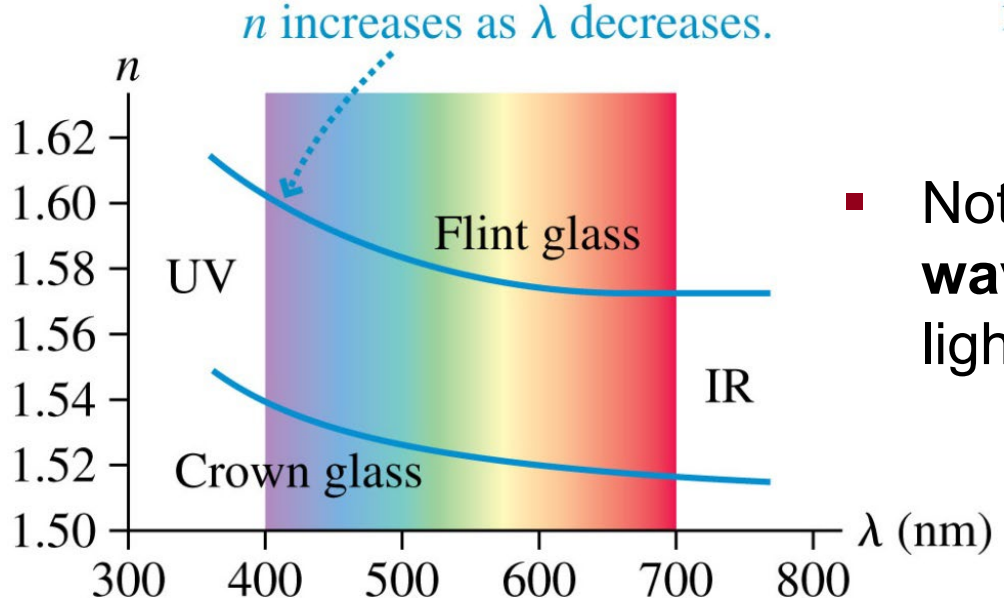
B. The top half only of the pencil.

C. The bottom half only of the pencil.

D. No pencil, only the paper.

Rainbows: Refraction + Reflection + Dispersion

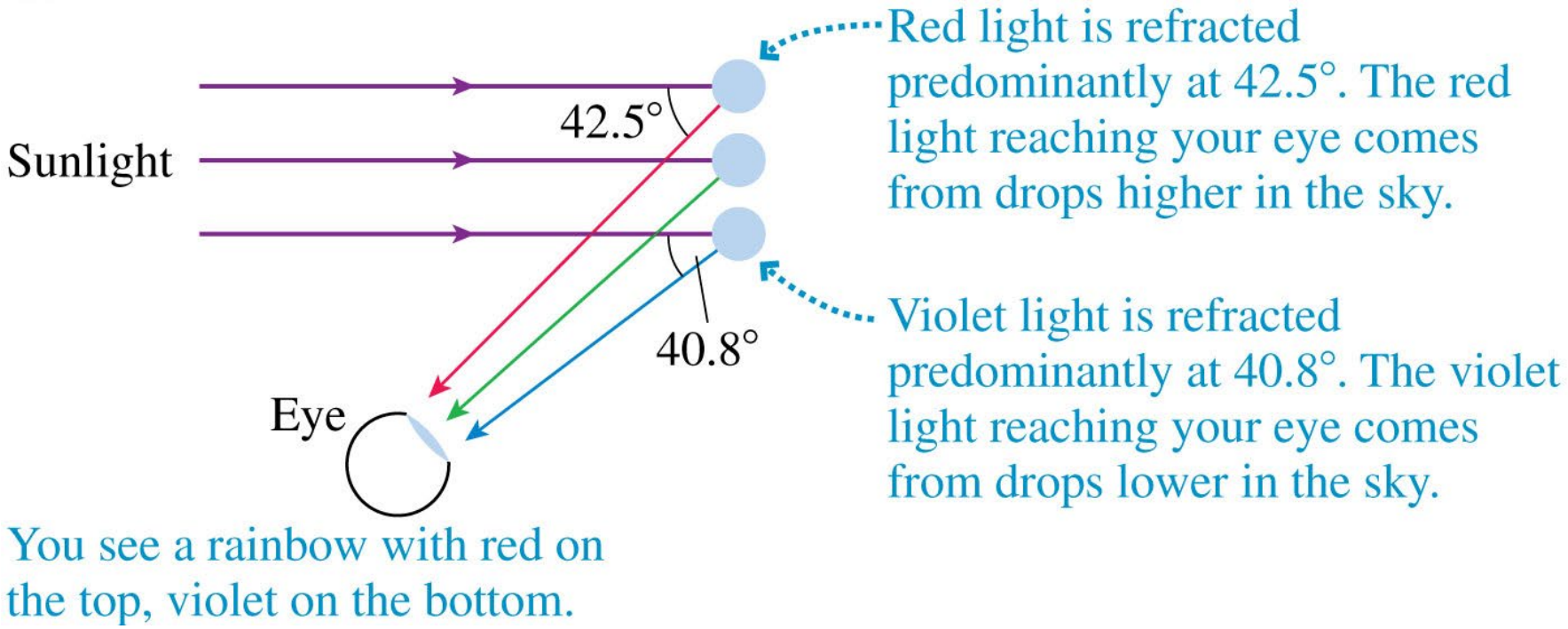




- Notice that ***n*** **is <u>larger</u> when the wavelength is <u>shorter</u>**, thus violet light refracts more than red light.

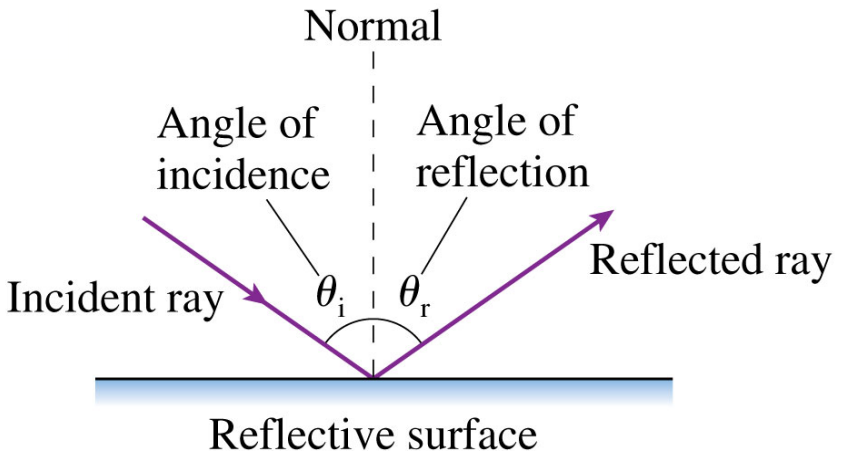
Getty

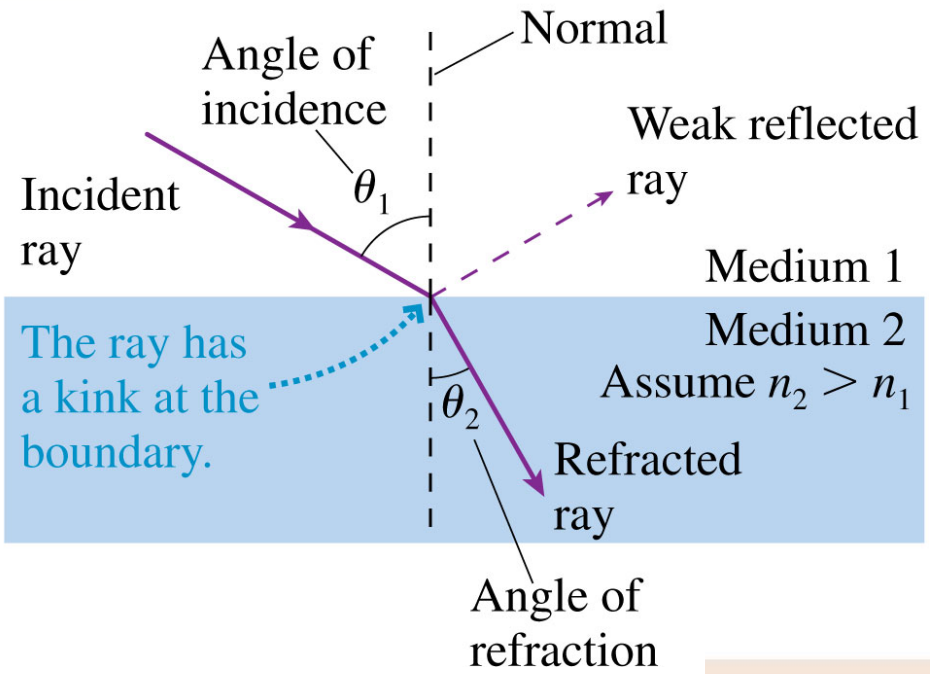
Rainbows

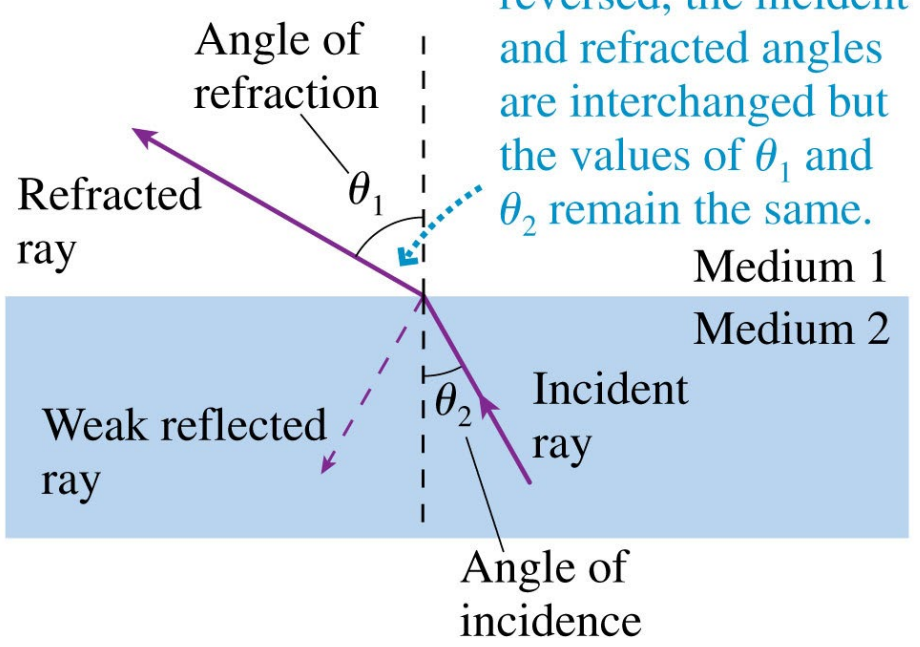

Sunlight
42.5°
40.8°
Eye
Red light is refracted predominantly at 42.5°. The red light reaching your eye comes from drops higher in the sky.
Violet light is refracted predominantly at 40.8°. The violet light reaching your eye comes from drops lower in the sky.
You see a rainbow with red on the top, violet on the bottom.

Reflection & Refraction

$\theta_i = \theta_r$







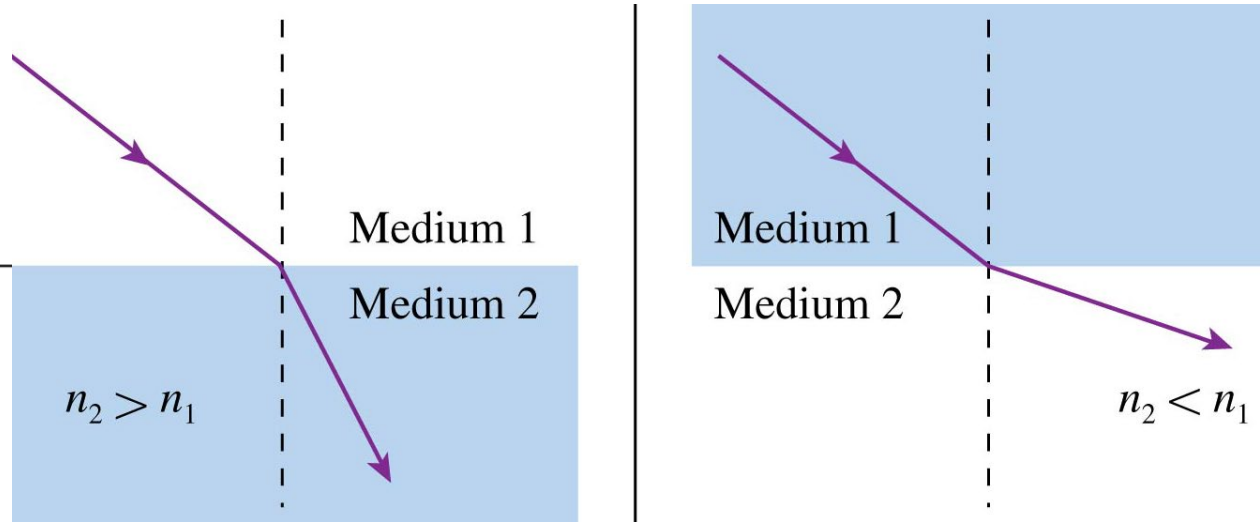
$$n_1 \sin\theta_1 = n_2 \sin\theta_2 \qquad \text{(Snell's law of refraction)}$$

Indices of Refraction

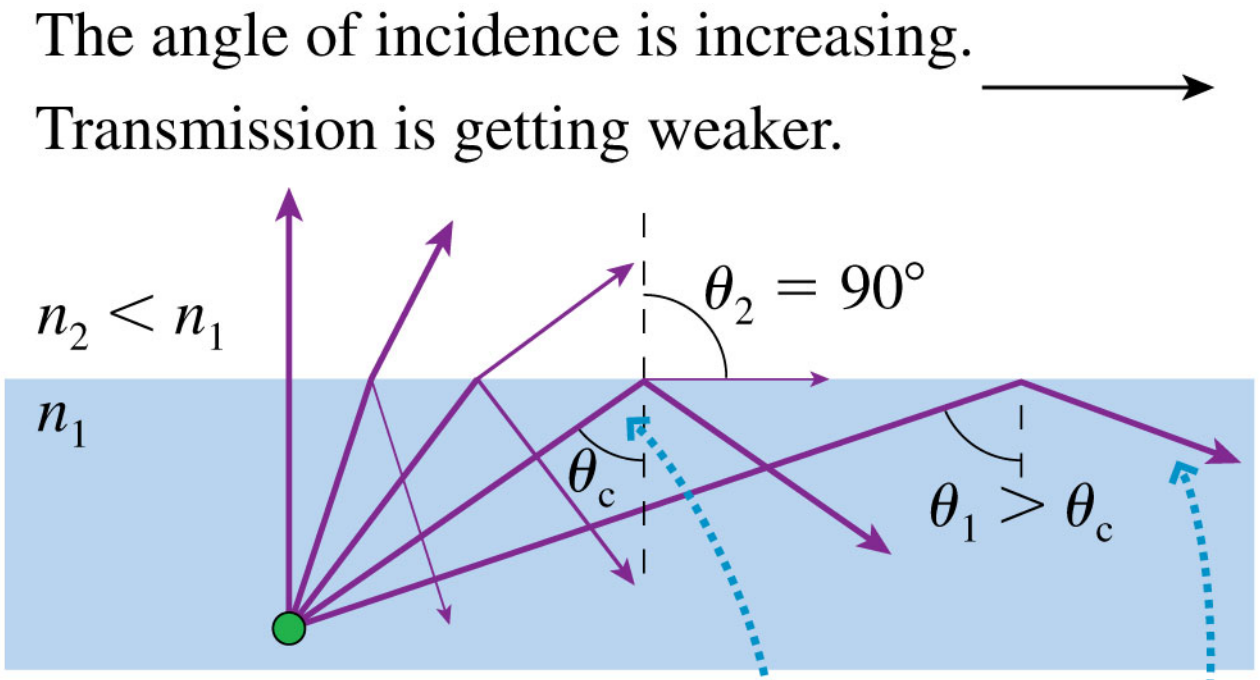
TABLE 23.1 Indices of refraction

Medium	n
Vacuum	1.00 exactly
Air (actual)	1.0003
Air (accepted)	1.00
Water	1.33
Ethyl alcohol	1.36
Oil	1.46
Glass (typical)	1.50
Polystyrene plastic	1.59
Cubic zirconia	2.18
Diamond	2.41
Silicon (infrared)	3.50

$$n = \frac{c}{v_{\text{medium}}}$$



Total Internal Reflection



$$\theta_c = \sin^{-1} \frac{n_2}{n_1}$$

Q&A 23.5

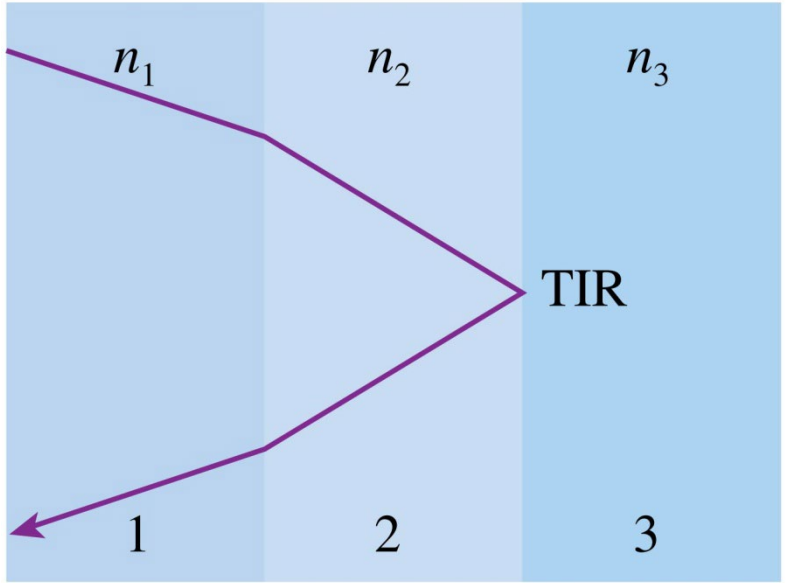
A laser beam undergoes two refractions plus total internal reflection at the interface between medium 2 and medium 3. Which is true?

A. $n_1 < n_3$.

✔ B. $n_1 > n_3$.

C. $n_1 = n_3$.

D. There's not enough information to compare n_1 and n_3.

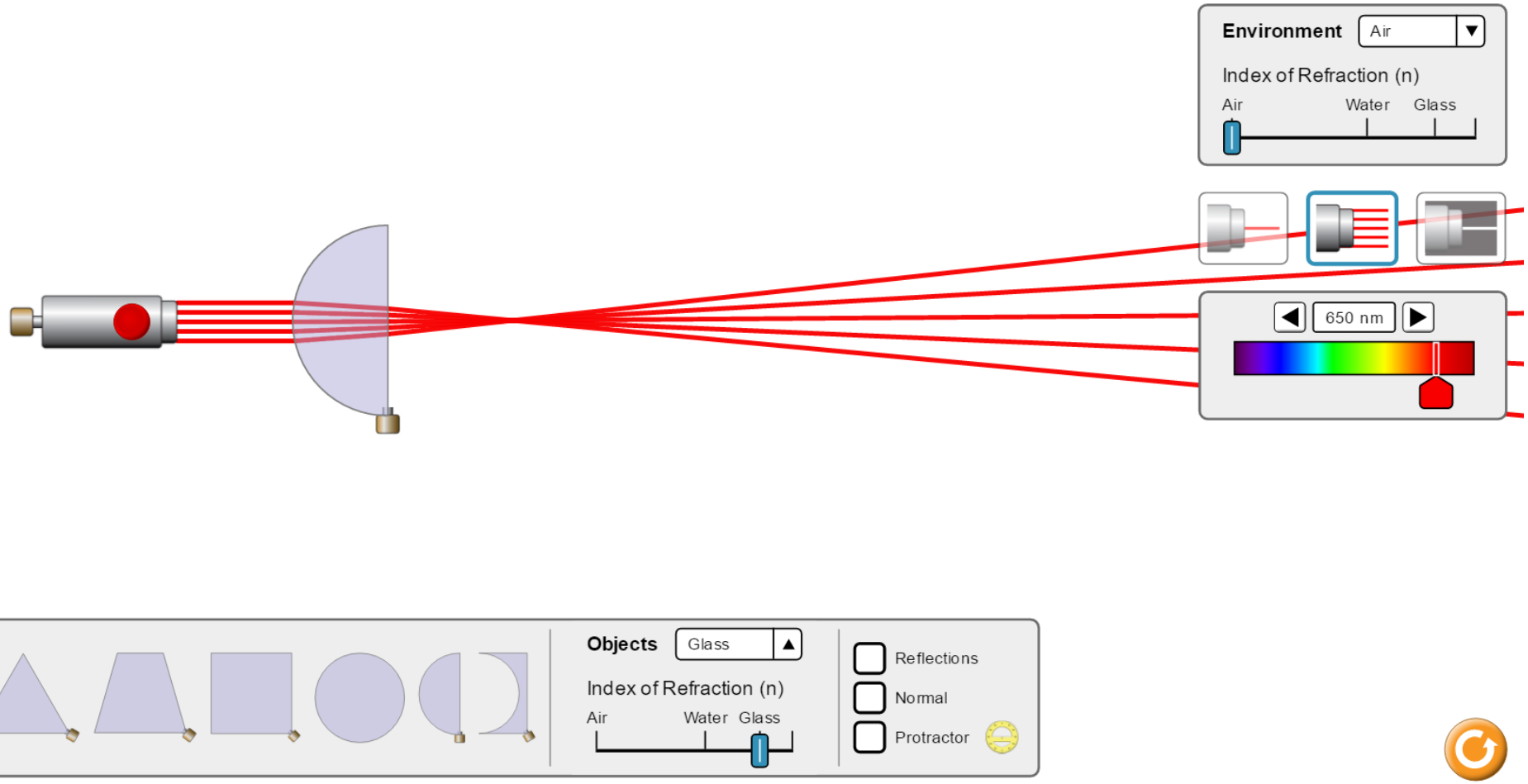


$n_1 \mid n_2$: Ray bending away from normal → $n_1 > n_2$.

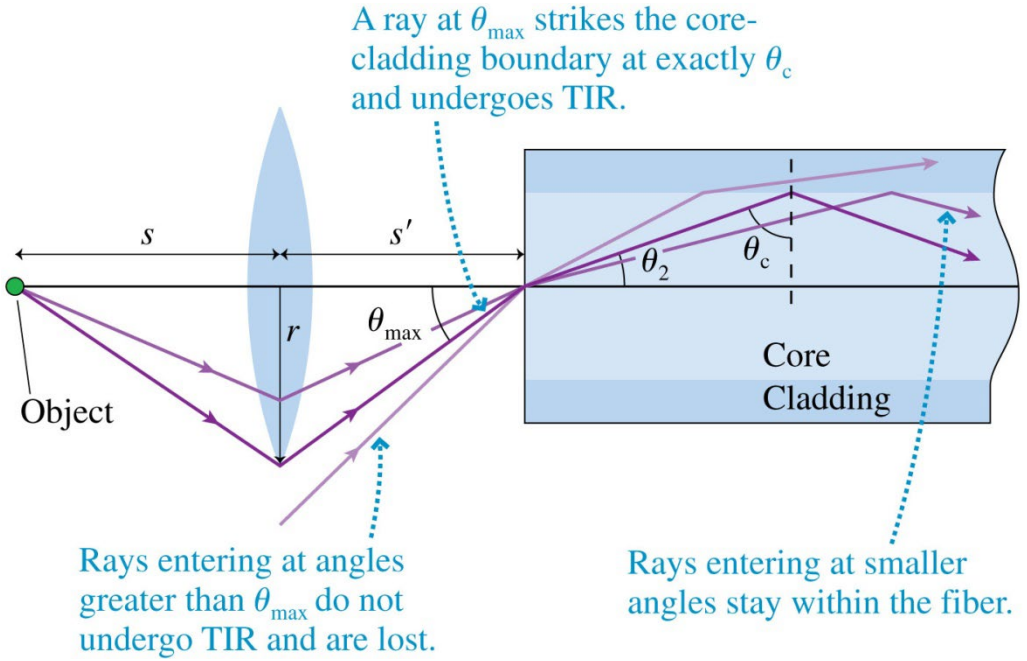
$n_2 \mid n_3$: Ray undergoing *total internal reflection* (TIR) → $n_2 > n_3$.

Thus: $n_1 > n_3$.

PhET: *Bending Light: Prisms*



Optical Fiber Imaging



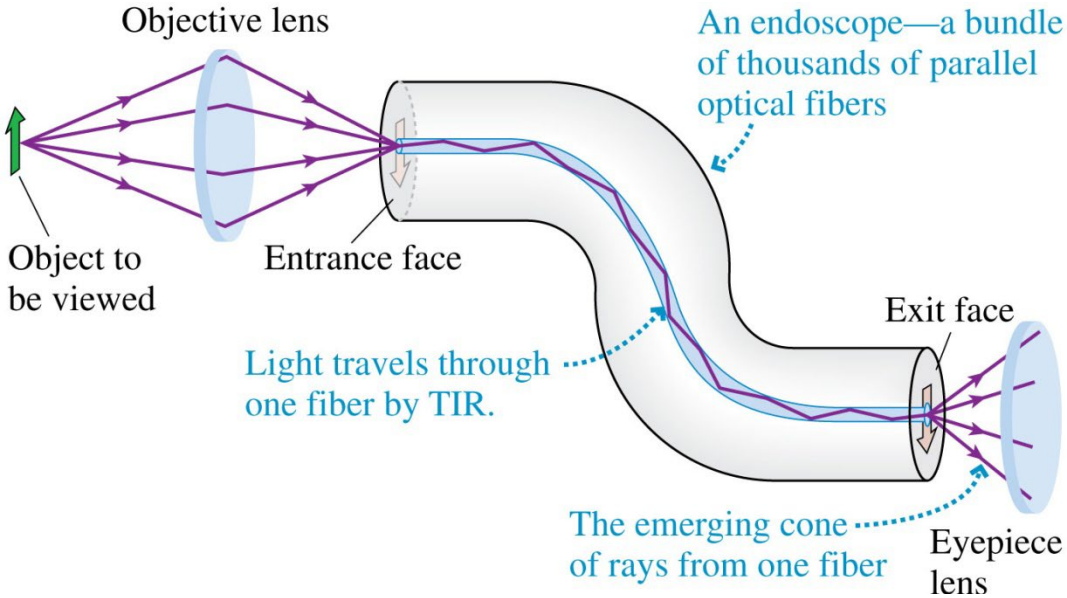
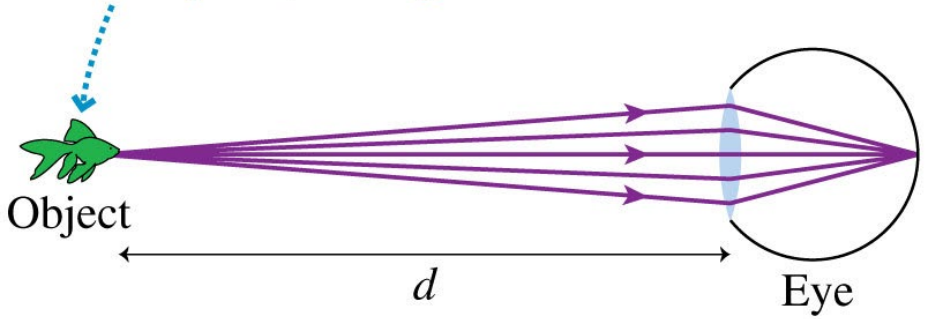


Image Formation by Refraction

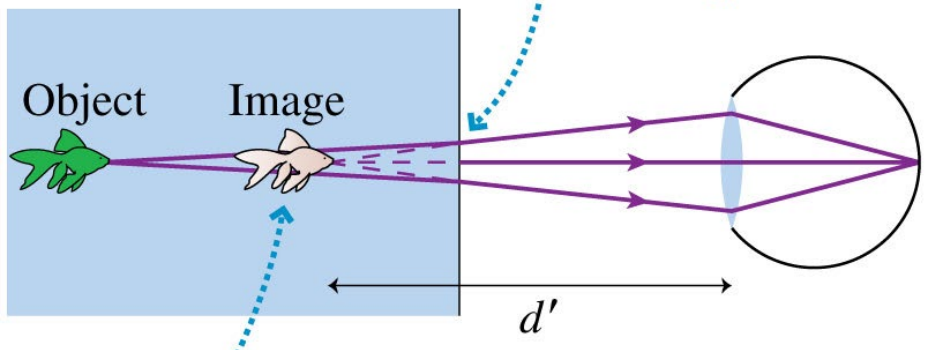
(a) A fish out of water

The rays that reach the eye are diverging from this point, the object.



(b) A fish in the aquarium

Refraction causes the rays to bend at the boundary.



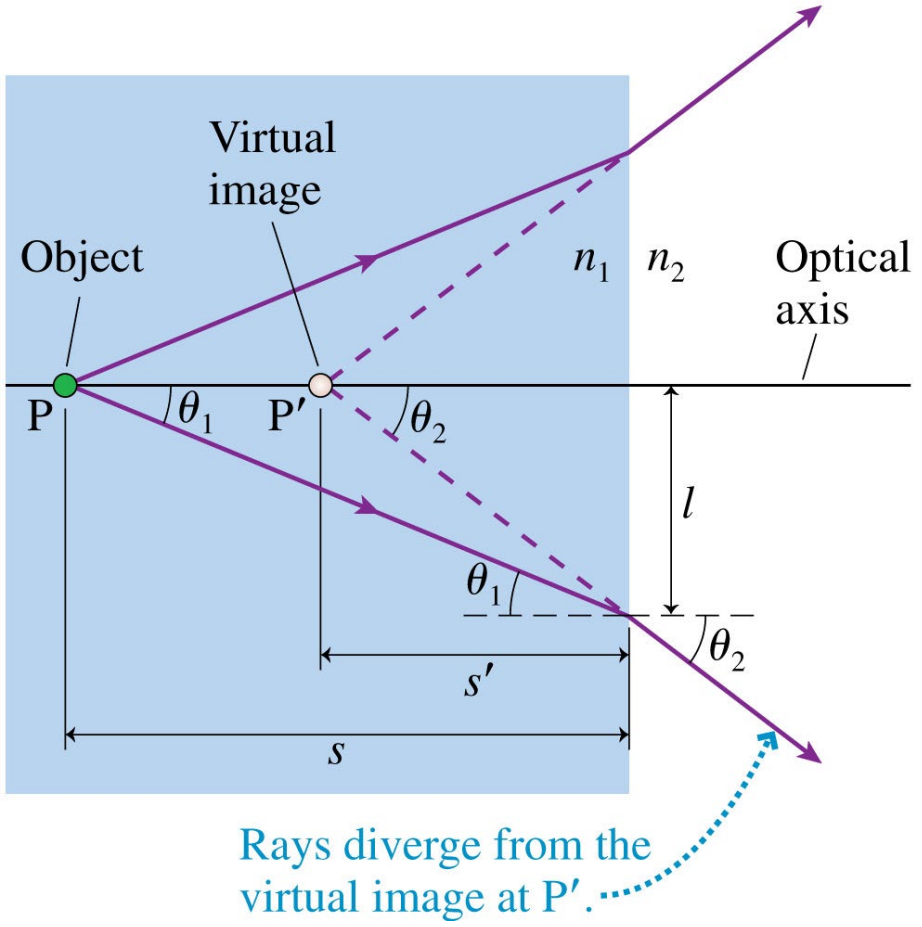
Now the rays that reach the eye are diverging from this point, the image.

Image Formation by Refraction

- Rays emerge from a material with $n_1 > n_2$.
- Consider only **paraxial rays**, for which θ_1 and θ_2 are quite small.
- In this case:

$$s' = \frac{n_2}{n_1} s$$

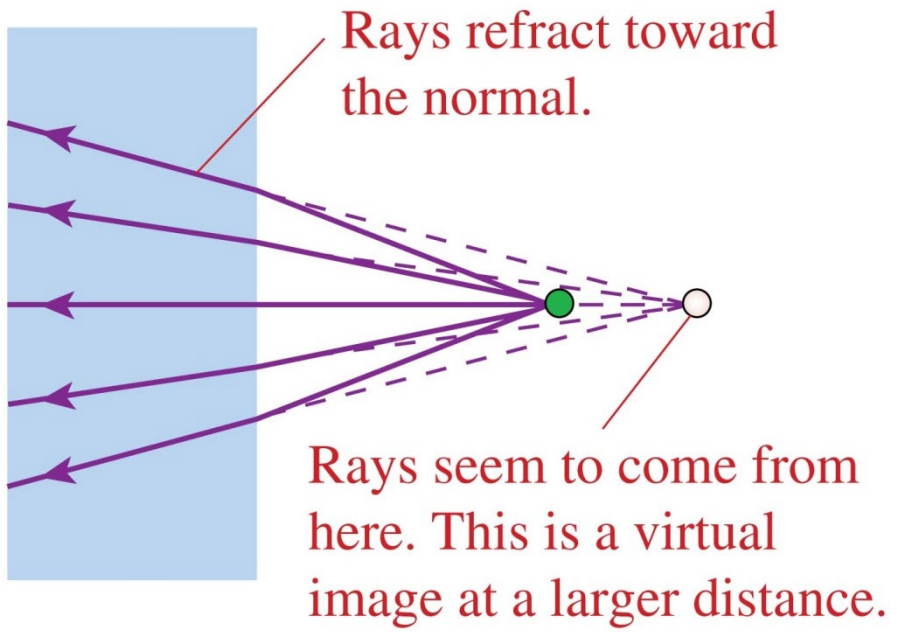
where s is the **object distance** and s' is the **image distance.**



Q&A 23.6

A fish in an aquarium with flat sides looks out at a hungry cat. To the fish, the distance to the cat appears to be

A. Less than the actual distance.
B. Equal to the actual distance.
✔ C. More than the actual distance.



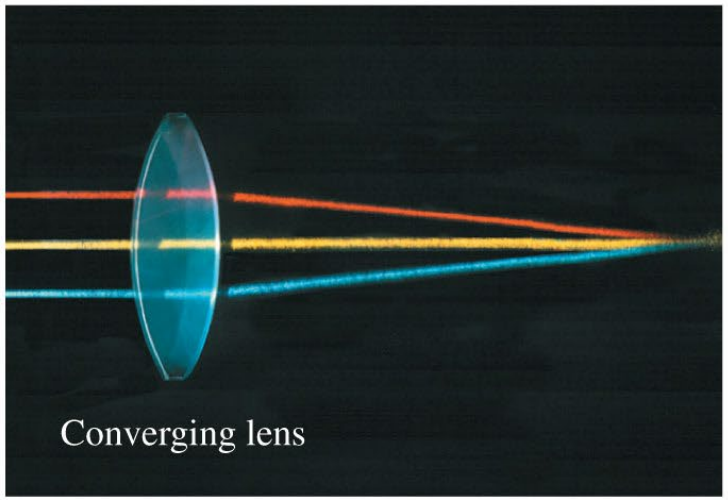
Q&A 23.8

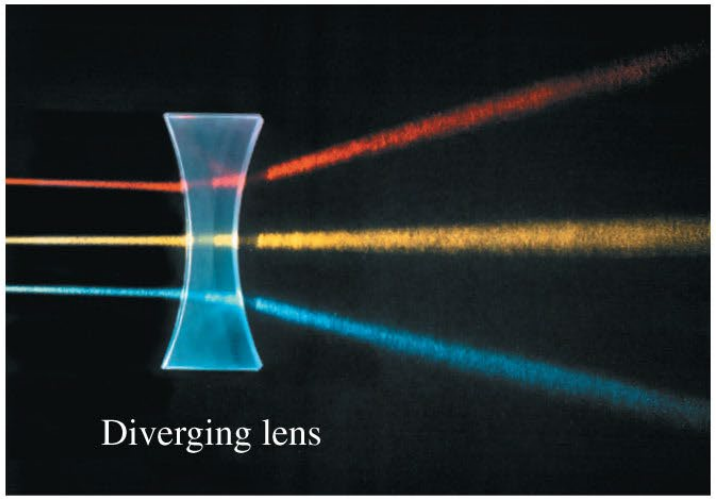
You can use the Sun's rays and a lens to start a fire. To do so, you should use

✔A. A converging lens.

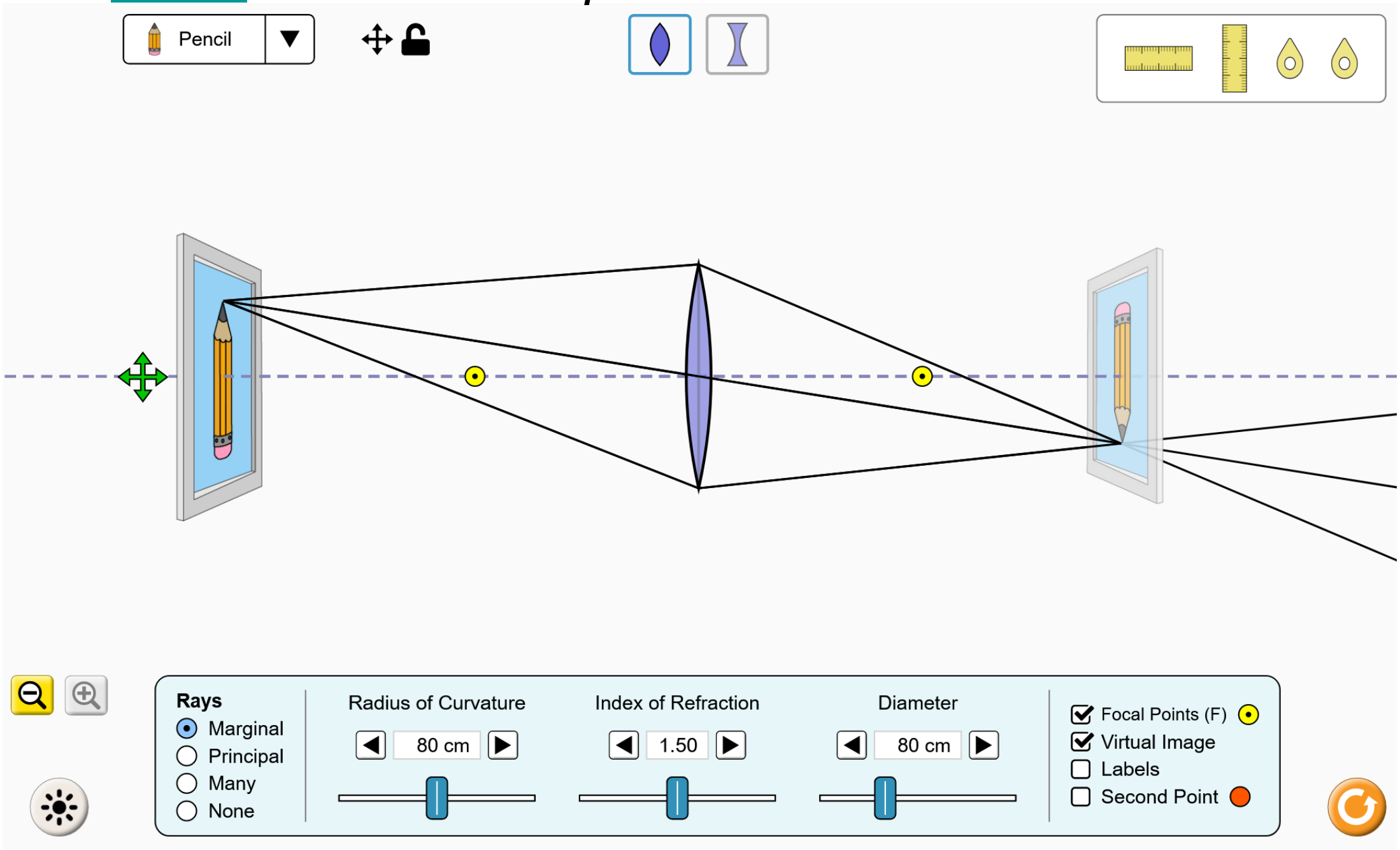
B. A diverging lens.

C. Either a converging or a diverging lens will work if you use it correctly.



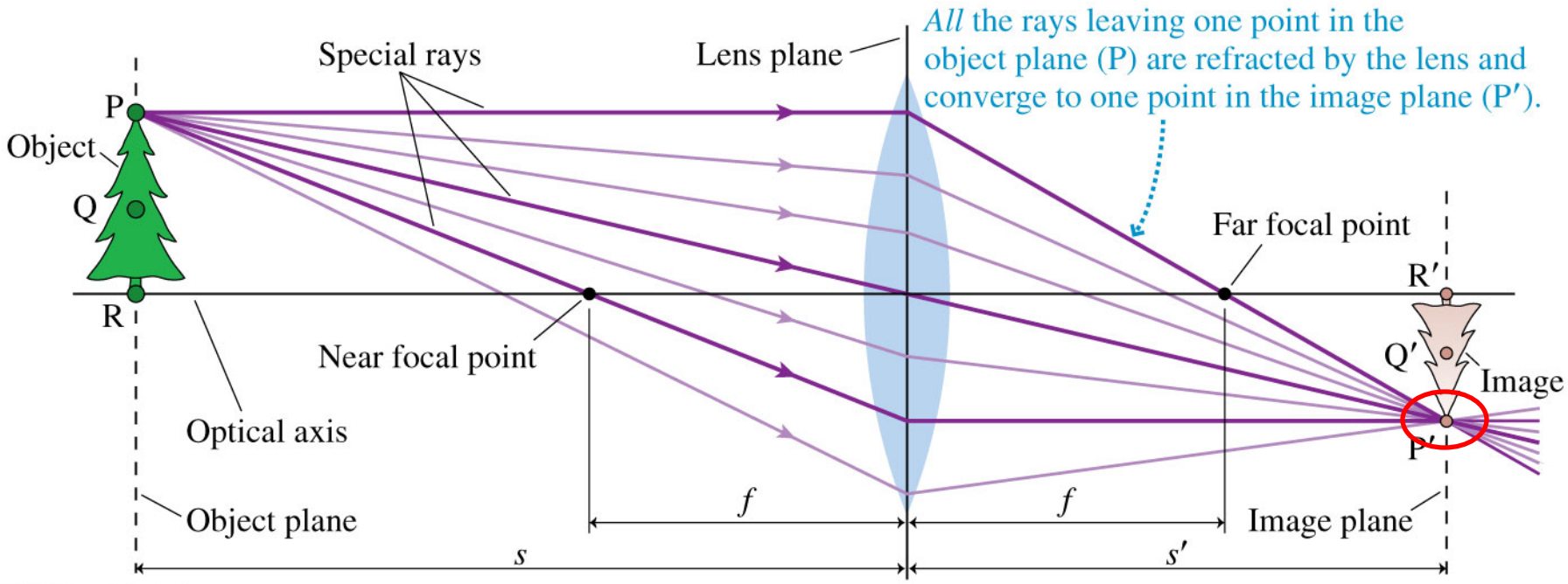


PhET: *Geometric Optics*



Thin Lenses: Ray Tracing

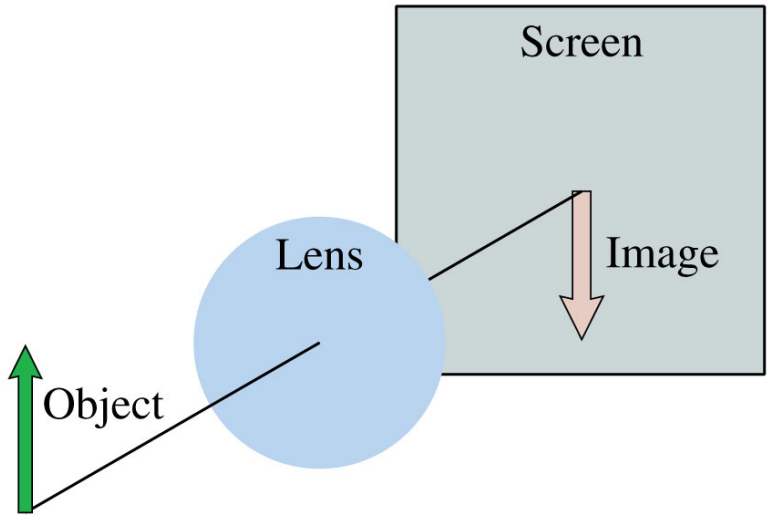
Rays from an object point P are refracted by the lens and converge to a real image at point P′.



$$\frac{1}{s} + \frac{1}{s'} = \frac{1}{f} \qquad \text{(thin-lens equation)}$$

Q&A 23.9

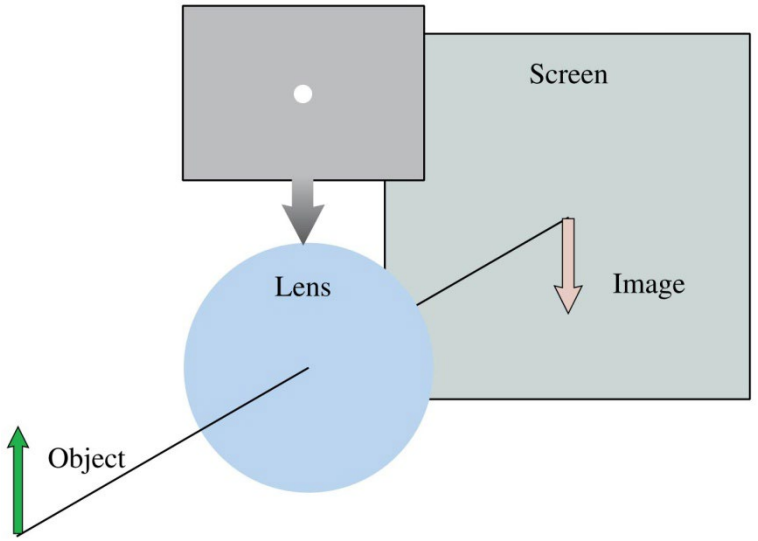
A lens produces a sharply focused, inverted image on a screen. What will you see on the screen if the lens is removed?



A. An inverted but blurry image.
B. An image that is dimmer but otherwise unchanged.
C. A sharp, upright image.
D. A blurry, upright image.
✔ E. No image at all.

Q&A 23.11

A lens produces a sharply focused, inverted image on a screen. What will you see on the screen if the lens is covered by a dark mask having only a small hole (2.44 mm) in the center?



A. An inverted but blurry image.

✔ B. An image that is dimmer but otherwise unchanged.

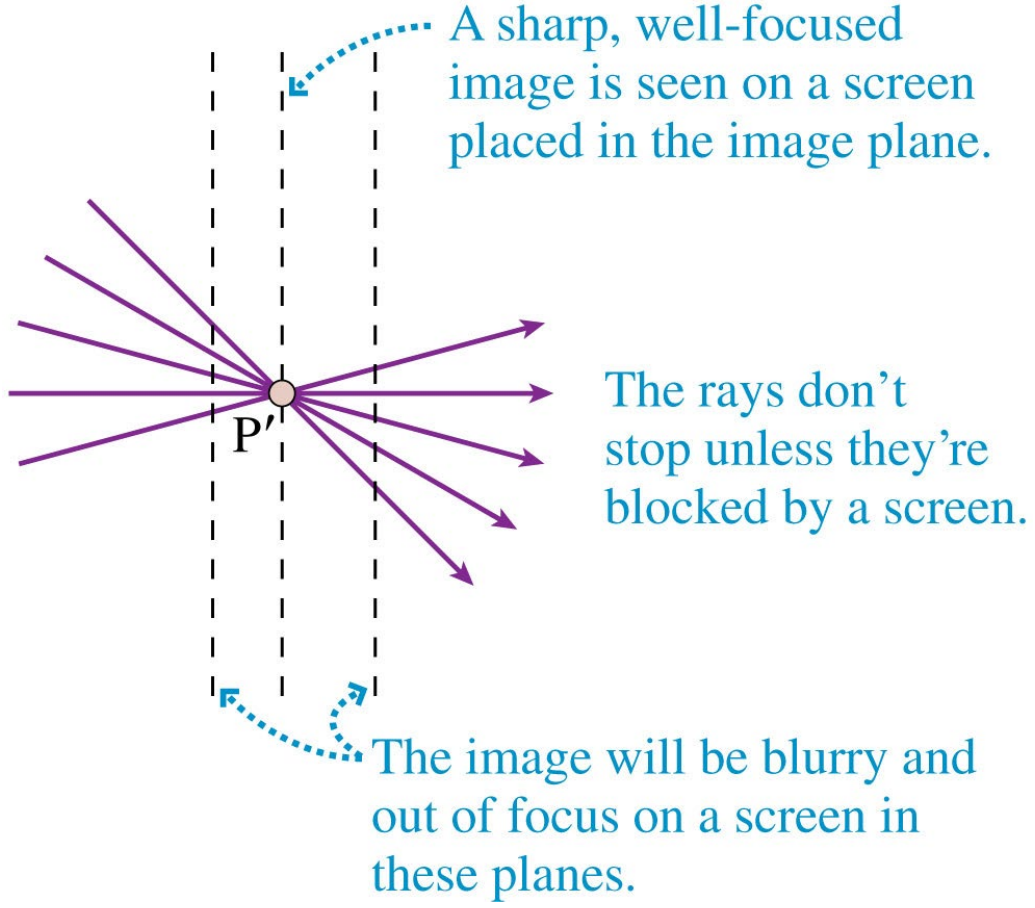
C. Only the middle piece of the image.

D. A circular diffraction pattern.

E. No image at all.

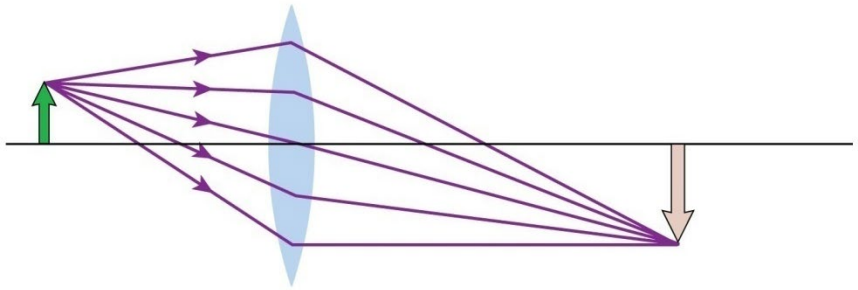
Image Formation

- The figure is a close-up view of the rays very near the **image plane**.
- To **focus** an image, you must either move the screen to coincide with the image plane or move the lens or object to make the image plane coincide with the screen.



Q 23.12

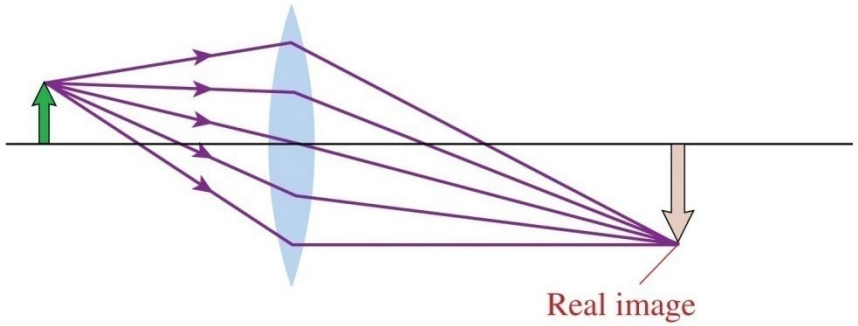
A lens creates an image as shown. In this situation, the **object distance** s is



A. Larger than the focal length f.
B. Equal to the focal length f.
C. Smaller than focal length f.
D. Impossible to determine.

Q&A 23.12

A lens creates an image as shown. In this situation, the **object distance** s is

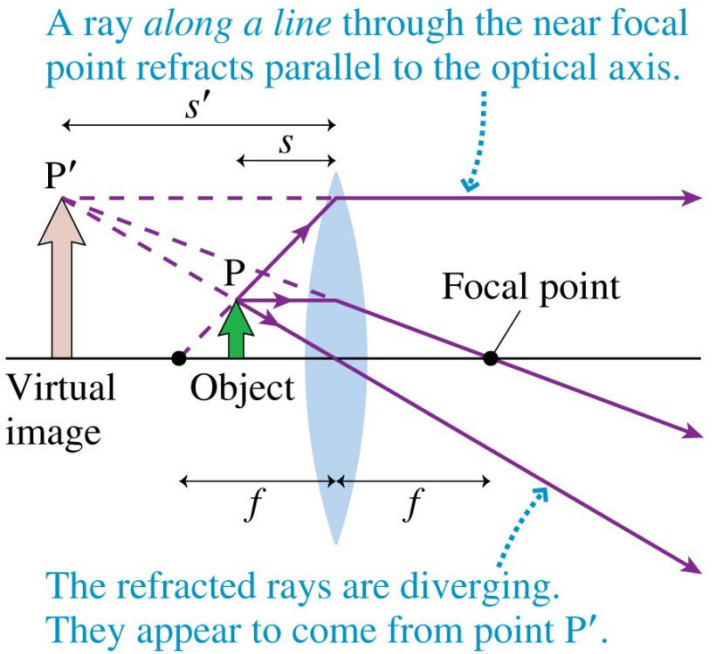


✔ **A. Larger than the focal length f.**

B. Equal to the focal length f.

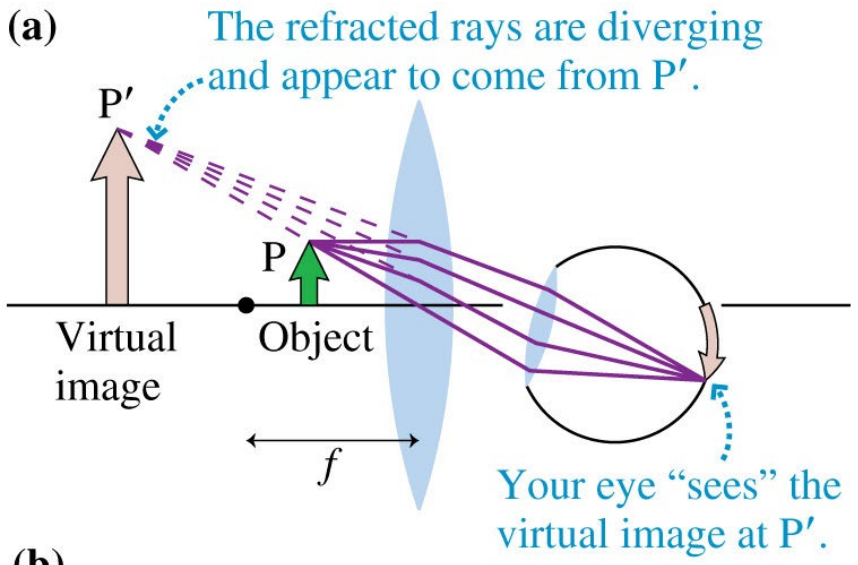
C. Smaller than focal length f.

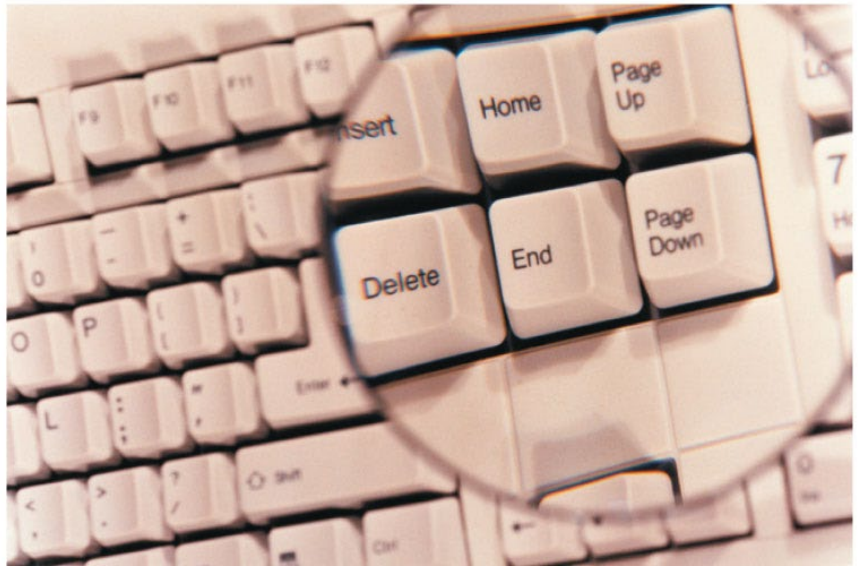
D. Impossible to determine.



Virtual Images

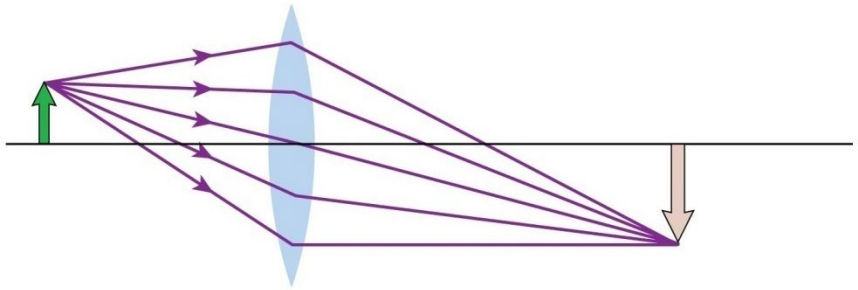
- You can “see” a virtual image by looking through the lens.
- This is exactly what you do with a magnifying glass, microscope or binoculars.





Q&A 23.13

A lens creates an image as shown. In this situation, the **image distance** s' is



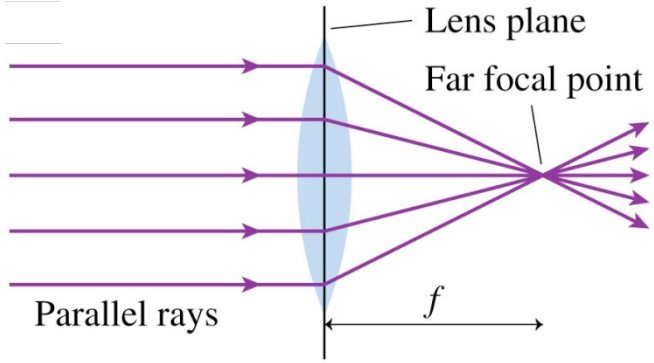
✔ A. Larger than the focal length f.

B. Equal to the focal length f.

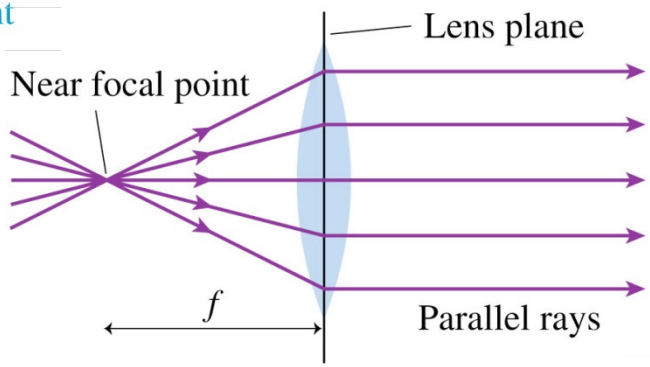
C. Smaller than focal length f.

D. Impossible to determine.

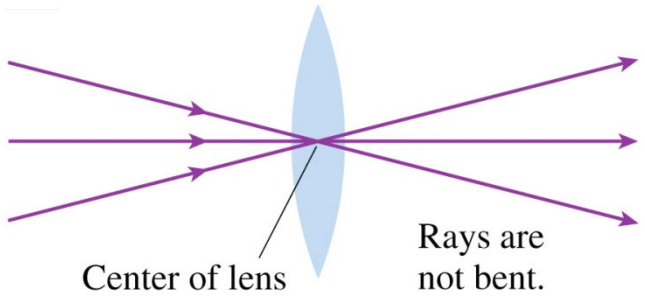
Ray Tracing for Converging Lenses



Any ray initially parallel to the optical axis will refract through the focal point on the far side of the lens.



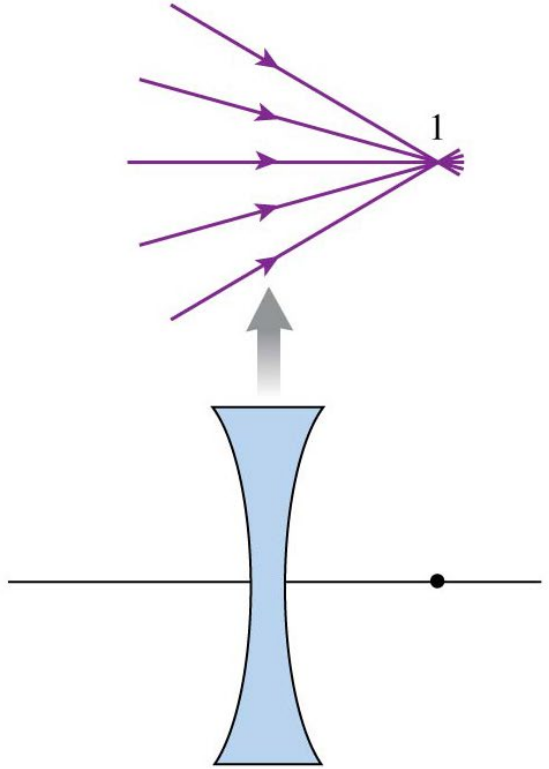
Any ray passing through the near focal point emerges from the lens parallel to the optical axis.

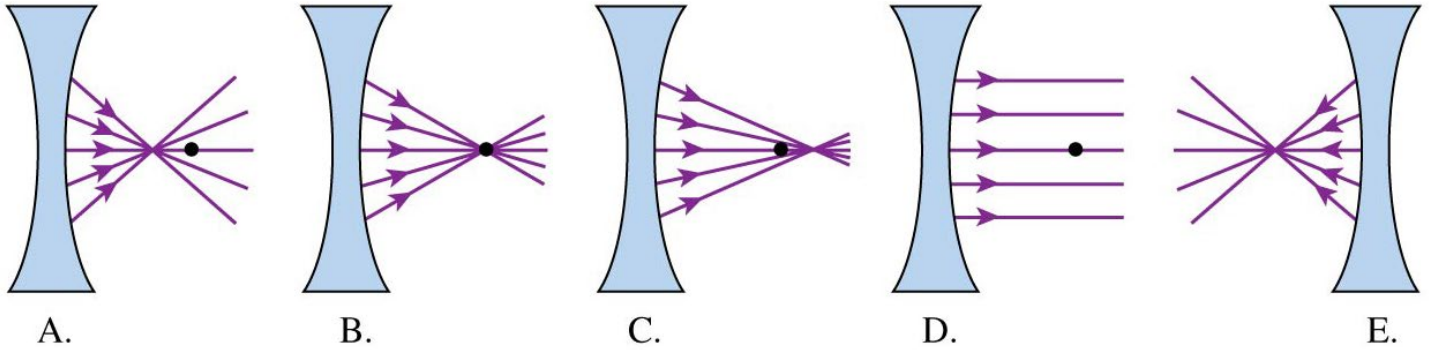


Any ray directed at the center of the lens passes through in a straight line.

Q 23.14

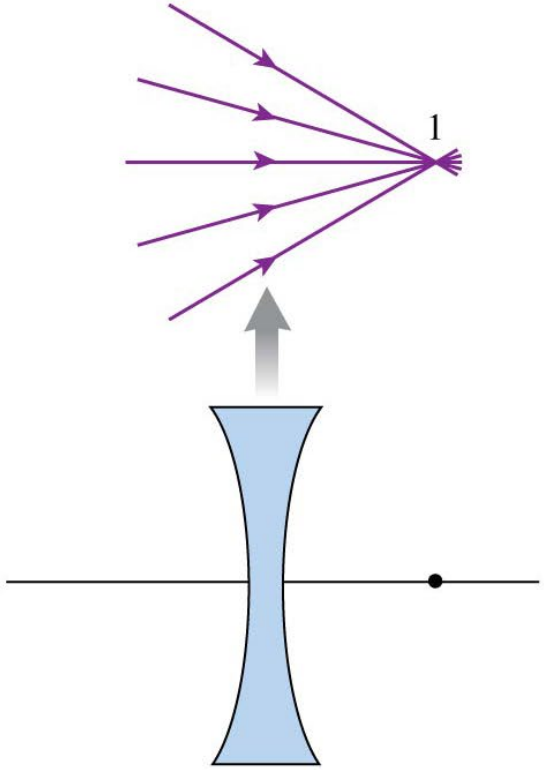
Light rays are converging to point 1. The lens is inserted into the rays with its focal point at point 1. Which picture shows the rays leaving the lens?

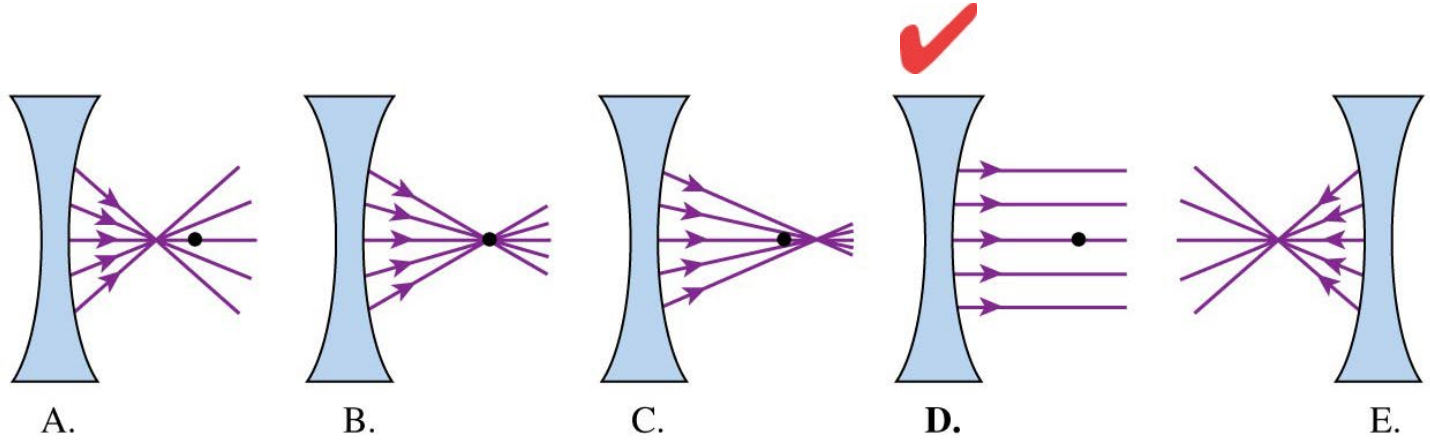




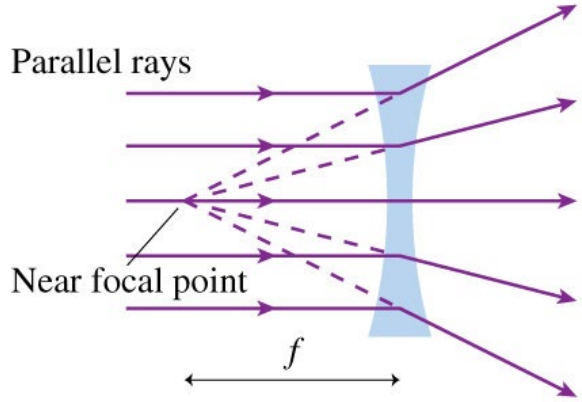
Q&A 23.14

Light rays are converging to point 1. The lens is inserted into the rays with its focal point at point 1. Which picture shows the rays leaving the lens?

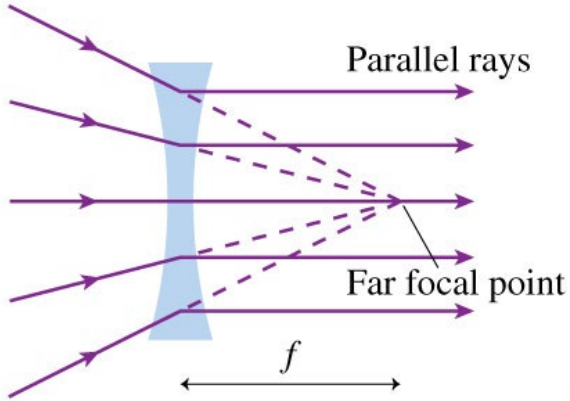




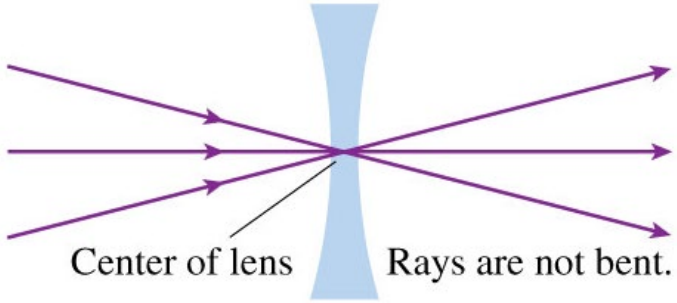
Ray Tracing for Diverging Lenses



Any ray initially parallel to the optical axis diverges along a line through the near focal point.



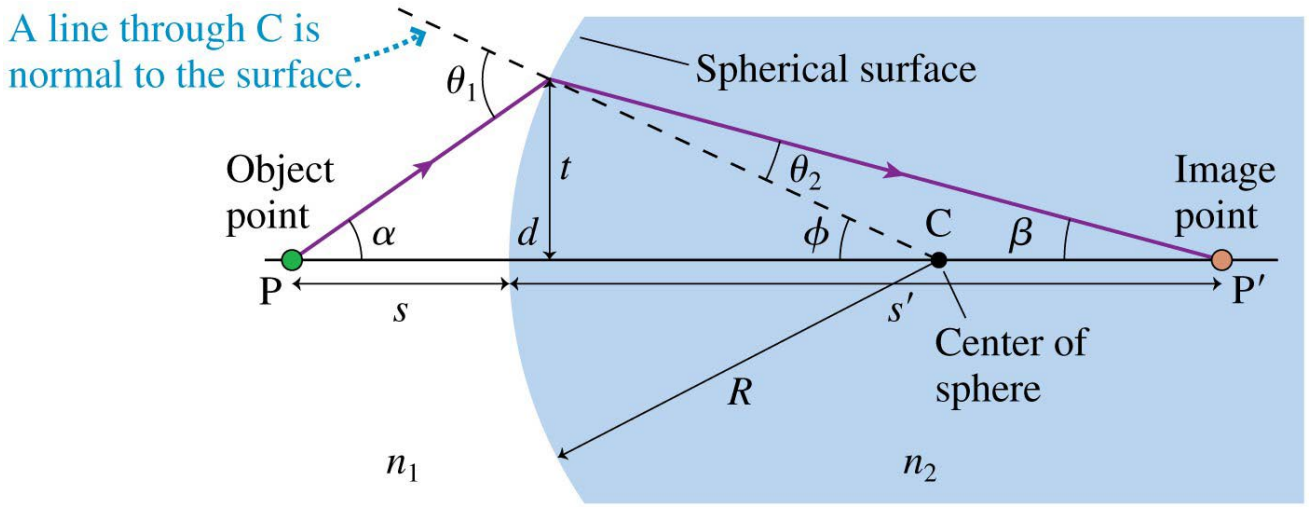
Any ray directed along a line toward the far focal point emerges from the lens parallel to the optical axis.



Any ray directed at the center of the lens passes through in a straight line.

Thin Lenses: Refraction Theory

- Consider a spherical boundary between two transparent media with indices of refraction n_1 and n_2.
- The sphere has radius of curvature R and is centered at point C.



Thin Lenses: Refraction Theory

- If an object is located at distance s from a spherical refracting surface, an image will be formed at distance s' given by

$$\frac{n_1}{s} + \frac{n_2}{s'} = \frac{n_2 - n_1}{R}$$

TABLE 34.2 Sign convention for refracting surfaces

	Positive	Negative
R	Convex toward the object	Concave toward the object
s'	Real image, opposite side from object	Virtual image, same side as object

Lenses

In an actual lens, rays refract *twice*, at spherical surfaces having **radii of curvature** R_1 and R_2.

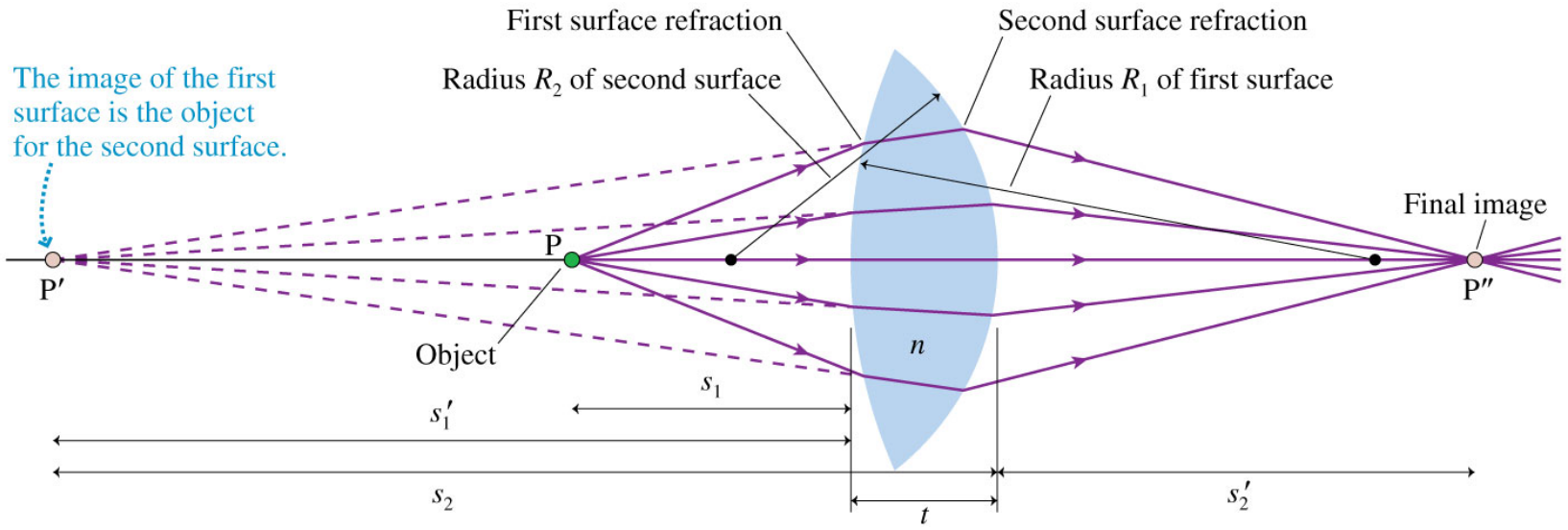


TABLE 23.4 Sign convention for thin lenses

	Positive	**Negative**
R_1, R_2	Convex toward the object	Concave toward the object
f	Converging lens, thicker in center	Diverging lens, thinner in center
s'	Real image, opposite side from object	Virtual image, same side as object

The Thin Lens Equation

The object distance s is related to the image distance s' by:

$$\frac{1}{s} + \frac{1}{s'} = \frac{1}{f} \qquad \text{(thin-lens equation)}$$

where f is the focal length of the lens, which can be found from:

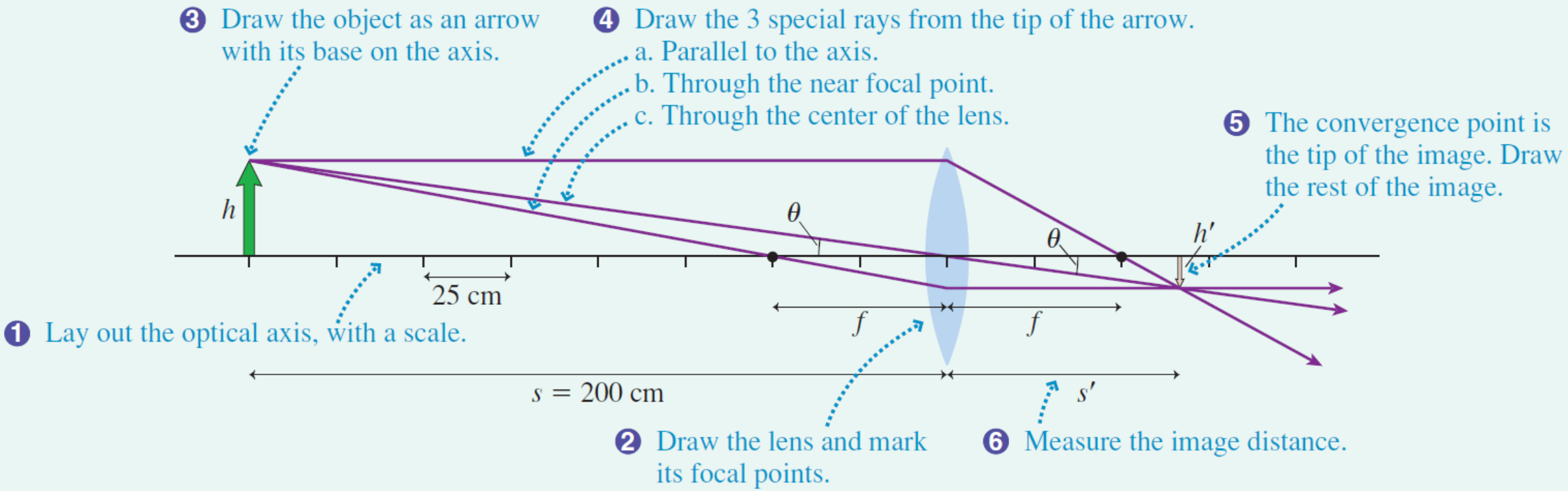
$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \qquad \text{(lens maker's equation)}$$

where R_1 is the radius of curvature of the first surface, and R_2 is the radius of curvature of the second surface, and the material surrounding the lens has $n = 1$.

Example: Problem 23.73

An object is 60 cm from a screen. What are the radii of curvature of a symmetric converging plastic lens ($n = 1.59$) that will form an image on the screen twice the height of the object?

FIGURE 23.38 Ray-tracing diagram for Example 23.8.



$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \quad \text{(lens maker's equation)}$$

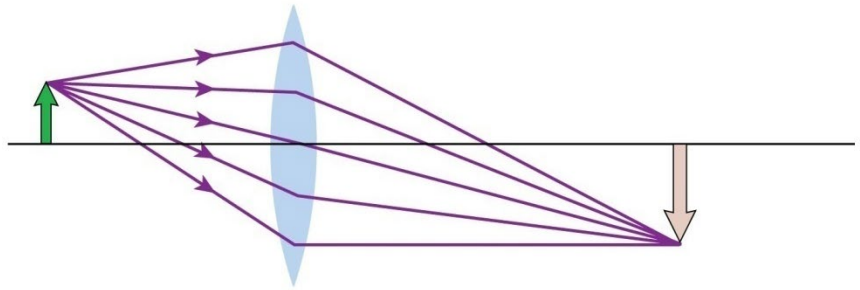
$$\frac{1}{s} + \frac{1}{s'} = \frac{1}{f} \quad \text{(thin-lens equation)}$$

$$m = -\frac{s'}{s}$$

$$\frac{h'}{s'} = \frac{h}{s}$$

Q&A 23.15

A lens creates an image as shown. In this situation,



A. $s < f$.

✔ B. $f < s < 2f$.

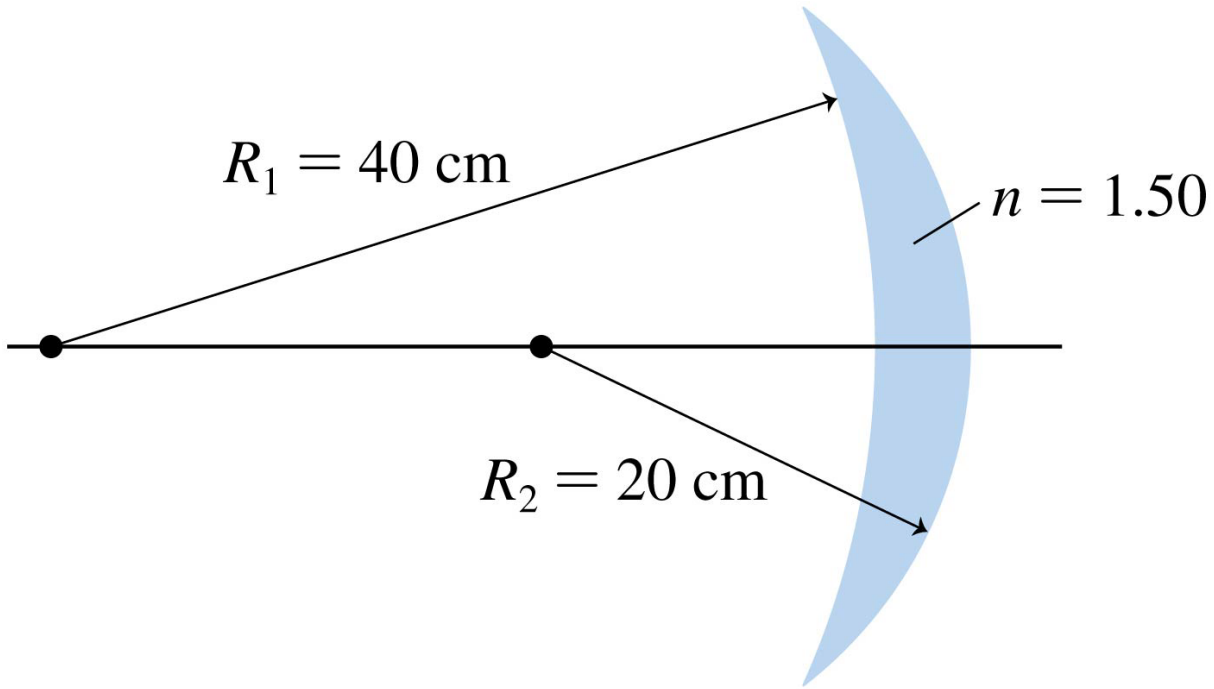
C. $s > 2f$.

D. There isn't enough information to compare s to f.

The image is real, which requires $s > f$.
The image is taller than the object, and $s' > s$ requires $s < 2f$.

Example: Focal Length of a Meniscus Lens

What is the focal length of the glass *meniscus lens* shown in FIGURE 34.42? Is this a converging or diverging lens?



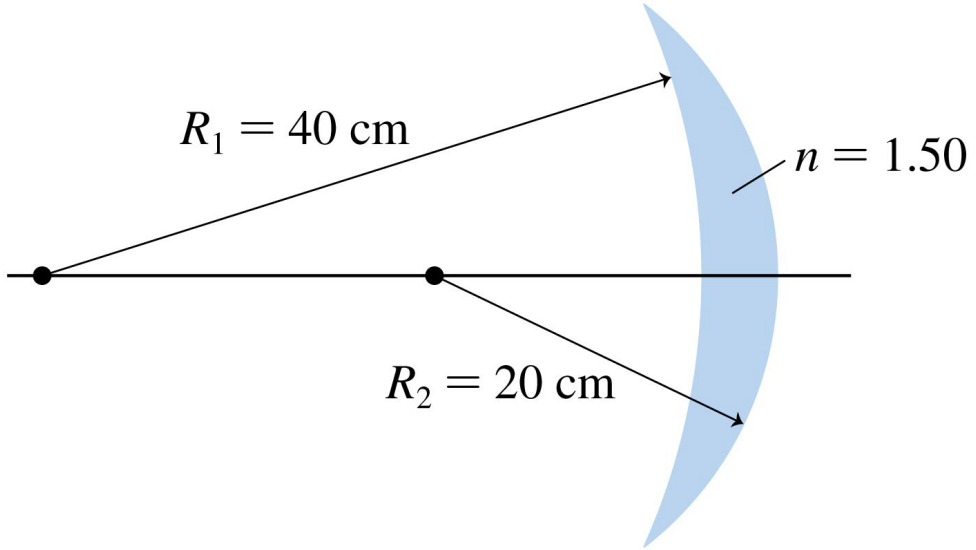
Example: Focal Length of a Meniscus Lens

SOLVE If the object is on the left, then the first surface has $R_1 = -40$ cm (concave toward the object) and the second surface has $R_2 = -20$ cm (also concave toward the object). The index of refraction of glass is $n = 1.50$, so the lens maker's equation is

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = (1.50-1)\left(\frac{1}{-40\text{ cm}} - \frac{1}{-20\text{ cm}}\right)$$

$$= 0.0125\text{ cm}^{-1}$$

Inverting this expression gives $f = 80$ cm. This is a converging lens, as seen both from the positive value of f and from the fact that the lens is thicker in the center.

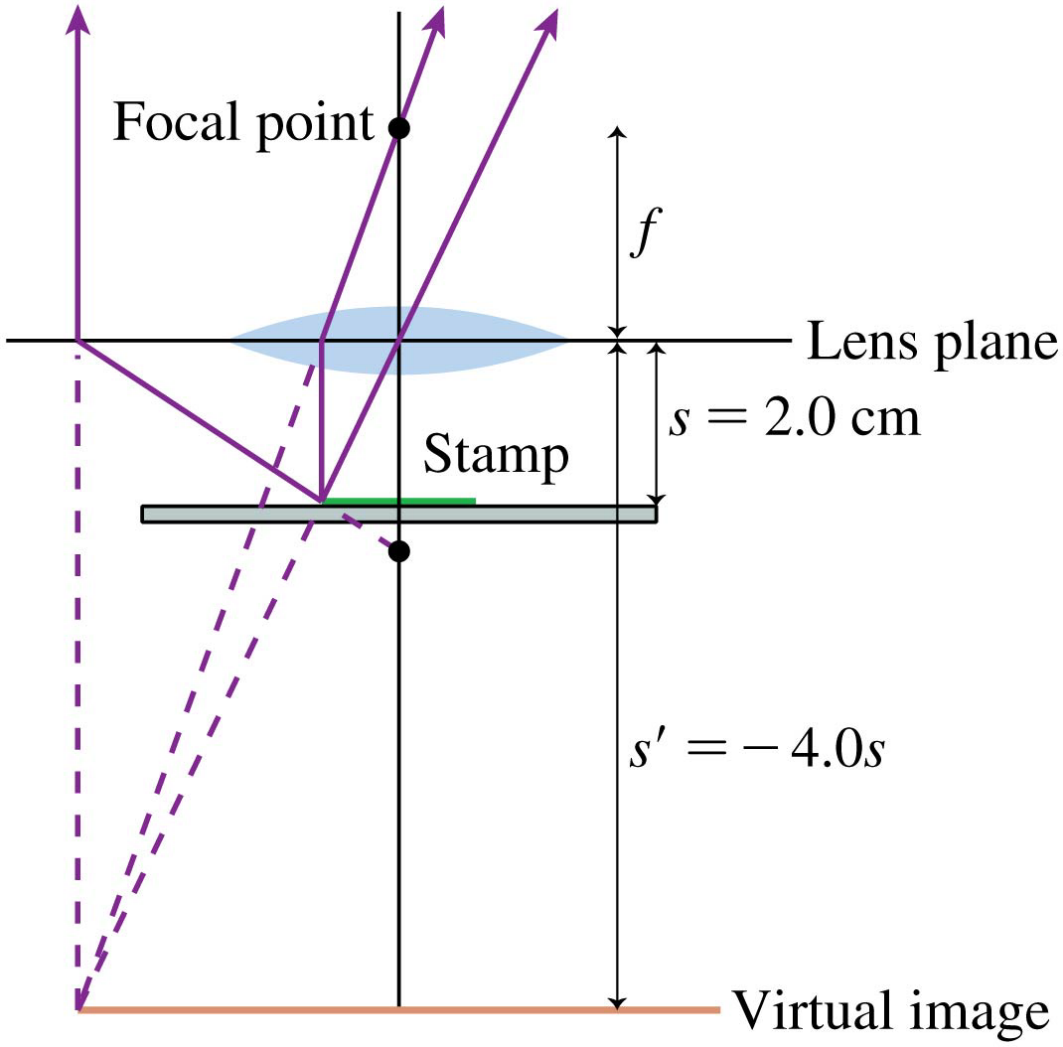


Example: A Magnifying Lens

A stamp collector uses a magnifying lens that sits 2.0 cm above the stamp. The magnification is 4.0. What is the focal length of the lens?

MODEL A magnifying lens is a converging lens with the object distance less than the focal length ($s < f$). Assume it is a thin lens.

VISUALIZE **FIGURE 34.44** shows the lens and a ray-tracing diagram. We do not need to know the actual shape of the lens, so the figure shows a generic converging lens.



Example 34.14 A Magnifying Lens

SOLVE A virtual image is upright, so $m = +4.0$. The magnification is $m = -s'/s$, thus

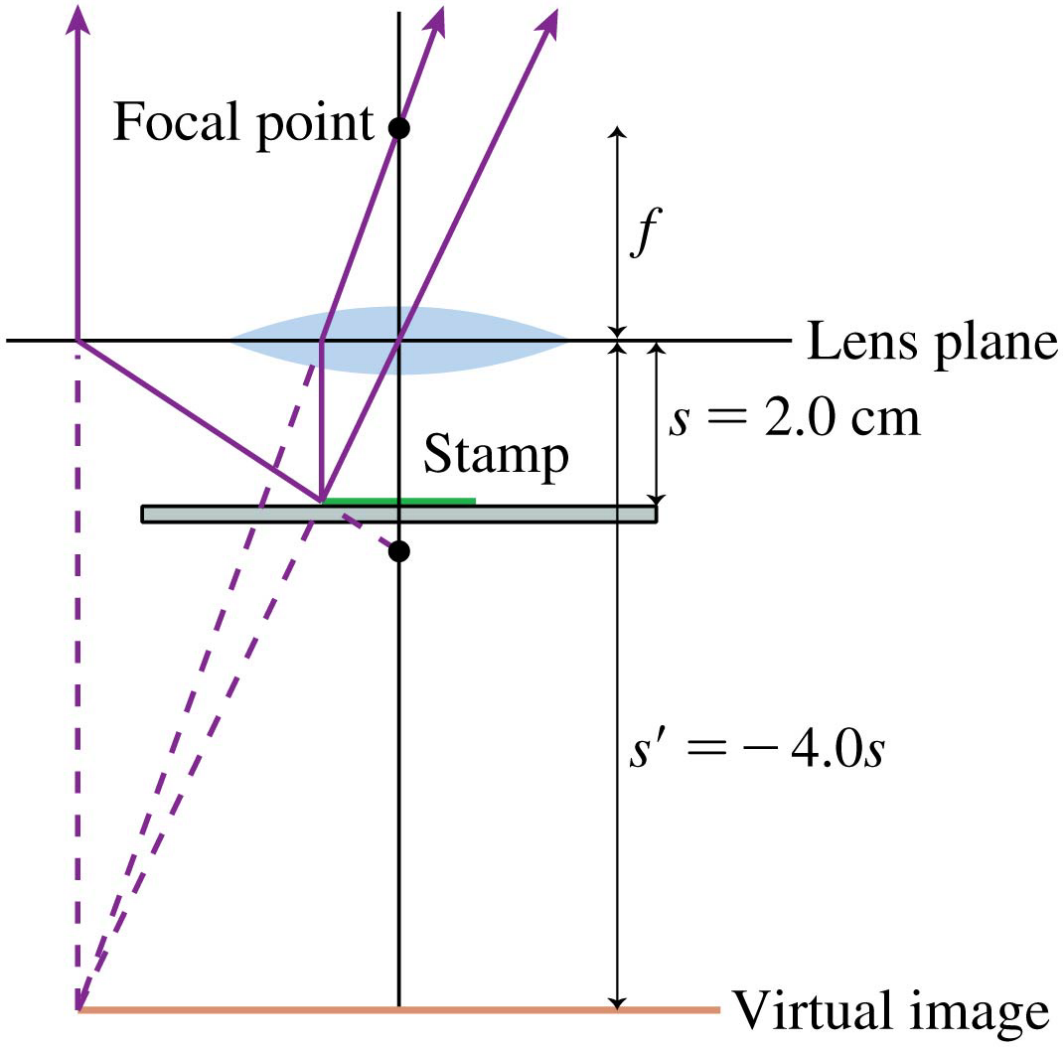
$$s' = -4.0s = -(4.0)(2.0\text{ cm}) = -8.0\text{ cm}$$

We can use s and s' in the thin-lens equation to find the focal length:

$$\frac{1}{f} = \frac{1}{s} + \frac{1}{s'} = \frac{1}{2.0\text{ cm}} + \frac{1}{-8.0\text{ cm}} = 0.375\text{ cm}^{-1}$$

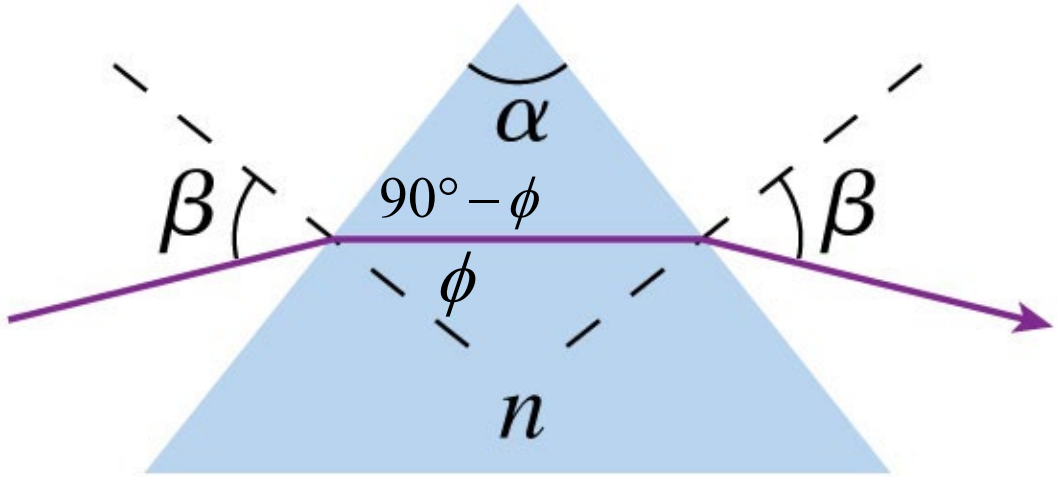
$$f = 2.7\text{ cm}$$

ASSESS $f > 2$ cm, as expected.



Example: Problem 23.54

There is one angle of incidence β onto a prism for which the light inside an isosceles prism travels parallel to the base and emerges at angle β. Find an expression for β in terms of the prism's apex angle α and index of refraction n.



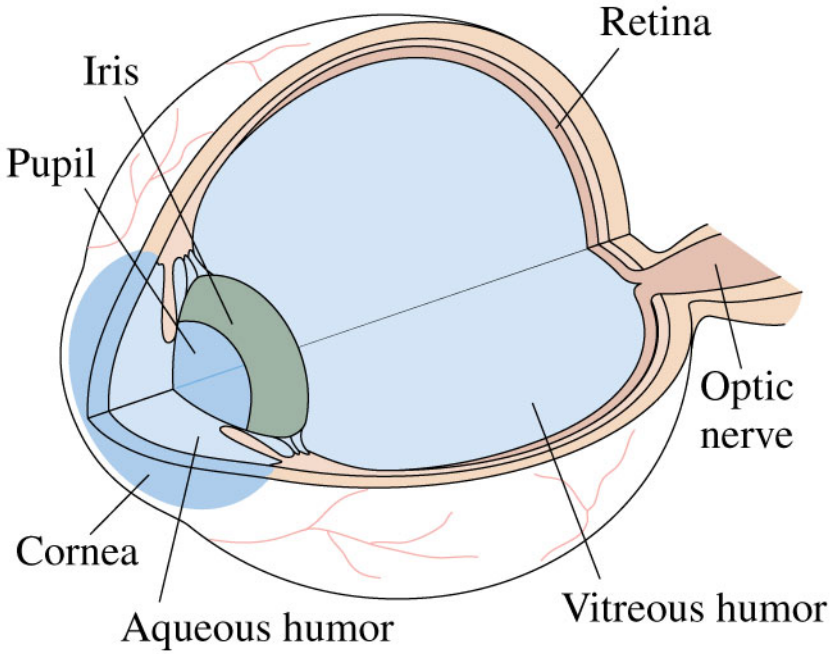
Snell's law: $n_{\text{air}} \sin\beta = n\sin\phi \Rightarrow \beta = \sin^{-1}\left[n\sin\phi\right]$

$$(90° - \phi) + \alpha + (90° - \phi) = 180° \Rightarrow \phi = \frac{\alpha}{2}$$

$$\therefore \beta = \sin^{-1}\left[n\sin\left(\tfrac{\alpha}{2}\right)\right]$$

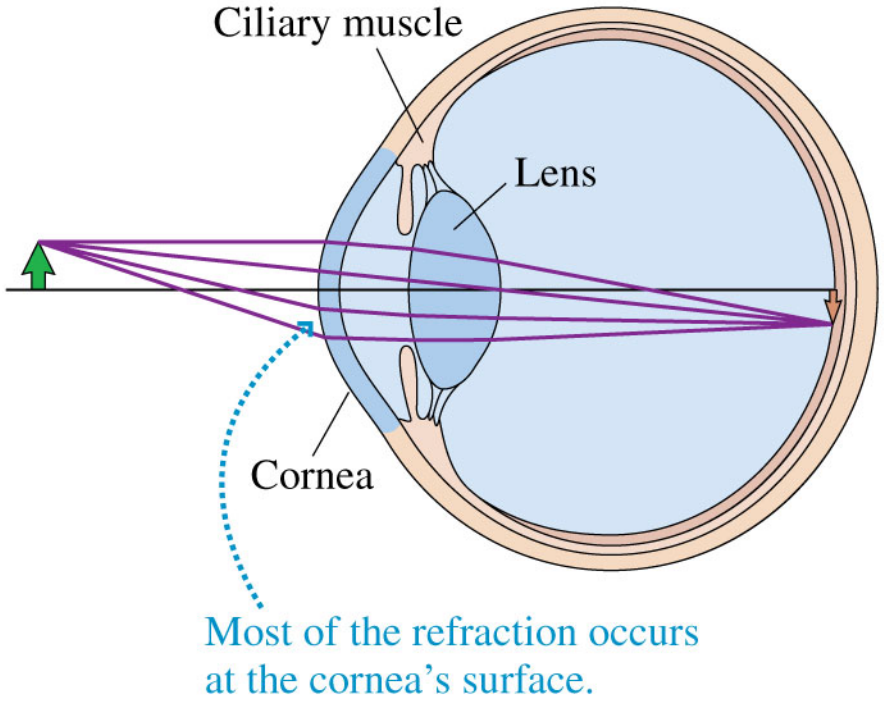
Vision

- The human eye is roughly spherical, about $2.4\ \text{cm}$ in diameter.
- The transparent **cornea** and the *lens* are the eye's refractive elements.
- The eye is filled with a clear, jellylike fluid called the *aqueous humor* and the *vitreous humor*.



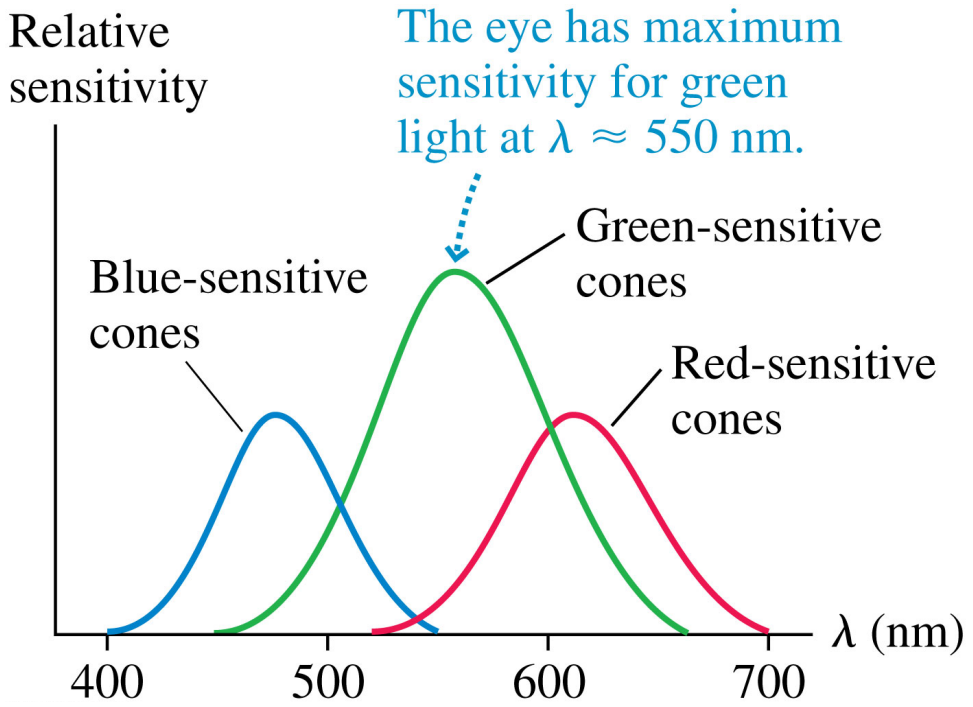
Vision

- The indices of refraction of the aqueous and vitreous humors are 1.34, only slightly different from water.
- The lens has an average index of 1.44.
- The **pupil**, a variable-diameter aperture in the **iris**, automatically opens and closes to control the light intensity.
- The f-number varies from roughly $f/3$ to $f/16$, very similar to a camera!



Color Vision

- The eye's detector, the retina, contains light-sensitive cells called cones.
- The figure shows the wavelength responses of the three types of cones in a human eye.



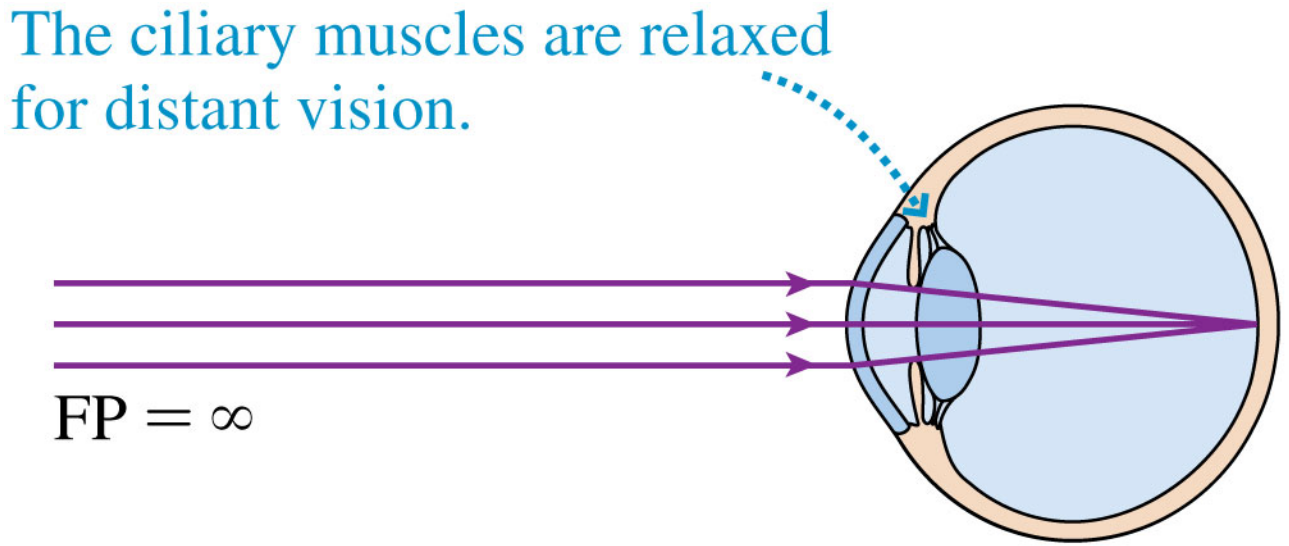
- The relative response of the different cones is interpreted by your brain as light of a particular color.
- Other animals, with slightly different retinal cells, can see ultraviolet or infrared wavelengths that we cannot see.

Focusing and Accommodation

- The eye focuses by changing the focal length of the lens by using the *ciliary muscles* to change the curvature of the lens surface.
- Tensing the ciliary muscles causes **accommodation**, which decreases the lens's radius of curvature and thus decreases its focal length.

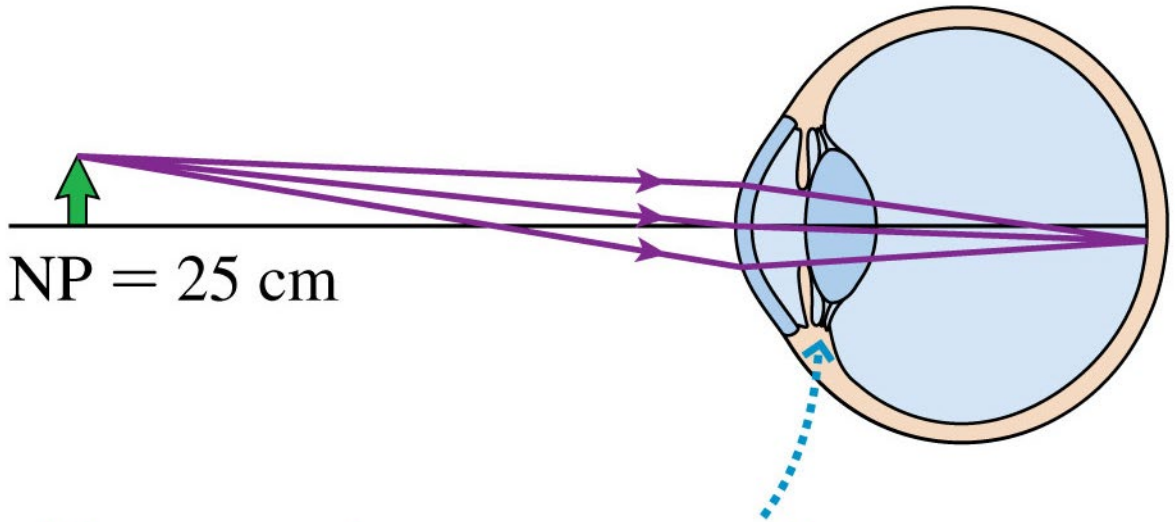
Focusing and Accommodation

- The farthest distance at which a relaxed eye can focus is called the eye's **far point** (FP).
- The far point of a normal eye is infinity; that is, the eye can focus on objects extremely far away.



Focusing and Accommodation

- The closest distance at which an eye can focus, using maximum accommodation, is the eye's **near point** (NP).



Corrective Lenses

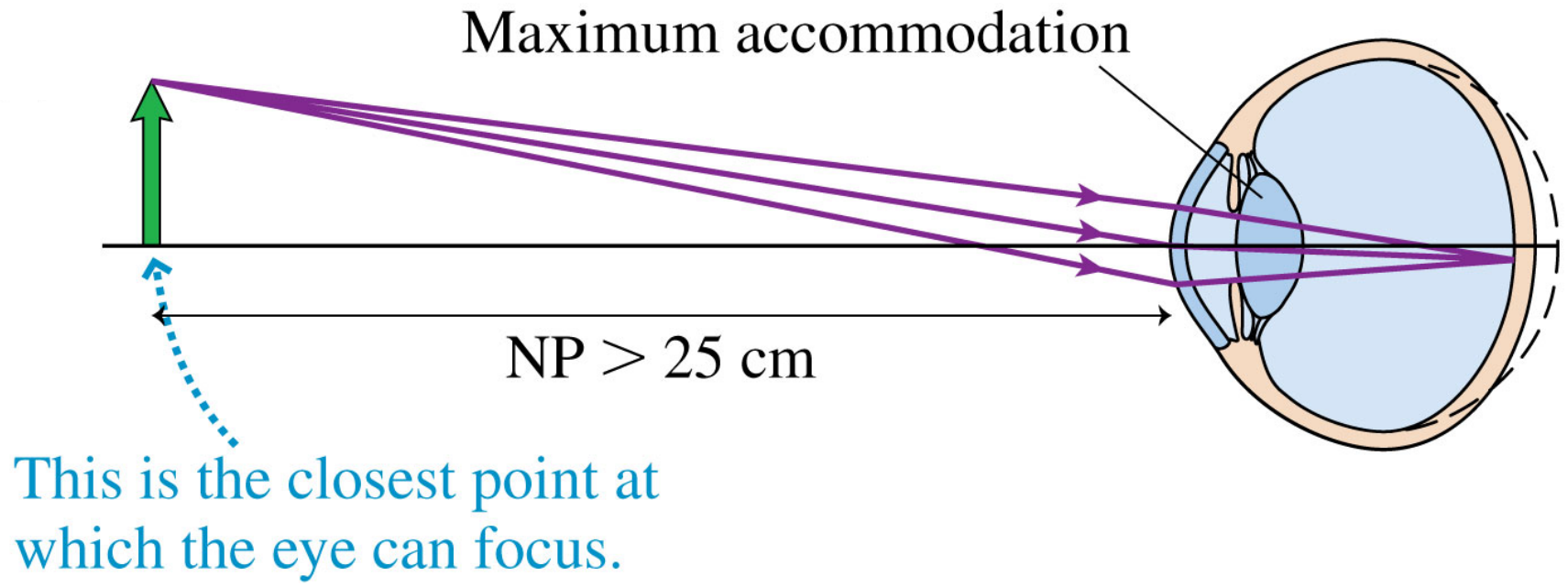
- Corrective lenses are prescribed not by their focal length but by their power.
- The power of a lens is the inverse of its focal length:

$$\text{Power of a lens} = P = \frac{1}{f}$$

- The SI unit of lens power is the diopter, abbreviated D, defined as $1\ \text{D} = 1\ \text{m}^{-1}$.
- Thus a lens with $f = 50\ \text{cm} = 0.50\ \text{m}$ has power $P = 2.0\ \text{D}$.

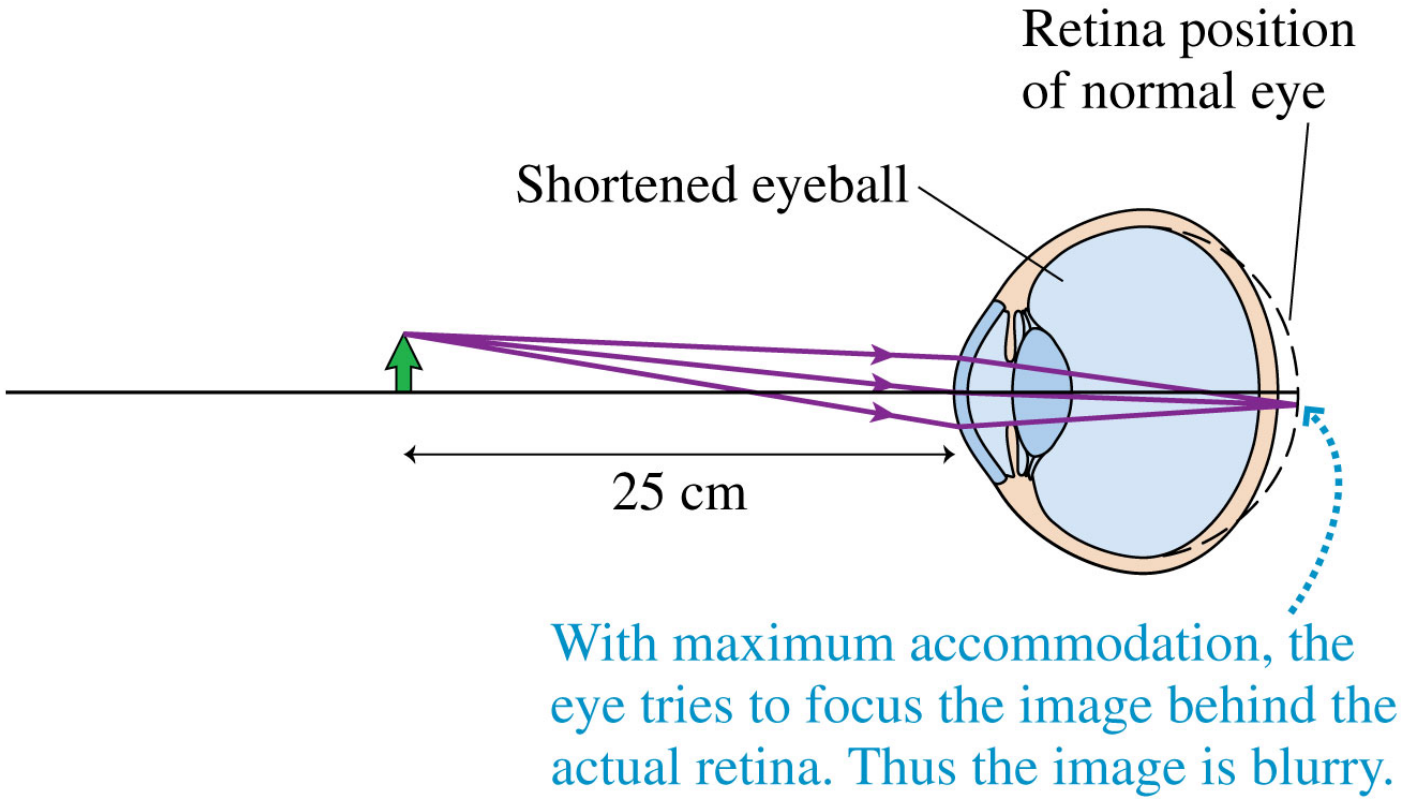
Hyperopia

- A person who is *farsighted* can see faraway objects (but even then must use some accommodation rather than a relaxed eye), but their near point is larger than $25\ \text{cm}$, often much larger, so they cannot focus on nearby objects.



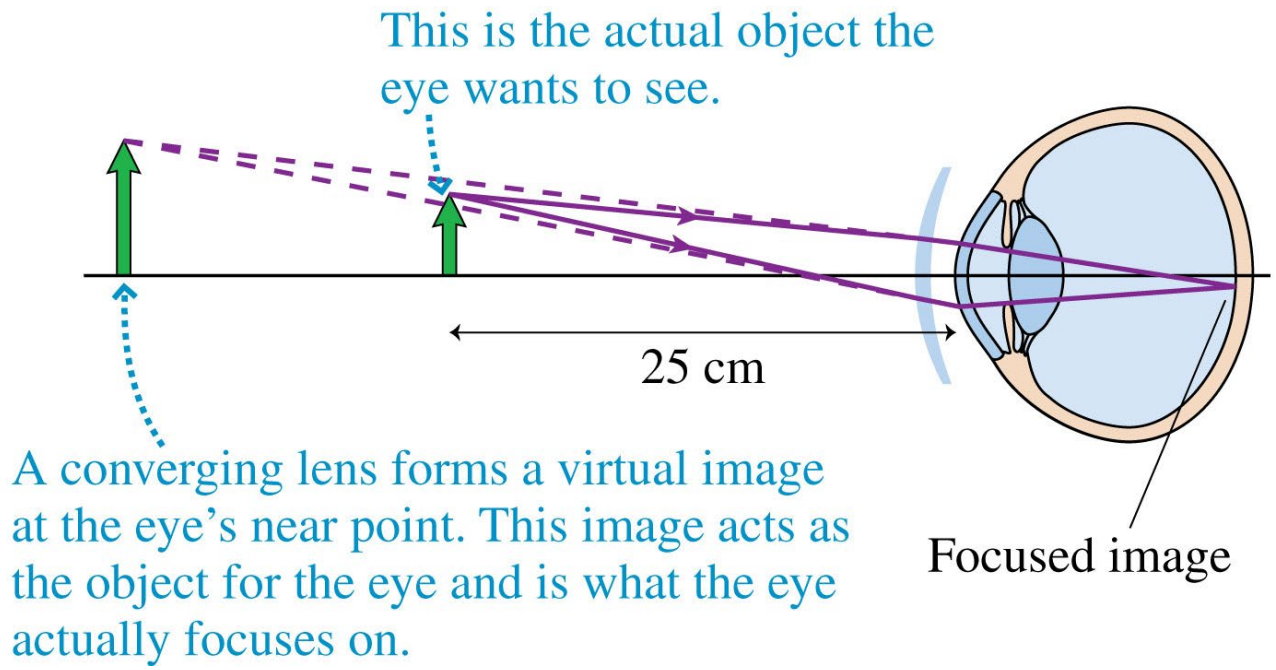
Hyperopia

- The cause of farsightedness—called **hyperopia**—is an eyeball that is too short for the refractive power of the cornea and lens.



Hyperopia

- With hyperopia, the eye needs assistance to focus the rays from a near object onto the closer-than-normal retina.
- This assistance is obtained by adding refractive power with the positive (i.e., converging) lens.



Example: Correcting Hyperopia

Sanjay has hyperopia. The near point of his left eye is 150 cm. What prescription lens will restore normal vision?

MODEL Normal vision will allow Sanjay to focus on an object 25 cm away. In measuring distances, we'll ignore the small space between the lens and his eye.

Example: Correcting Hyperopia

SOLVE Because Sanjay can see objects at 150 cm, using maximum accommodation, we want a lens that creates a virtual image at position $s' = -150$ cm (negative because it's a virtual image) of an object held at $s = 25$ cm. From the thin-lens equation,

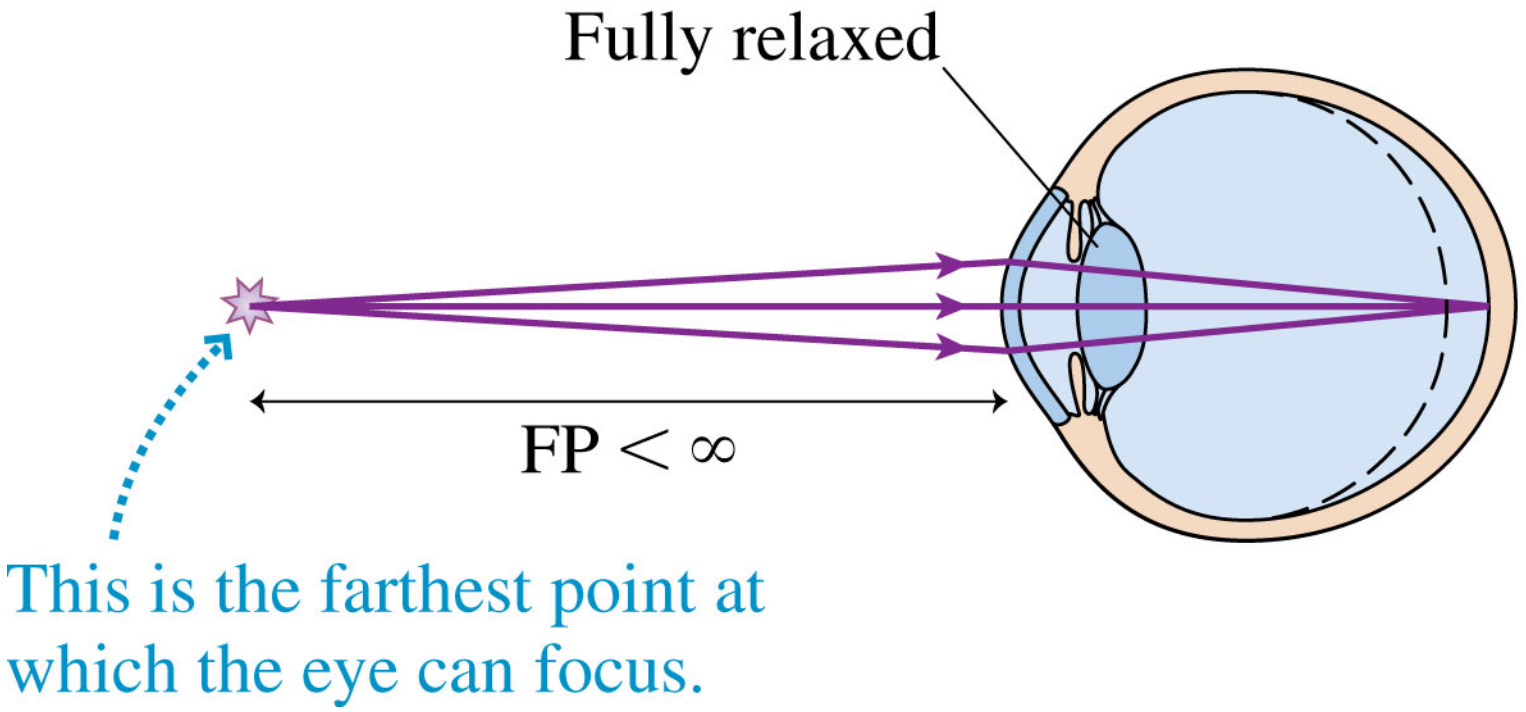
$$\frac{1}{f} = \frac{1}{s} + \frac{1}{s'} = \frac{1}{0.25\ \text{m}} + \frac{1}{-1.50\ \text{m}} = 3.3\ \text{m}^{-1}$$

$1/f$ is the lens power, and m^{-1} are diopters. Thus the prescription is for a lens with power $P = 3.3$ D.

ASSESS Hyperopia is always corrected with a converging lens.

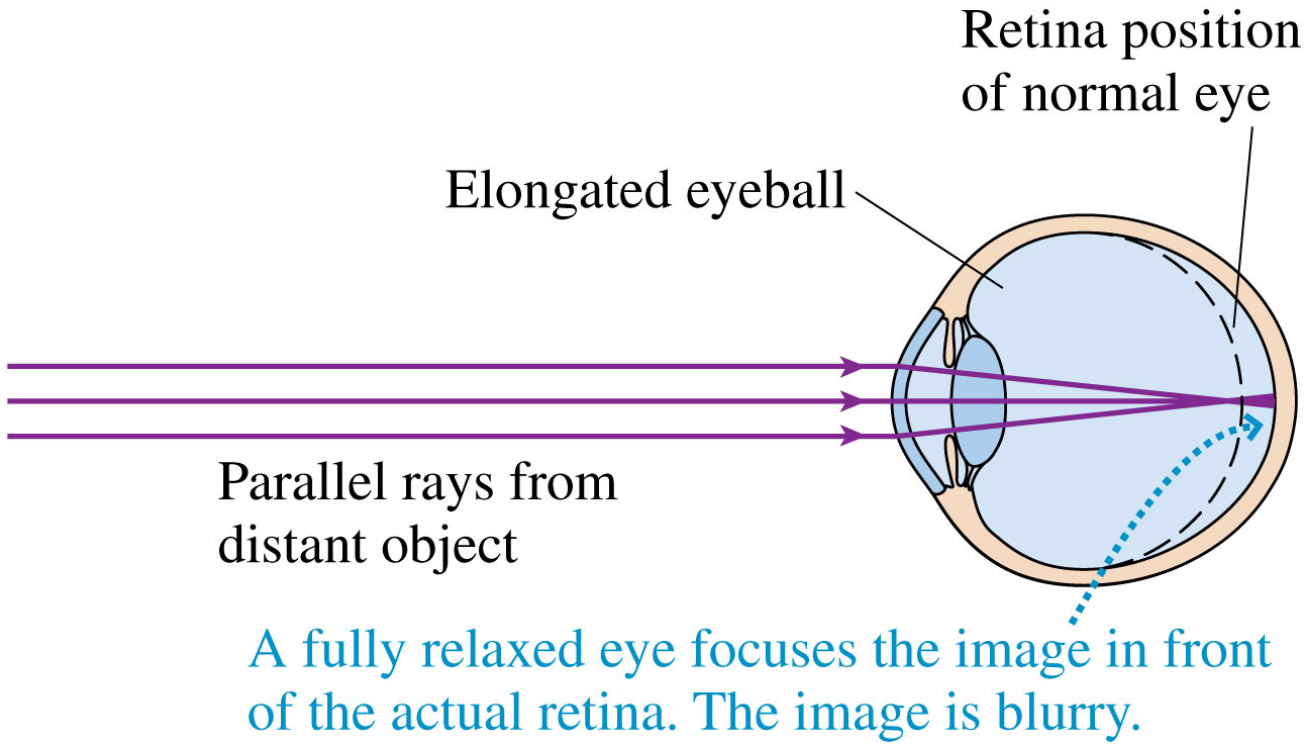
Myopia

- A person who is *nearsighted* can clearly see nearby objects when the eye is relaxed (and extremely close objects by using accommodation), but no amount of relaxation allows them to see distant objects.



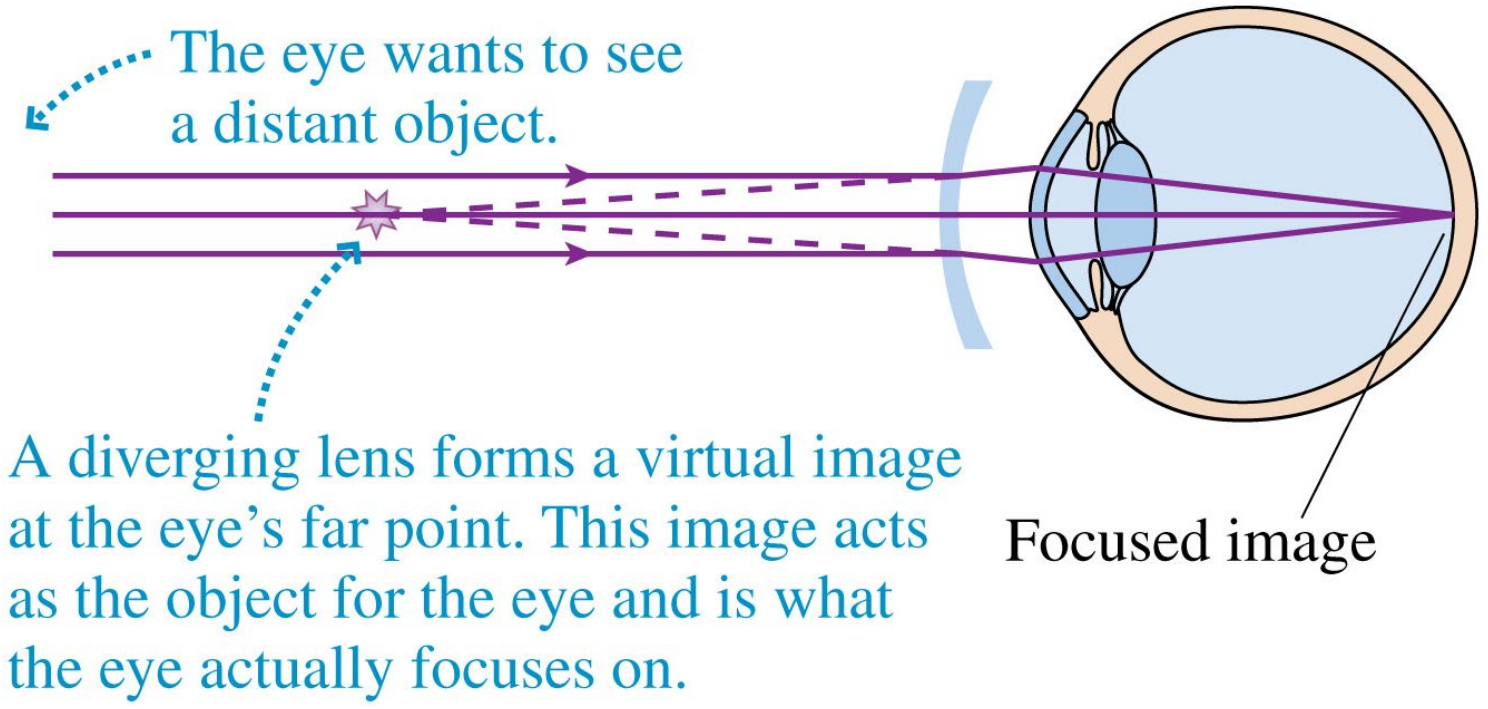
Myopia

- Nearsightedness—called **myopia**—is caused by an eyeball that is too long.
- Rays from a distant object come to a focus in front of the retina and have begun to diverge by the time they reach the retina.



Myopia

- To correct myopia, we needed a diverging lens to slightly defocus the rays and move the image point back to the retina.



Example: Correcting Myopia

Martina has myopia. The far point of her left eye is 200 cm. What prescription lens will restore normal vision?

MODEL Normal vision will allow Martina to focus on a very distant object. In measuring distances, we'll ignore the small space between the lens and her eye.

Example: Correcting Myopia

SOLVE Because Martina can see objects at 200 cm with a fully relaxed eye, we want a lens that will create a virtual image at position $s' = -200$ cm (negative because it's a virtual image) of a distant object at $s = \infty$ cm. From the thin-lens equation,

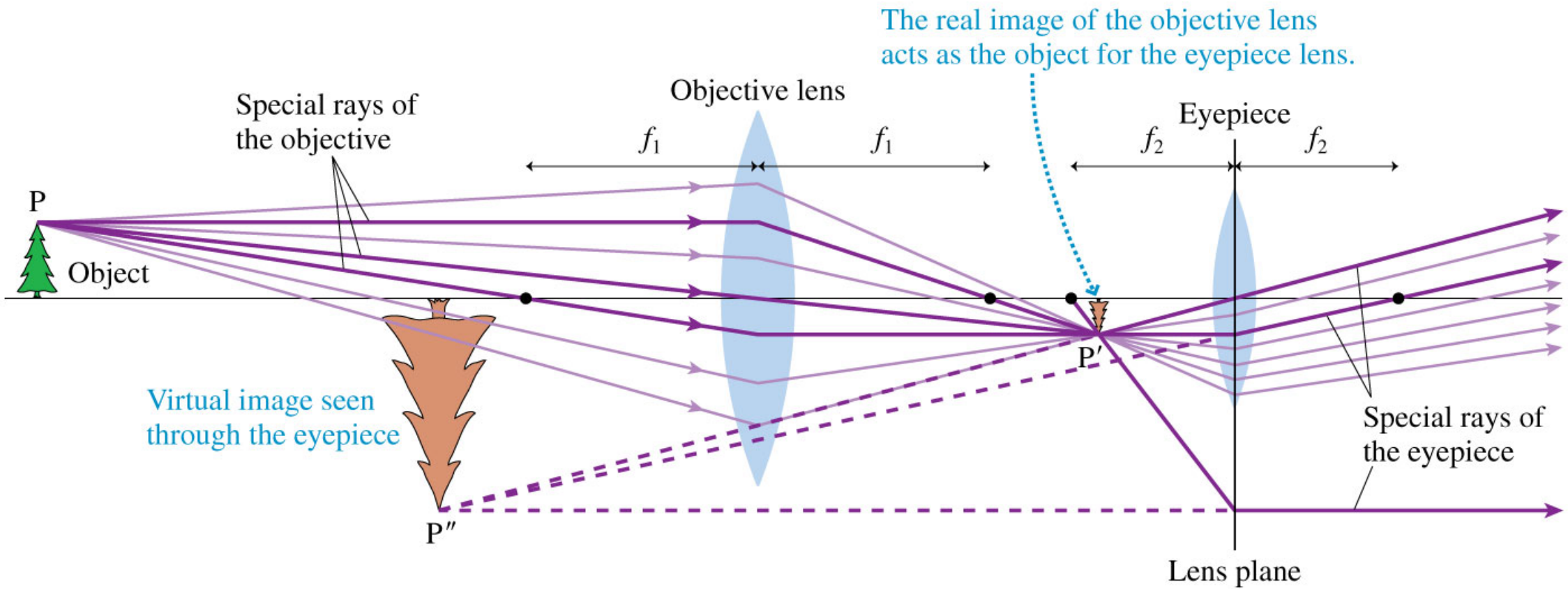
$$\frac{1}{f} = \frac{1}{s} + \frac{1}{s'} = \frac{1}{\infty\,\text{m}} + \frac{1}{-2.0\,\text{m}} = -0.5\,\text{m}^{-1}$$

Thus the prescription is for a lens with power $P = -0.5$ D.

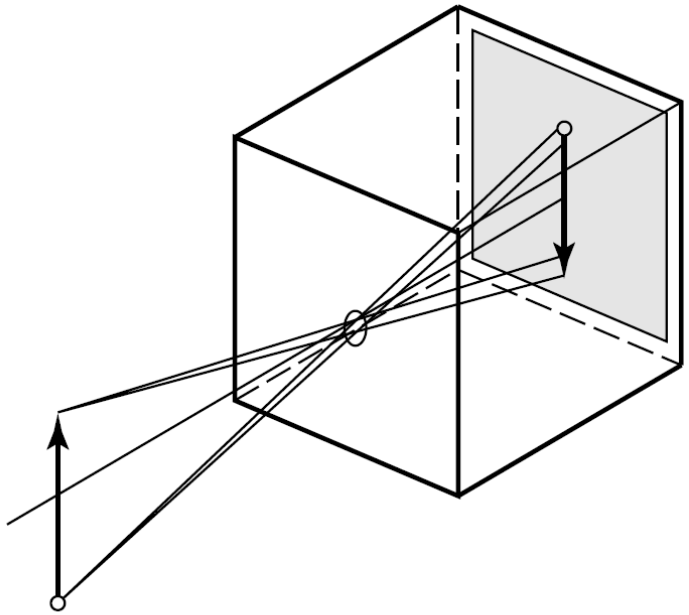
ASSESS Myopia is always corrected with a diverging lens.

Lenses in Combination

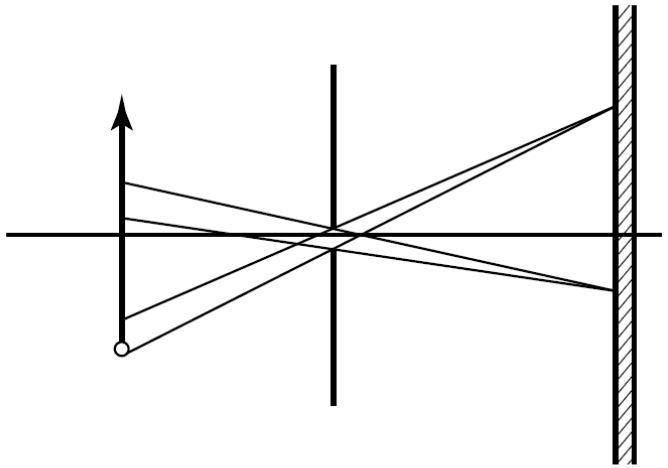
- The analysis of multi-lens systems requires only one new rule: **The image of the first lens acts as the object for the second lens.**
- Below is a ray-tracing diagram of a simple astronomical telescope.



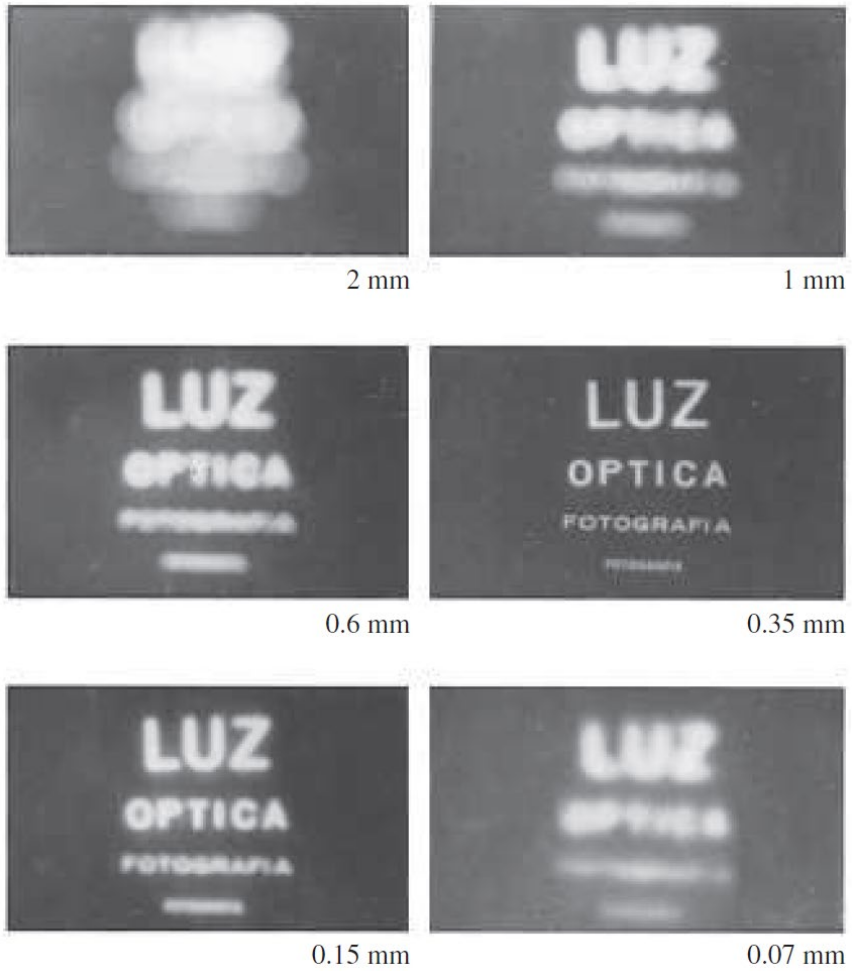
Pinhole Camera



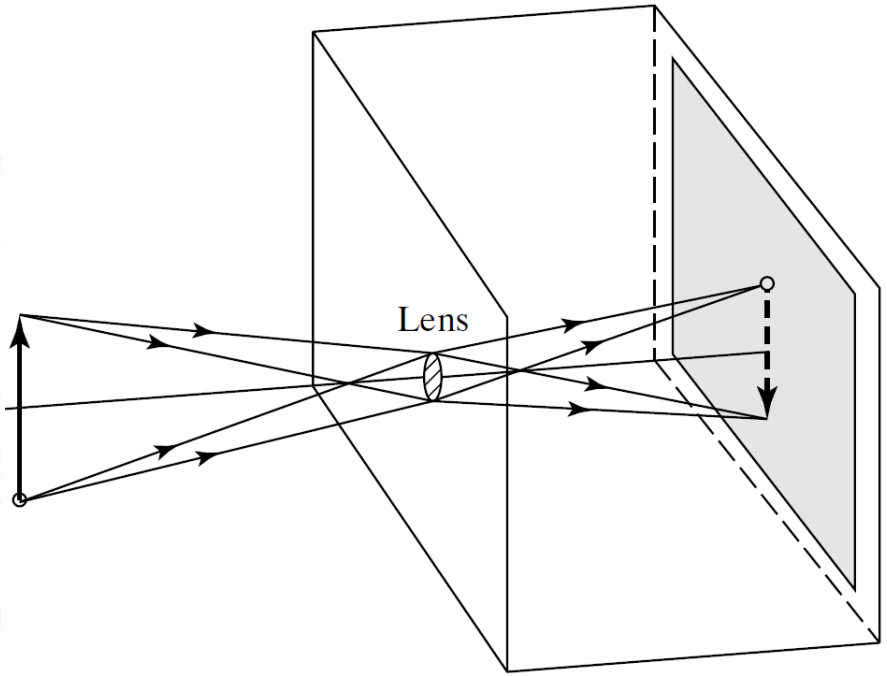
(a)



(b)

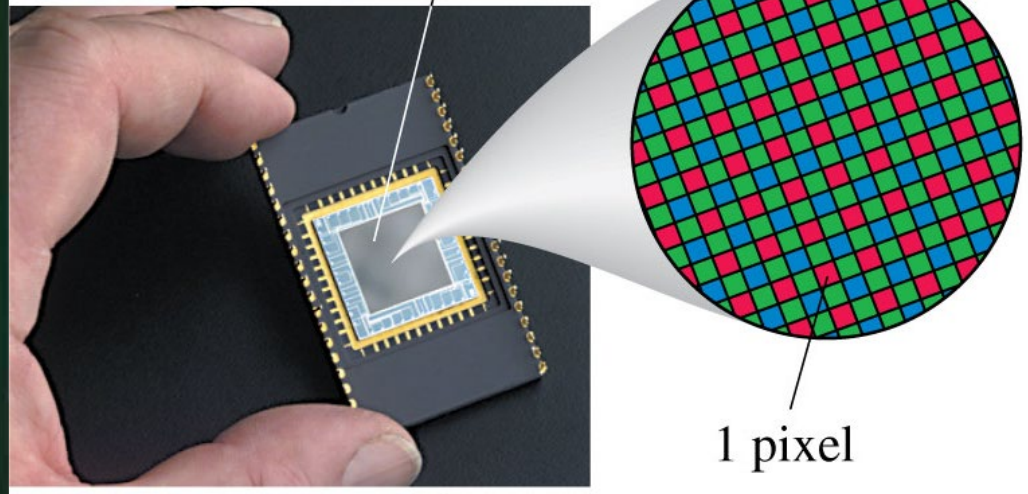


Simple Lens Camera



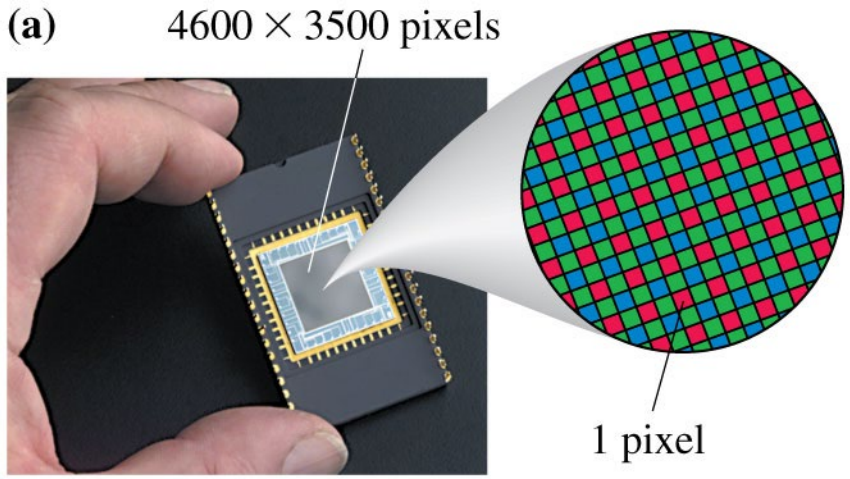
Not-So-Simple Camera

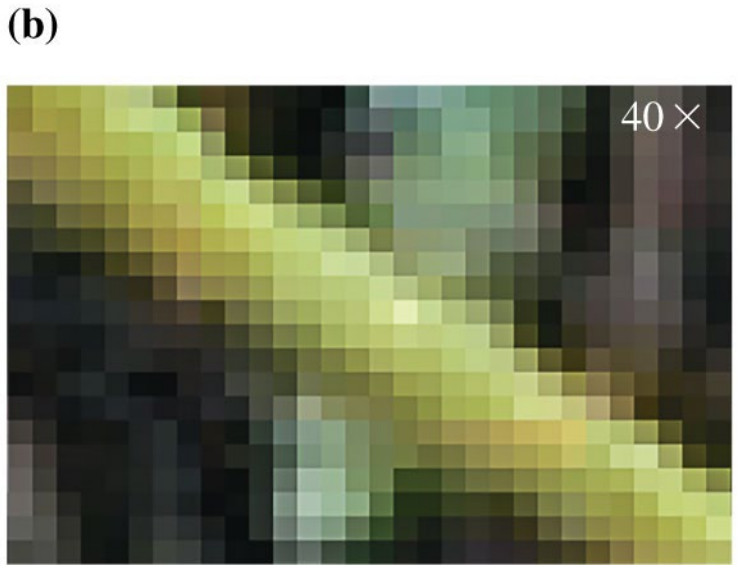
(a)



The Detector

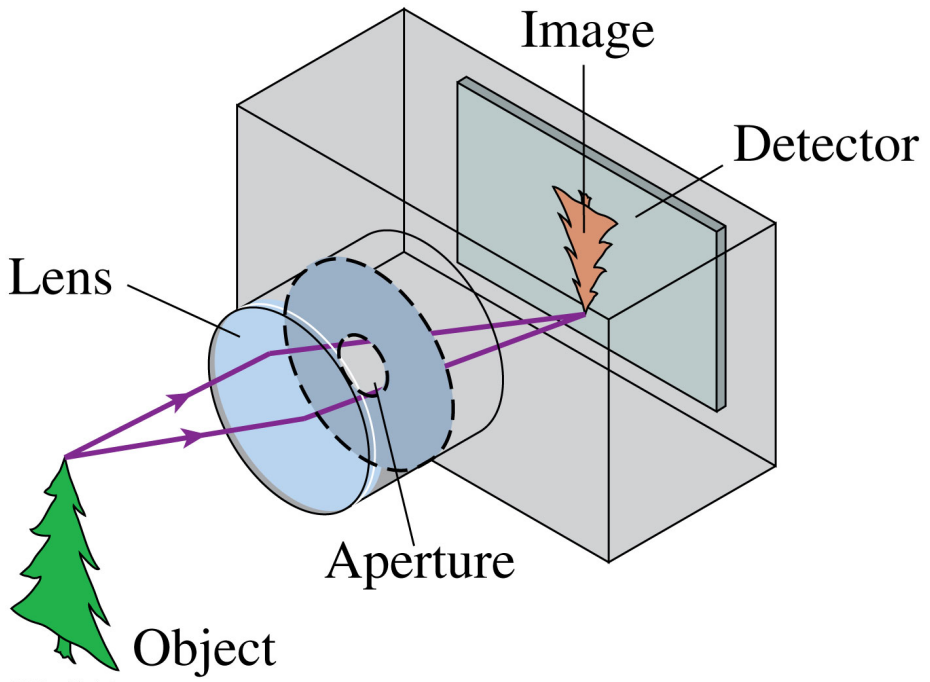
- Figure (a) shows a CCD "chip."
- To record color information, different pixels are covered by red, green, or blue filters.
- The pixels are so small that the picture looks "smooth" even after some enlargement.
- As you can see in figure (b), sufficient magnification reveals the individual pixels.



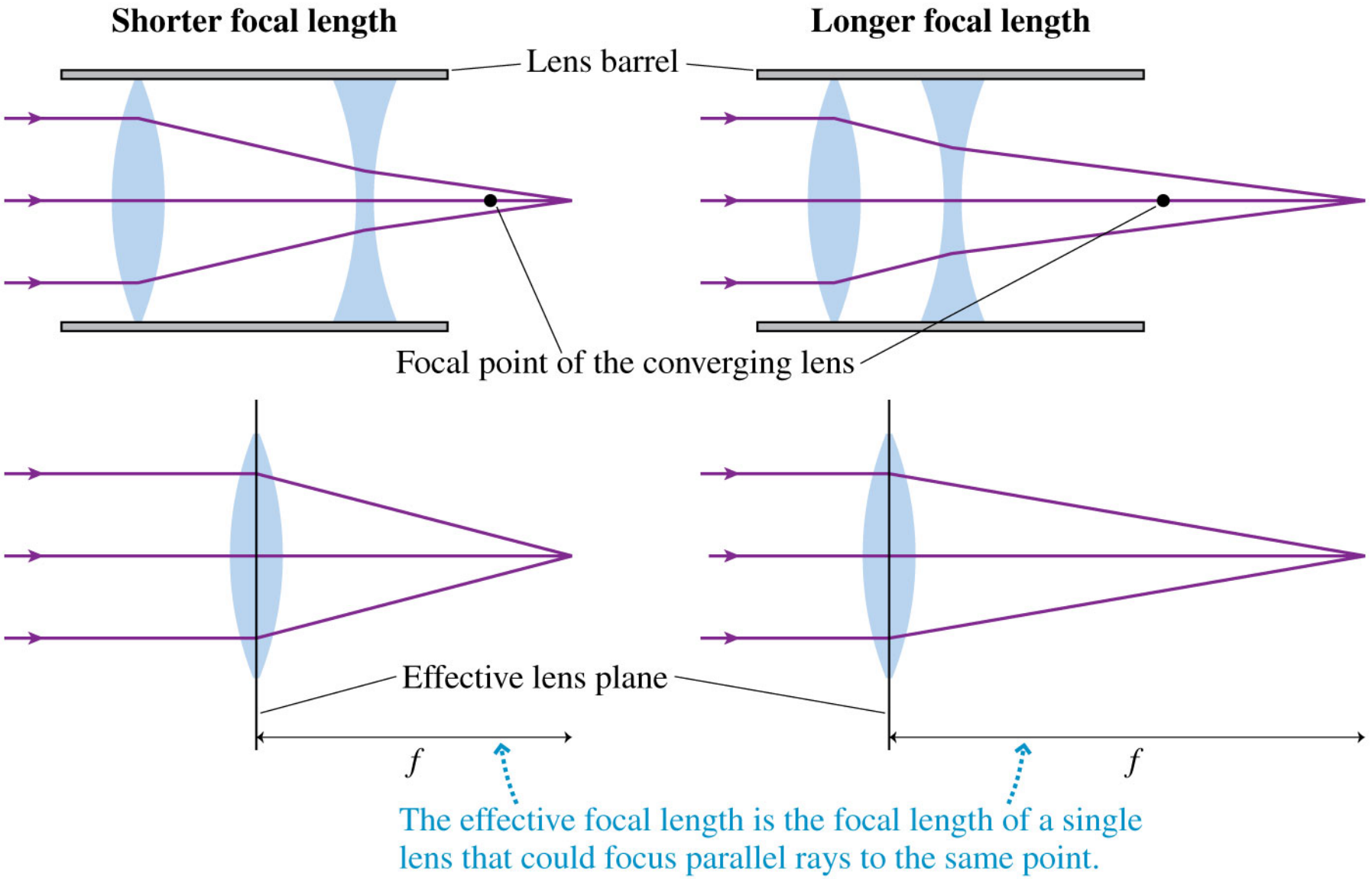


The Camera

- A **camera** “takes a picture” by using a lens to form a real, inverted image on a light-sensitive detector in a light-tight box.
- We can model a combination lens as a single lens with an **effective focal length** (usually called simply “the focal length”).
- A **zoom lens** changes the effective focal length by varying the spacing between the converging lens and the diverging lens.

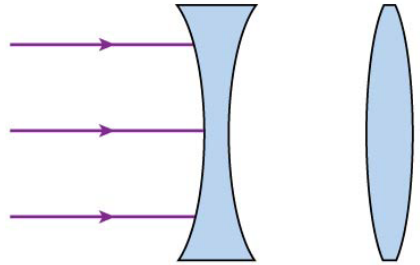


A Simple Camera Lens Is a Combination Lens



Q 35.1

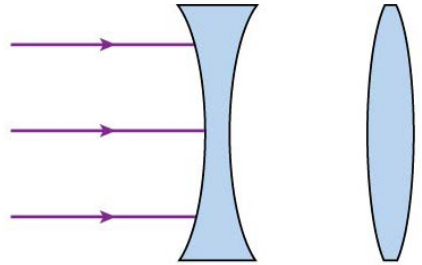
The parallel light rays will be focused at a point _____ the second lens than would light focused by the second lens acting alone.



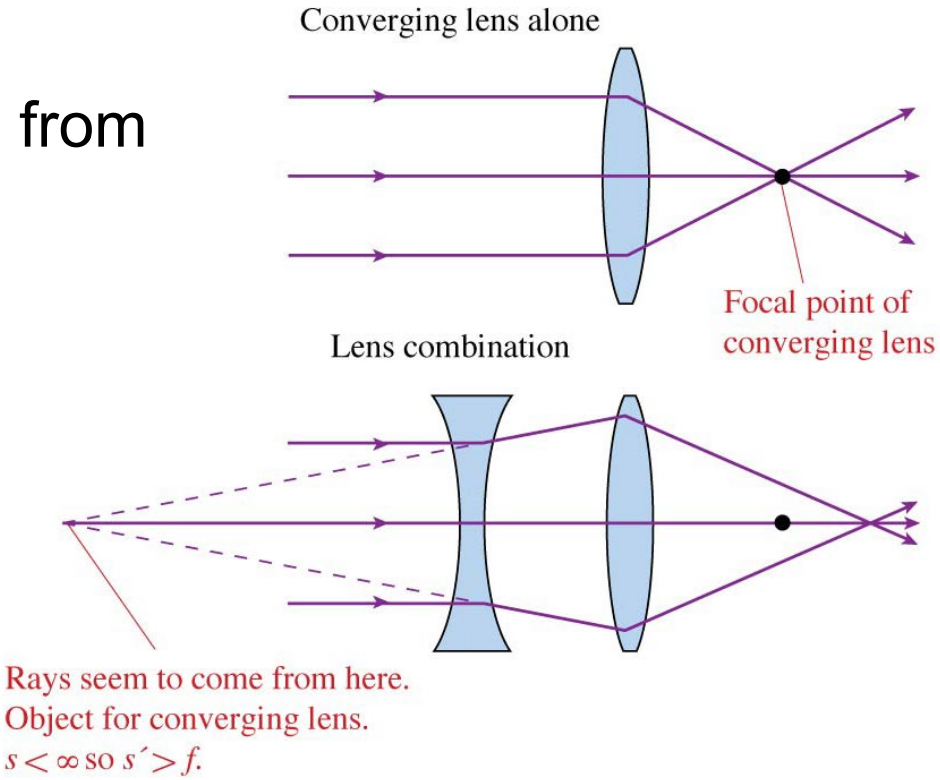
A. closer to
B. the same distance from
C. farther from

Q&A 35.1

The parallel light rays will be focused at a point _____ the second lens than would light focused by the second lens acting alone.



A. closer to
B. the same distance from
✔ **C. farther from**



Example: Focusing a Camera

Your digital camera lens, with an effective focal length of 10.0 mm, is focused on a flower 20.0 cm away. You then turn to take a picture of a distant landscape. How far, and in which direction, must the lens move to bring the landscape into focus?

MODEL Model the camera's combination lens as a single thin lens with $f = 10.0$ mm. Image and object distances are measured from the effective lens plane. Assume all the lenses in the combination move together as the camera refocuses.

SOLVE The flower is at object distance $s = 20.0\text{ cm} = 200\text{ mm}$. When the camera is focused, the image distance between the effective lens plane and the detector is found by solving the thin-lens equation $1/s + 1/s' = 1/f$ to give

$$s' = \left(\frac{1}{f} - \frac{1}{s}\right)^{-1} = \left(\frac{1}{10.0\text{ mm}} - \frac{1}{200\text{ mm}}\right)^{-1} = 10.5\text{ mm}$$

The distant landscape is effectively at object distance $s = \infty$, so its image distance is $s' = f = 10.0$ mm. To refocus as you shift scenes, the lens must move 0.5 mm closer to the detector.

ASSESS The required motion of the lens is very small, about the diameter of the lead used in a mechanical pencil.

Zoom Lenses

- When cameras focus on objects that are more than 10 focal lengths away (roughly $s > 20\ \text{cm}$ for a typical digital camera), the object is essentially "at infinity" and $s' \approx f$.
- The lateral magnification of the image is

$$m = -\frac{s'}{s} \approx -\frac{f}{s}$$

- The magnification is much less than 1, because $s >> f$, so the image on the detector is much *smaller* than the object itself.
- More important, **the size of the image is directly proportional to the focal length of the lens**.

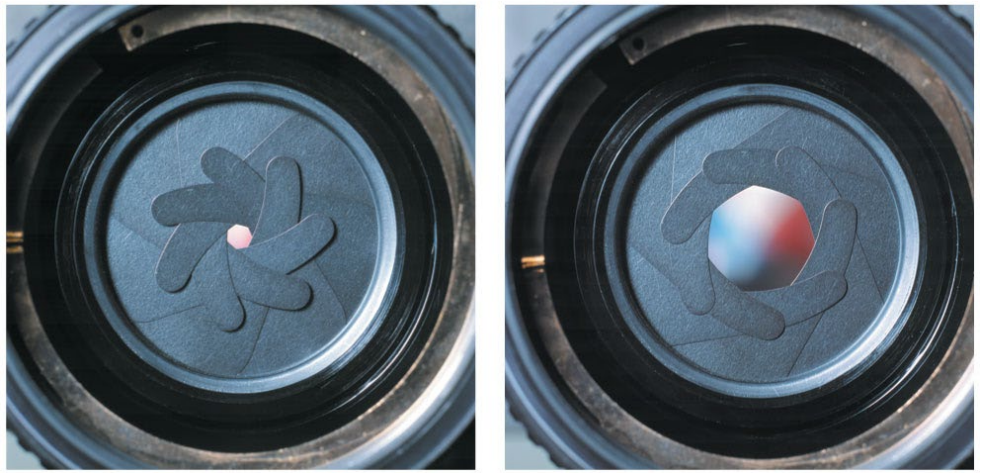
Controlling the Exposure

- The amount of light passing through the lens is controlled by an adjustable **aperture**, shown in the photos.
- The aperture sets the effective diameter *D* of the lens.
- The light-gathering ability of a lens is specified by its ***f*-number**, defined as:

$$f\text{-number} = \frac{f}{D}$$

- The light intensity on the detector is related to the lens's *f*-number by:

$$I \propto \frac{D^2}{f^2} = \frac{1}{(f\text{-number})^2}$$



$$I \propto \frac{\text{area of aperture}}{\text{area of image}} = \frac{D^2}{d^2} \propto \frac{D^2}{f^2}$$

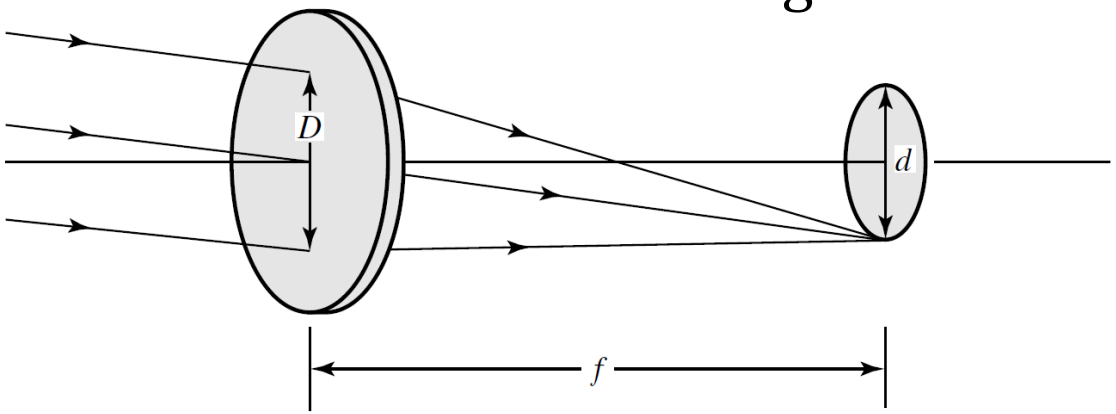


Figure 21 Illumination of image. The aperture (not shown) determines the useful diameter *D* of the lens.

Q&A 35.2

If the f-number of a camera lens is doubled, say from F4.0 to F8.0, that means the diameter of the lens aperture is

A. Quadrupled (increased by a factor of 4).

B. Doubled (increased by a factor of 2).

✔C. Halved (decreased by a factor of 2).

D. Quartered (decreased by a factor of 4).

$f\text{-number} = f/D$

Controlling the Exposure

- Focal length and *f*-number information is stamped on a camera lens.
- This lens is labeled 5.8–23.2 mm 1:2.6–5.5.
- The first numbers are the range of focal lengths.

- They span a factor of 4, so this is a 4× zoom lens.
- The second numbers show that the minimum *f*-number ranges from *f*/2.6 (for the $f = 5.8$ mm focal length) to *f*/5.5 (for the $f = 23.2$ mm focal length).

Q&A 35.3

A camera gives a proper exposure when set to a shutter speed of 1/250 s at *f*-number F8.0. The photographer wants to change the shutter speed to 1/1000 s to prevent motion blur. To maintain proper exposure, she should also change the *f*-number to

A. F2.0

✔ B. F4.0

C. F8.0

D. F16

E. F32

Light intensity goes up a factor of 2 when the *f*-number decreases by the square root of 2.

Example 35.3 Capturing the Action

EXAMPLE 35.3 | **Capturing the action**

Before a race, a photographer finds that she can make a perfectly exposed photo of the track while using a shutter speed of 1/250 s and a lens setting of $f/8.0$. To freeze the sprinters as they go past, she plans to use a shutter speed of 1/1000 s. To what f-number must she set her lens?

MODEL The exposure depends on $I\Delta t_{\text{shutter}}$, and the light intensity depends inversely on the square of the f-number.

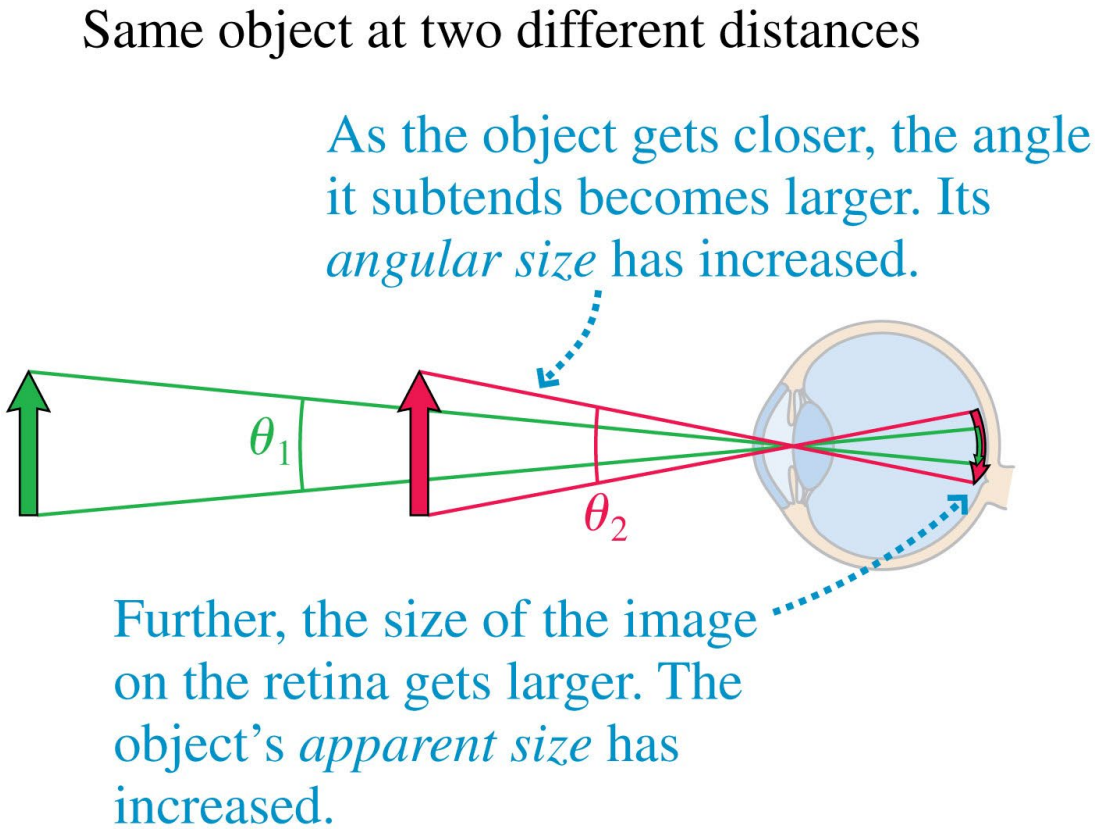
Example 35.3 Capturing the Action

EXAMPLE 35.3 | **Capturing the action**

SOLVE Changing the shutter speed from 1/250 s to 1/1000 s will reduce the light reaching the detector by a factor of 4. To compensate, she needs to let 4 times as much light through the lens. Because $I \propto 1/(f\text{-number})^2$, the intensity will increase by a factor of 4 if she *decreases* the f-number by a factor of 2. Thus the correct lens setting is $f/4.0$.

Optical Systems That Magnify

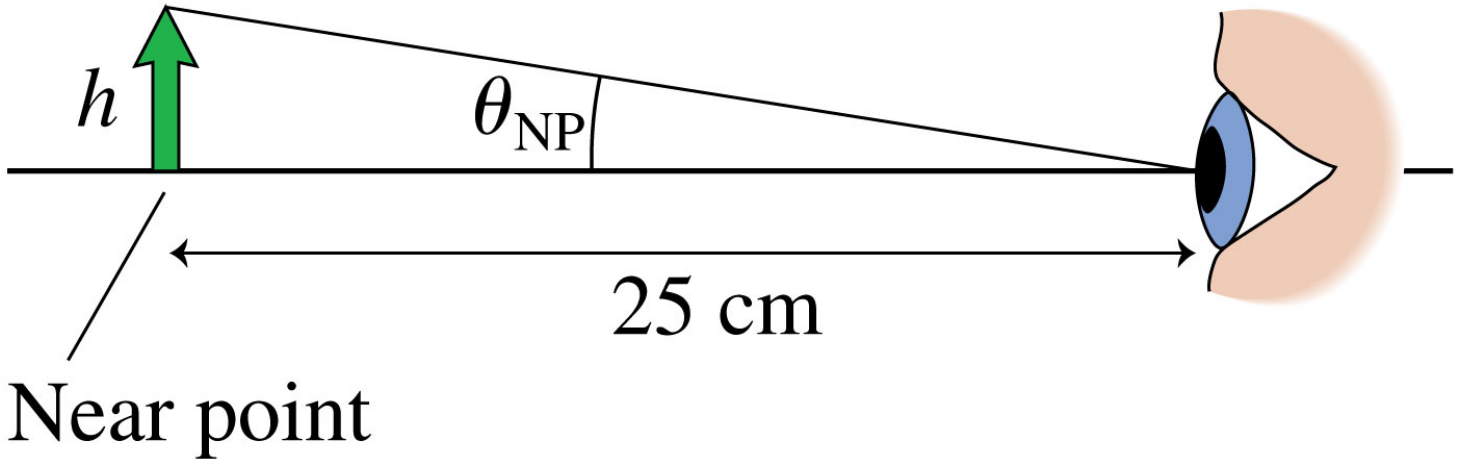
- The easiest way to magnify an object requires no extra optics at all: Simply get closer!
- Closer objects look larger because they subtend a larger angle θ, called the **angular size** of the object.



Optical Systems That Magnify

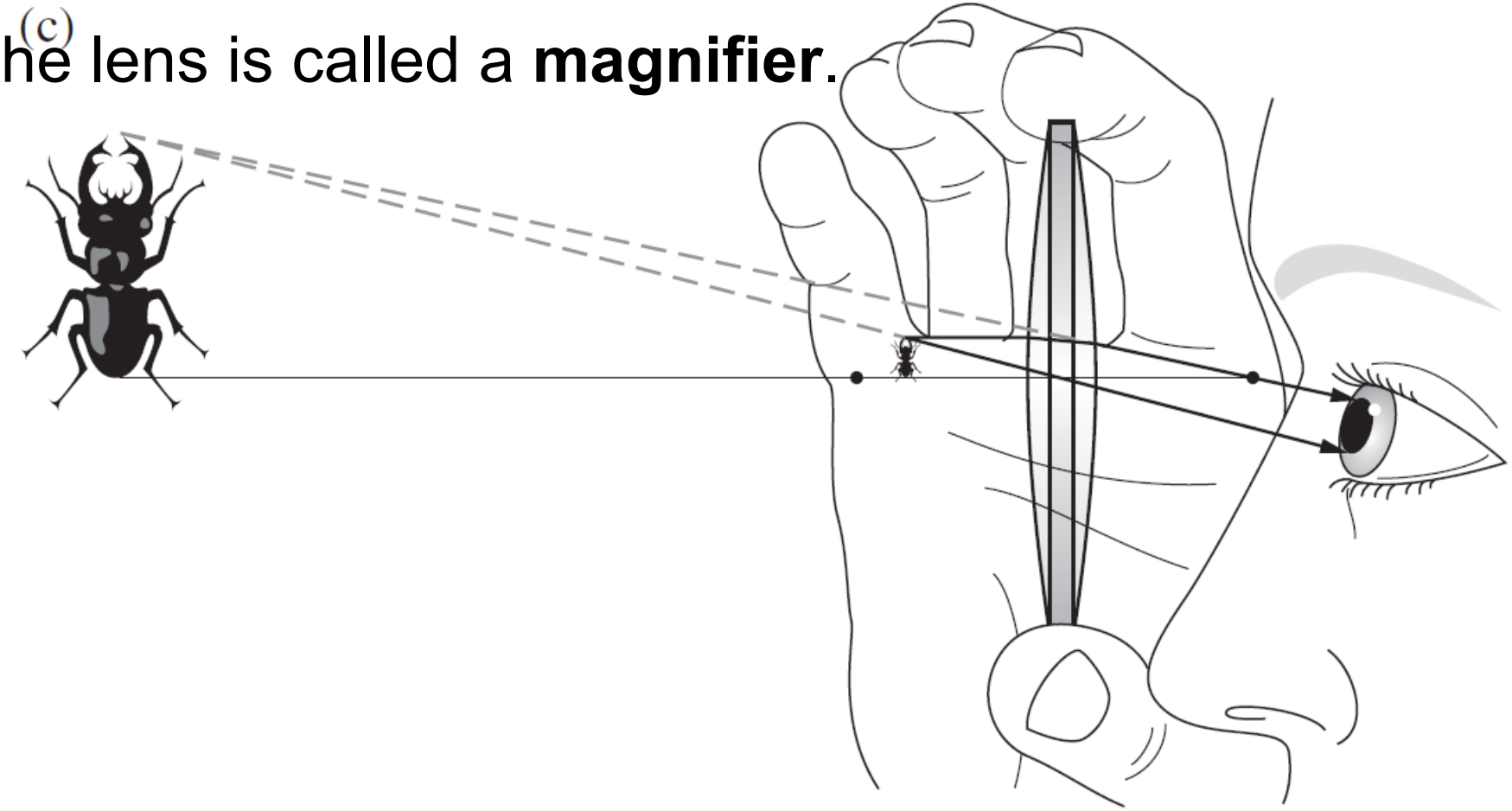
- You can't keep increasing an object's angular size because you can't focus on the object if it's closer than your near point, which is $\approx 25\ \text{cm}$.
- The maximum angular size viewable by your unaided eye is

$$\theta_{\text{NP}} = \frac{h}{25\ \text{cm}}$$

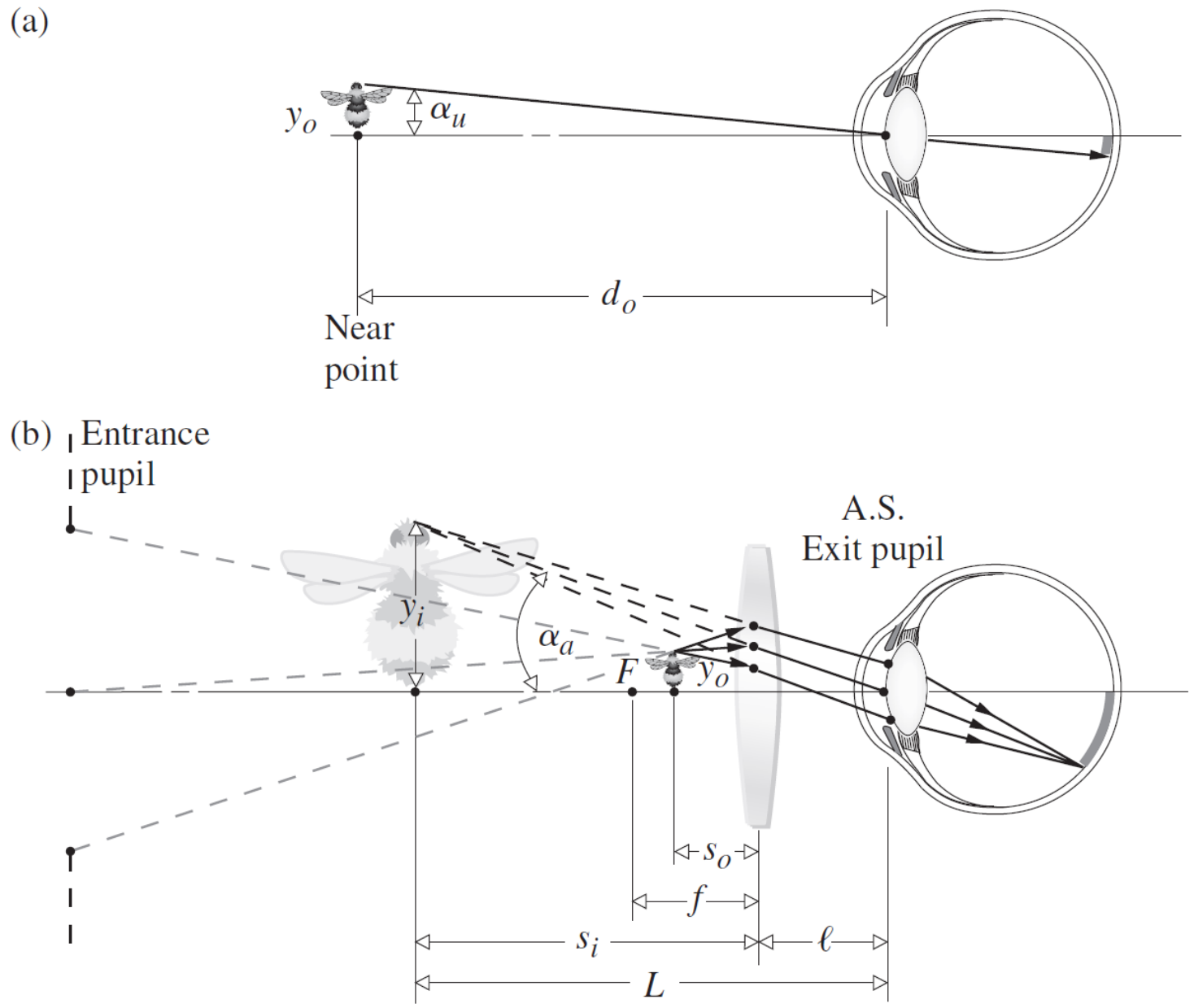


The Magnifier

- Suppose we view an object of height h through a single converging lens.
- If the object's distance from the lens is less than the lens's focal length, we'll see an enlarged, upright image.
- Used in this way, the lens is called a **magnifier**.



Unaided Eye vs. Magnifier



The Magnifier

- When using a magnifier, your eye sees a virtual image subtending an angle $\theta = h/s$.
- If we place the image at a distance $s' \approx \infty$ the object distance is $s \approx f$, so:

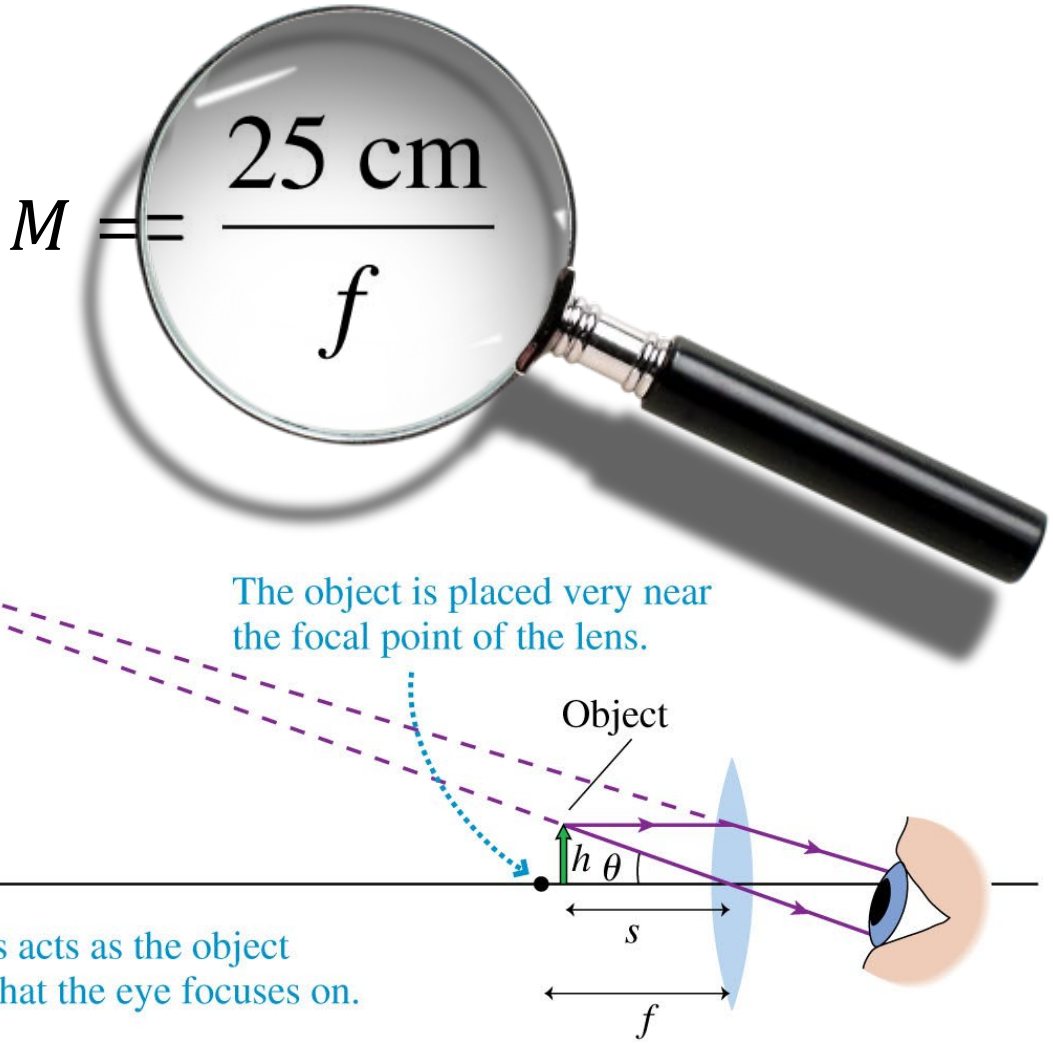
$$\theta = \frac{h}{s} \approx \frac{h}{f}$$

- Angular magnification is the ratio of the apparent size of the object when using a magnifying lens rather than simply holding the object at your near point:

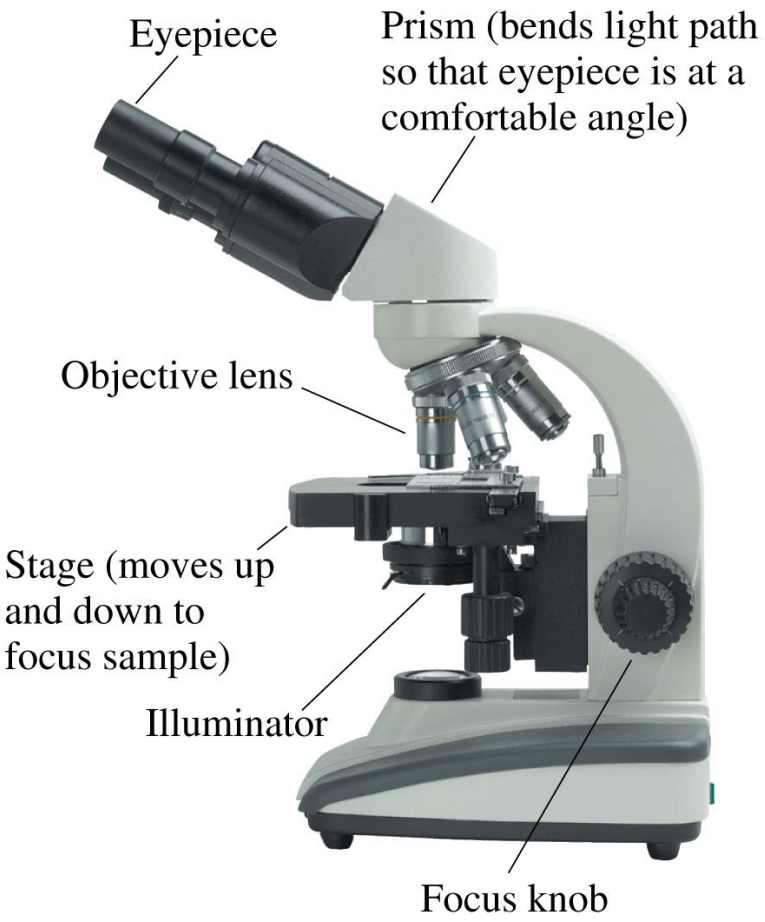
$$M = \theta/\theta_{\text{NP}} \qquad \theta_{\text{NP}} = \frac{h}{25\text{ cm}}$$

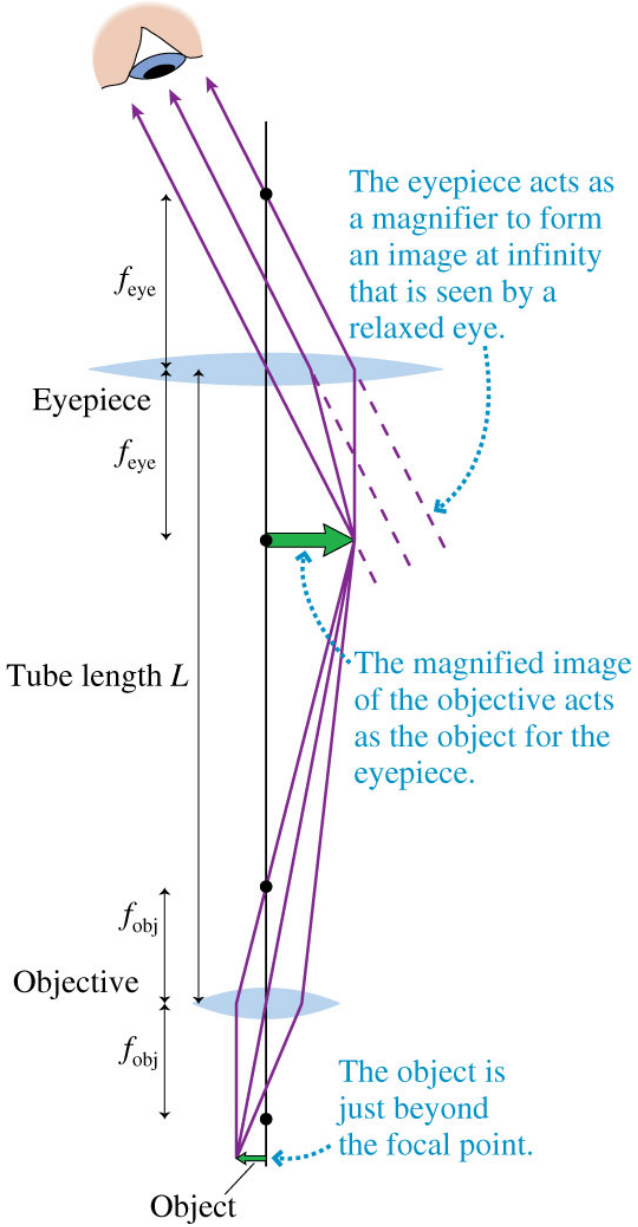
- Combining these equations, we find the **angular magnification** of a magnifying glass is:

$$M = \frac{25\text{ cm}}{f}$$



The Microscope

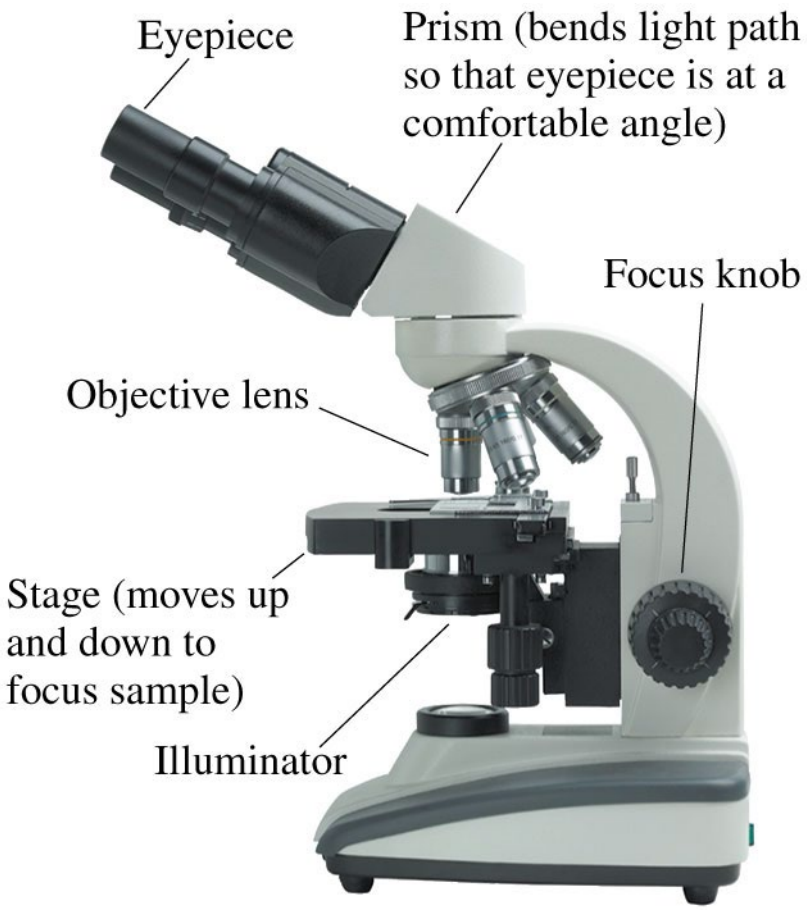




Angular magnification:

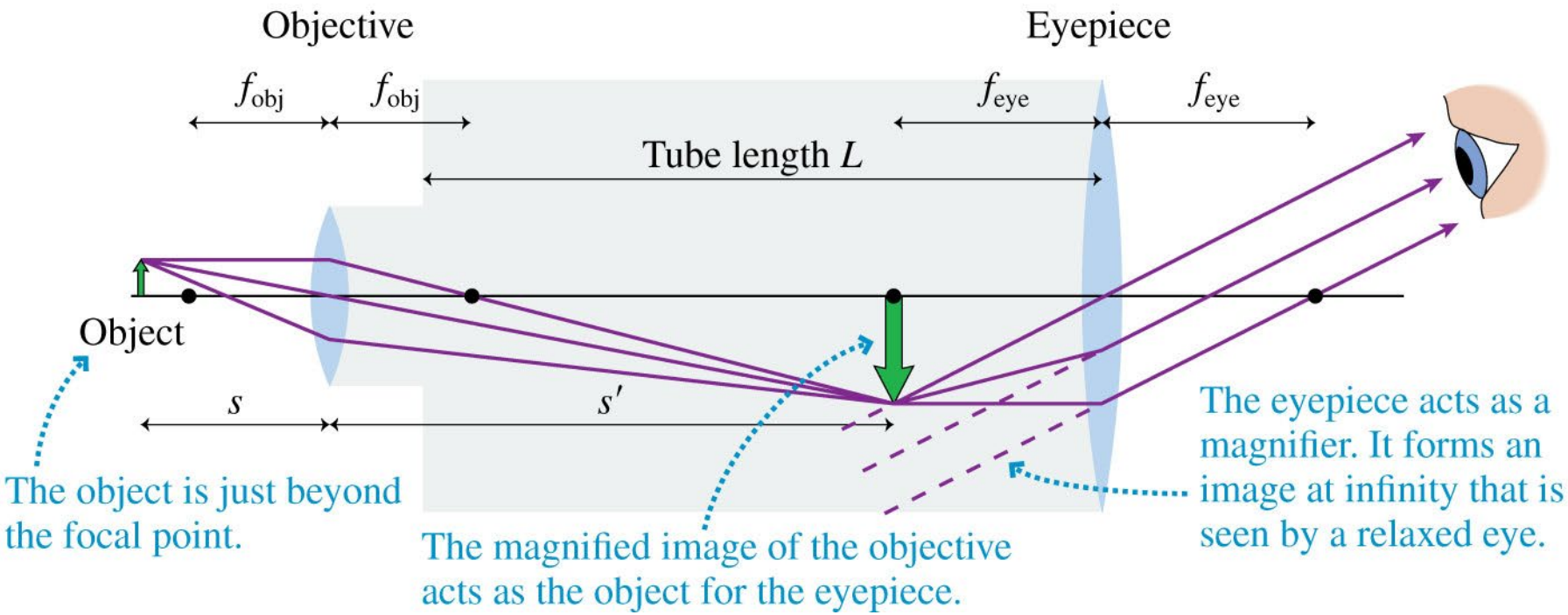
$$M = m_{\text{objective}} M_{\text{eyepiece}} = -\frac{L}{f_{\text{obj}}} \frac{25\text{ cm}}{f_{\text{eye}}}$$

The Microscope



- A microscope, whose major parts are shown in the figure, can attain a magnification of up to $1000\times$ by a *two-step* magnification process.
- A specimen to be observed is placed on the *stage* of the microscope, directly beneath the **objective**, a converging lens with a relatively short focal length.
- The objective creates a magnified real image that is further enlarged by the **eyepiece**.

The Microscope



- This is a simple two-lens model of a microscope.
- The object is placed just outside the focal point of the objective, which creates a highly magnified real image with lateral magnification $m = -s'/s$.

The Microscope

- The lateral magnification of the objective is

$$m_{\text{obj}} = -\frac{s'}{s} \approx -\frac{L}{f_{\text{obj}}}$$

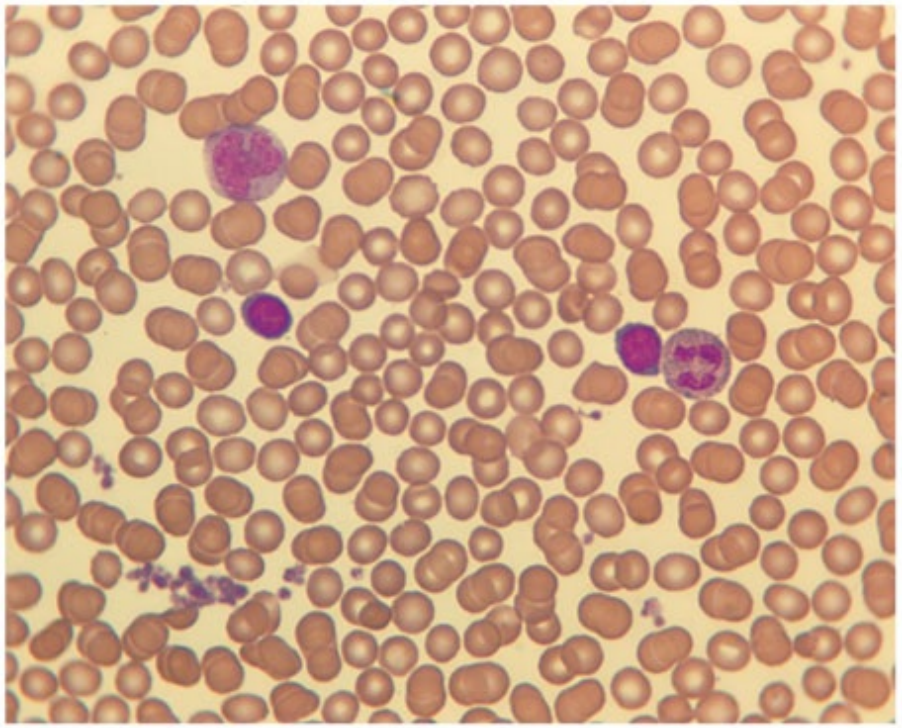
- Together, the objective and eyepiece produce a total angular magnification:

$$M = m_{\text{obj}} M_{\text{eye}} = -\frac{L}{f_{\text{obj}}} \frac{25\ \text{cm}}{f_{\text{eye}}}$$

- The minus sign shows that the image seen in a microscope is inverted.
- Most biological microscopes are standardized with a tube length $L = 160\ \text{mm}$.

Example: Viewing Blood Cells

A pathologist inspects a sample of 7-μm-diameter human blood cells under a microscope. She selects a 40× objective and a 10× eyepiece. What size object, viewed from 25 cm, has the same apparent size as a blood cell seen through the microscope?



Example: Viewing Blood Cells

MODEL Angular magnification compares the magnified angular size to the angular size seen at the near-point distance of 25 cm.

SOLVE The microscope's angular magnification is $M = -(40) \times (10) = -400$. The magnified cells will have the same apparent size as an object $400 \times 7\ \mu\text{m} \approx 3$ mm in diameter seen from a distance of 25 cm.

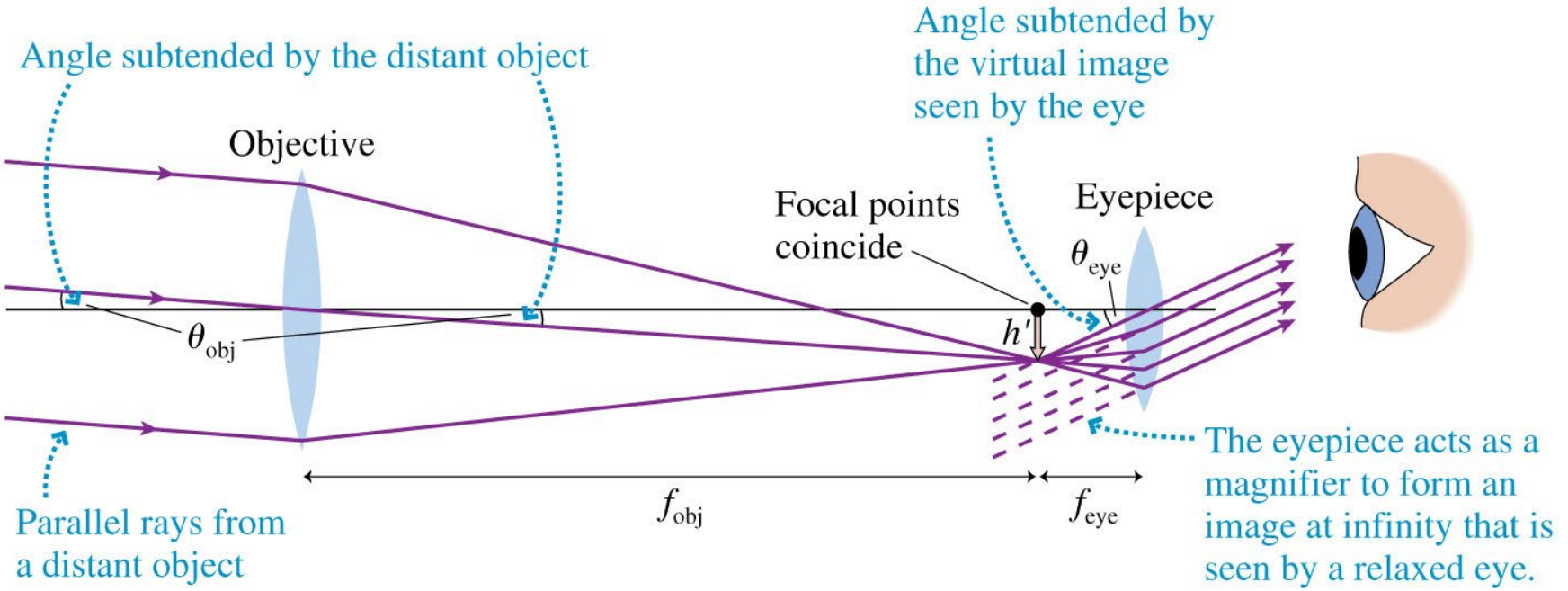
ASSESS 3 mm is about the size of a capital O in this textbook, so a blood cell seen through the microscope will have about the same apparent size as an O seen from a comfortable reading distance.

The Telescope

- A simple telescope contains a large-diameter objective lens that collects parallel rays from a distant object and forms a real, inverted image at distance $s' = f_{\text{obj}}$.
- The focal length of a telescope objective is very nearly the length of the telescope tube.
- The eyepiece functions as a simple magnifier.
- The viewer observes an inverted image.
- The angular magnification of a telescope is

$$M = \frac{\theta_{\text{eye}}}{\theta_{\text{obj}}} = -\frac{f_{\text{obj}}}{f_{\text{eye}}}$$

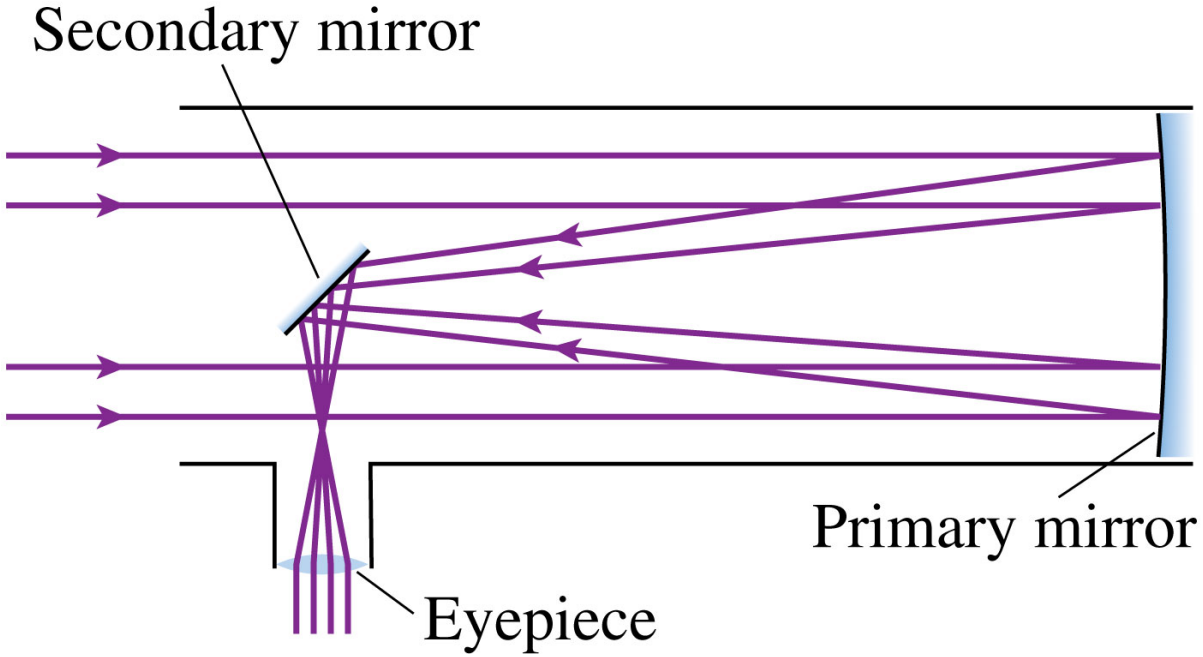
The Refracting Telescope



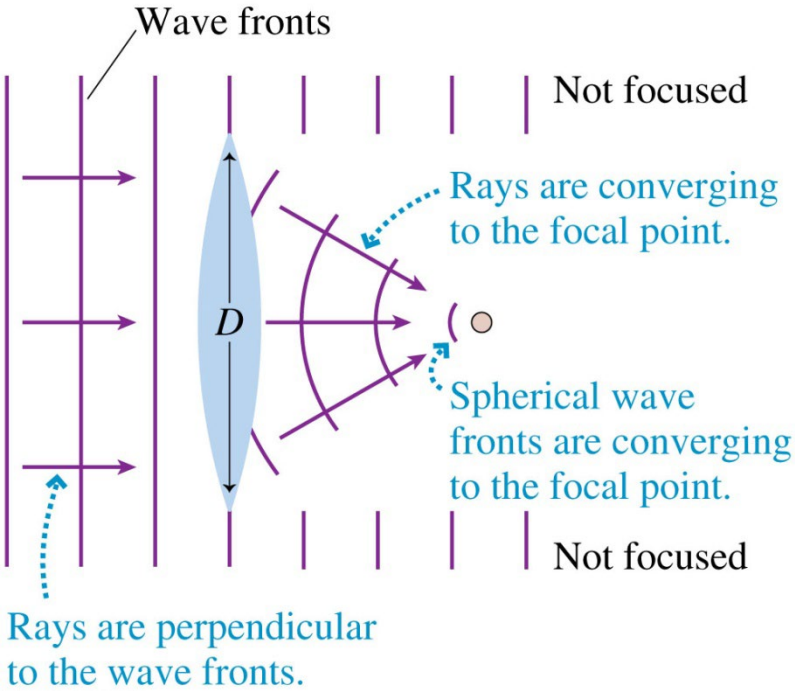
Angular magnification: $M = -\dfrac{\theta_{\text{eyepiece}}}{\theta_{\text{objective}}} = -\dfrac{f_{\text{obj}}}{f_{\text{eye}}}$

Telescopes

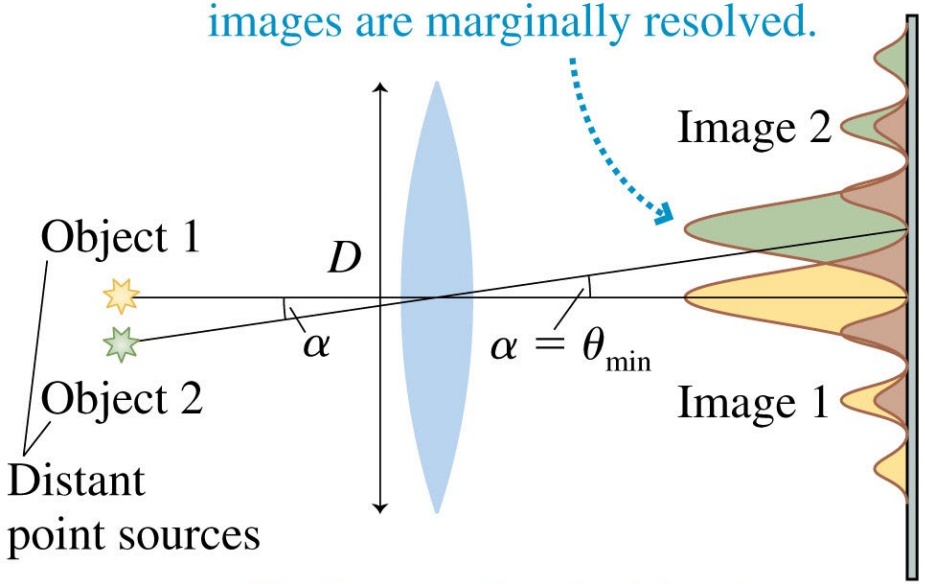
- Large light-gathering power requires a large-diameter objective lens, but large lenses are not practical; they begin to sag under their own weight.
- Thus **refracting telescopes**, with two lenses, are relatively small.
- Serious astronomy is done with a **reflecting telescope**, such as the one shown in the figure.



Circular Lens Diffraction & Rayleigh's Criterion

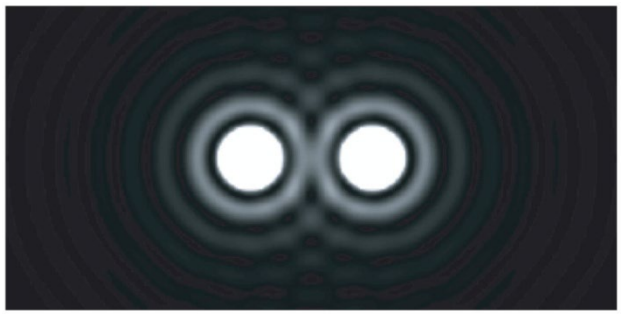


The maximum of image 2 falls on the first dark fringe of image 1. The images are marginally resolved.

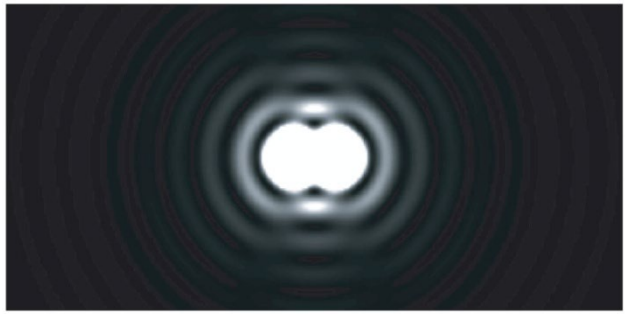


$$\theta_{\text{min}} = \frac{1.22\lambda}{D} \qquad \text{(angular resolution of a lens)}$$

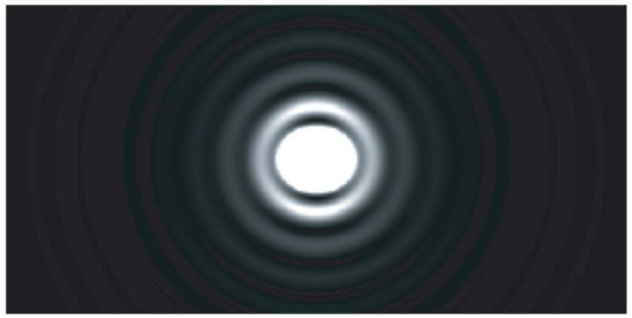
The image of each object is not a perfect point, but a small circular diffraction pattern.



$\alpha > \theta_{\text{min}}$
Resolved



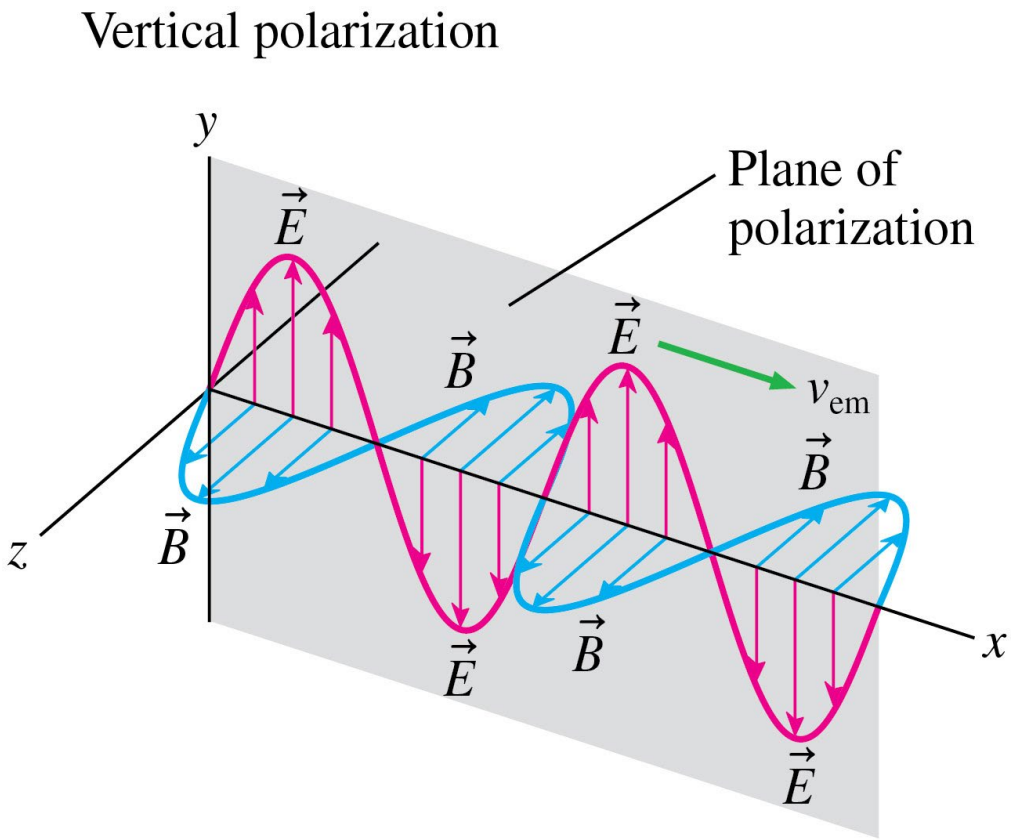
$\alpha = \theta_{\text{min}}$
Marginally resolved



$\alpha < \theta_{\text{min}}$
Not resolved

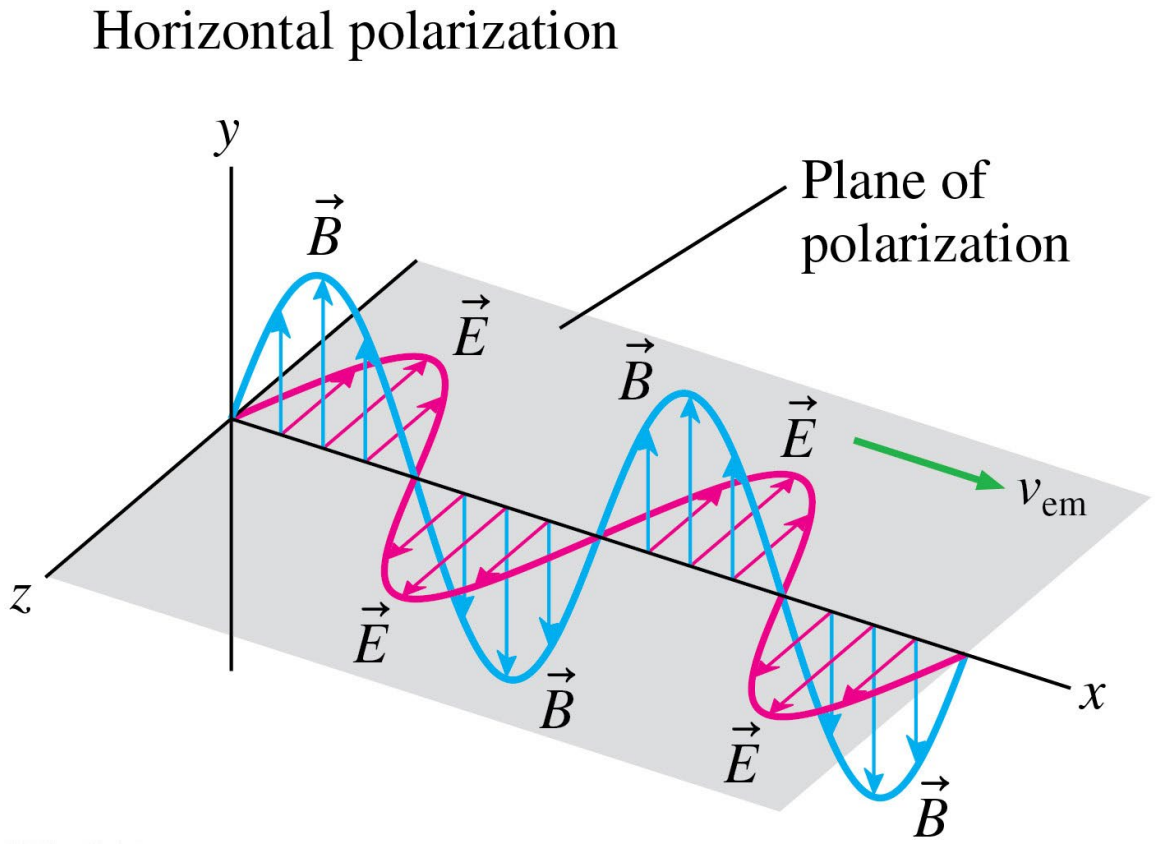
Polarization

- The plane of the electric field vector $\vec{E}$ and the Poynting vector $\vec{S}$ is called the **plane of polarization**.
- The electric field in the figure below oscillates vertically, so this wave is *vertically polarized*.



Polarization

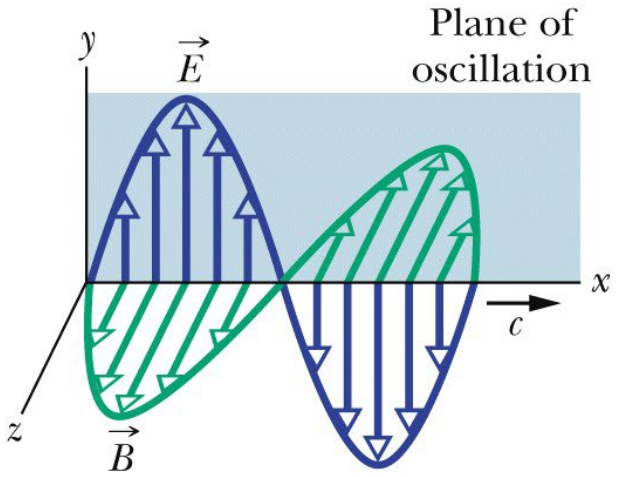
- The electric field in the figure below is *horizontally polarized.*

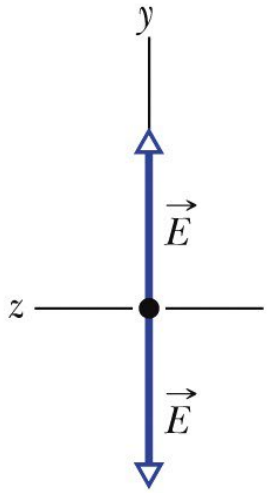


- Most natural sources of light are unpolarized, emitting waves whose electric fields oscillate randomly with all possible orientations.

Polarization

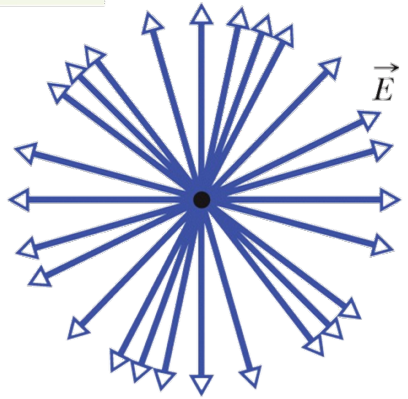
Electromagnetic waves are polarized if their electric field vectors are all in a single plane, called the plane of oscillation. Light waves from common sources are not polarized; that is, they are unpolarized, or polarized randomly.





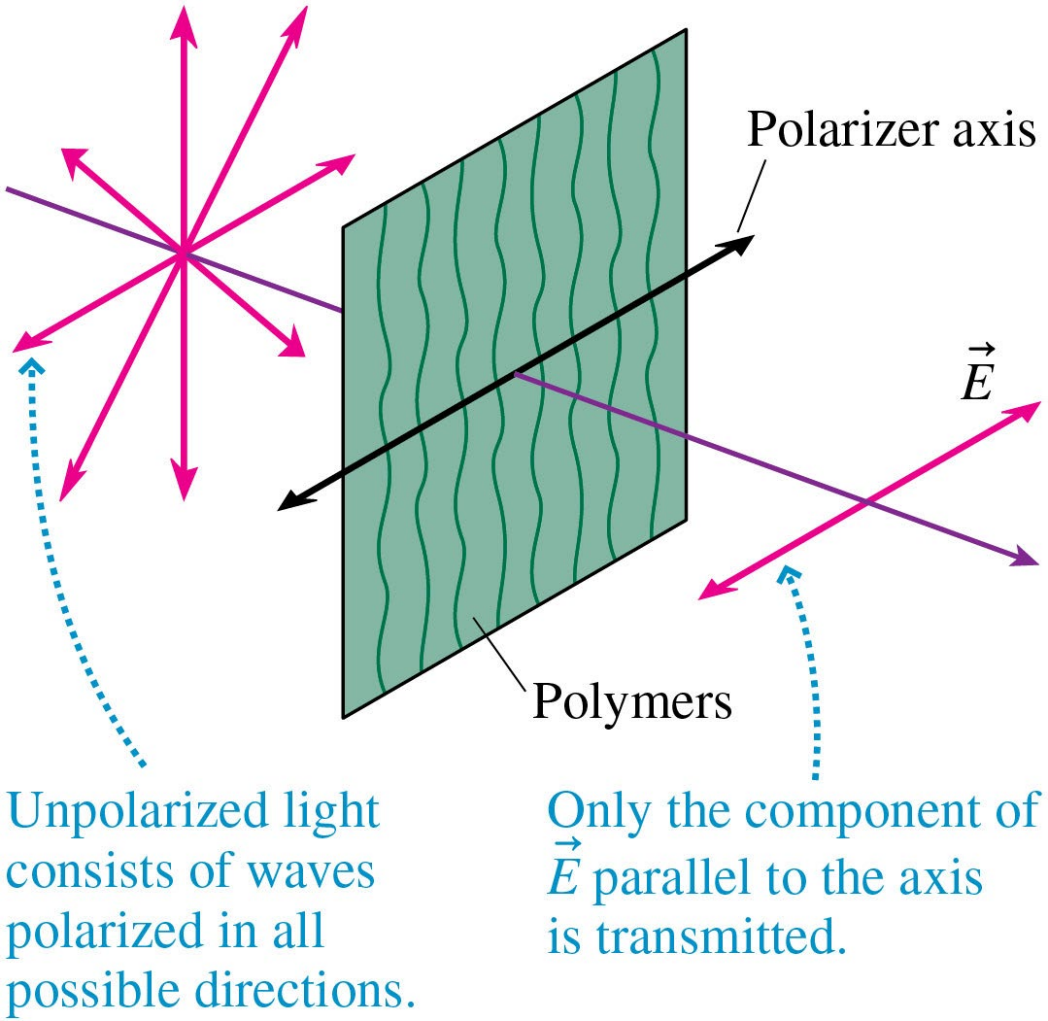
Vertically polarized light headed toward you—the electric fields are all vertical.

Unpolarized light headed toward you—the electric fields are in all directions in the plane.



An electric field component parallel to the polarizing direction is passed (transmitted) by a polarizing sheet; a component perpendicular to it is absorbed.

Polarization



- The most common way of artificially generating polarized visible light is to send unpolarized light through a *polarizing filter*.

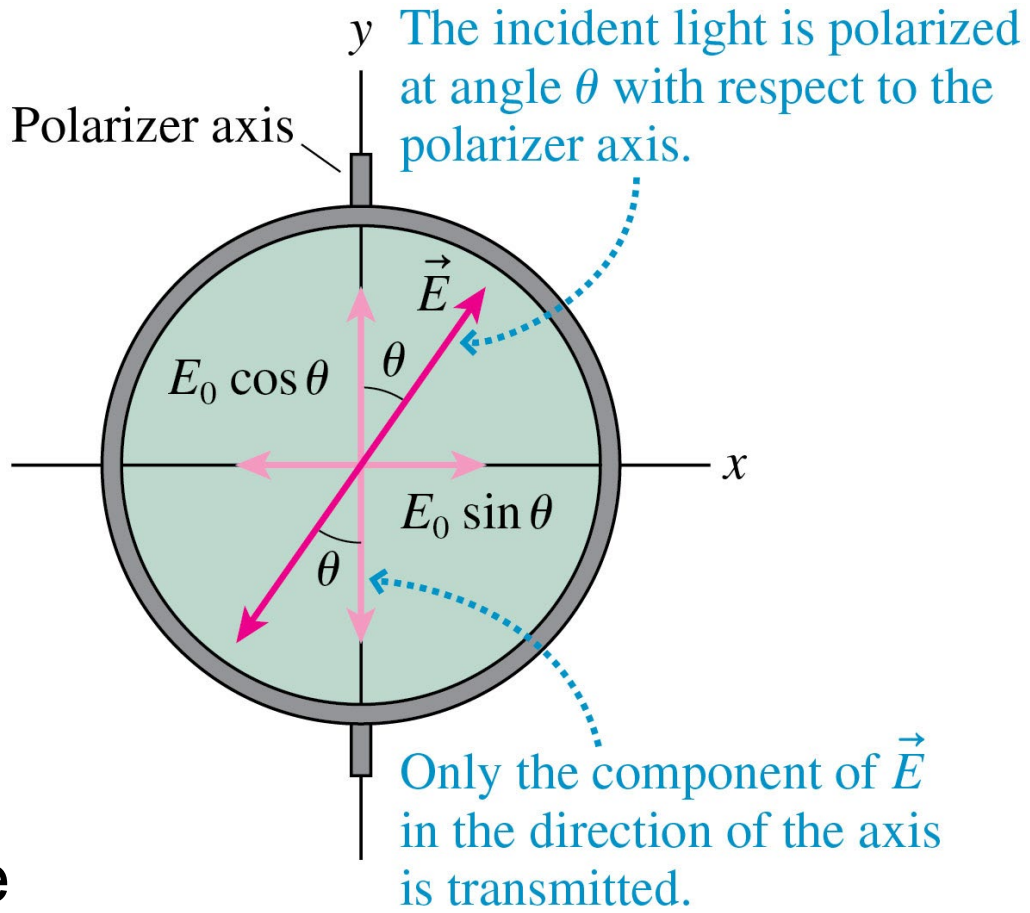
Malus's Law

- Suppose *polarized* light of intensity I_0 approaches a polarizing filter.
- The component of the incident electric field that is polarized *parallel* to the axis is transmitted:

$$\vec{E}_{\text{transmitted}} = E_{\parallel}\,\hat{j} = E_0 \cos\theta\,\hat{j}$$

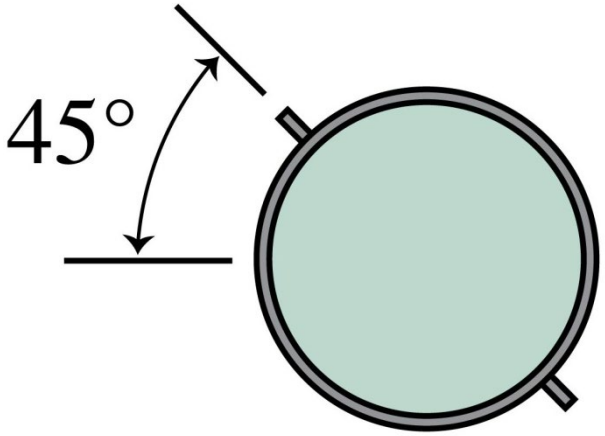
- The transmitted intensity depends on the square of the electric field amplitude:

$$I_{\text{transmitted}} = I_0 \cos^2\theta \qquad \text{(incident light polarized)}$$



Q 33.8

A vertically polarized light wave of intensity $1000\ \mathrm{mW/m^2}$ is coming toward you, out of the screen. After passing through this polarizing filter, the wave's intensity is:



A. $707\ \mathrm{mW/m^2}$

B. $500\ \mathrm{mW/m^2}$

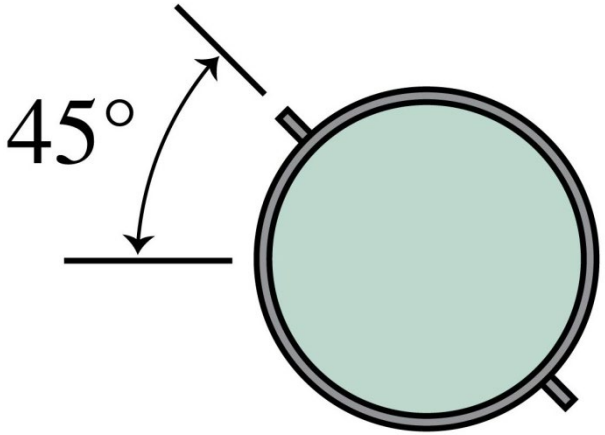
C. $333\ \mathrm{mW/m^2}$

D. $250\ \mathrm{mW/m^2}$

E. $0\ \mathrm{mW/m^2}$

Q&A 33.8

A vertically polarized light wave of intensity $1000\ \text{mW/m}^2$ is coming toward you, out of the screen. After passing through this polarizing filter, the wave's intensity is:



A. $707\ \text{mW/m}^2$

✔ **B. $500\ \text{mW/m}^2$** $I = I_0 \cos^2\theta$

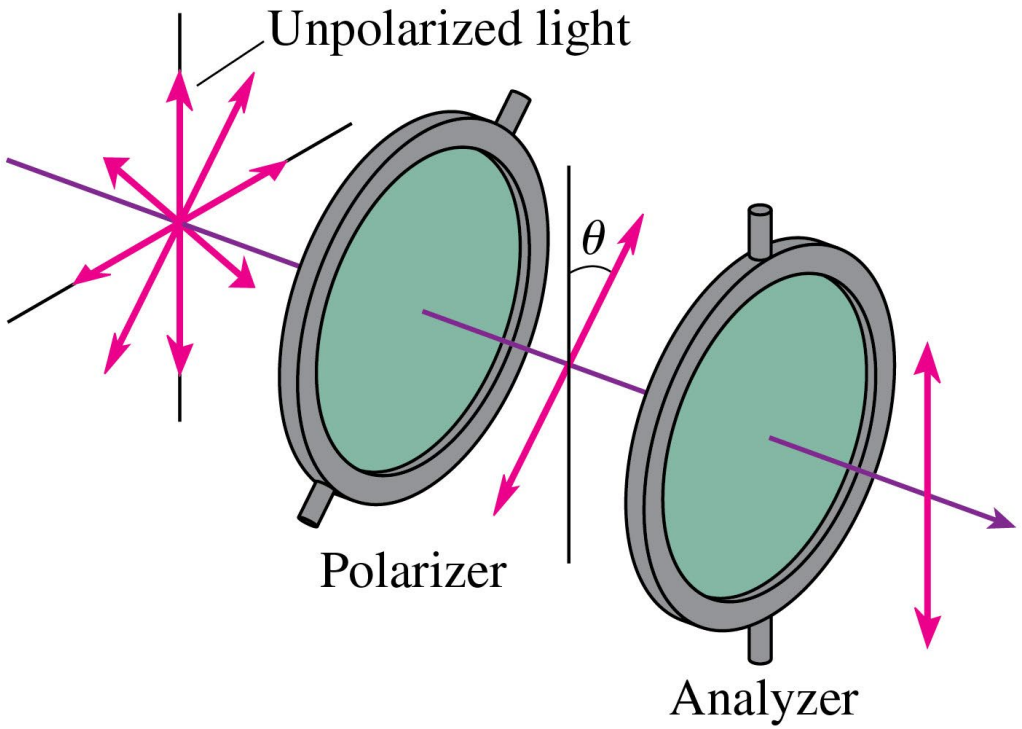
C. $333\ \text{mW/m}^2$

D. $250\ \text{mW/m}^2$

E. $0\ \text{mW/m}^2$

Polarizers and Analyzers

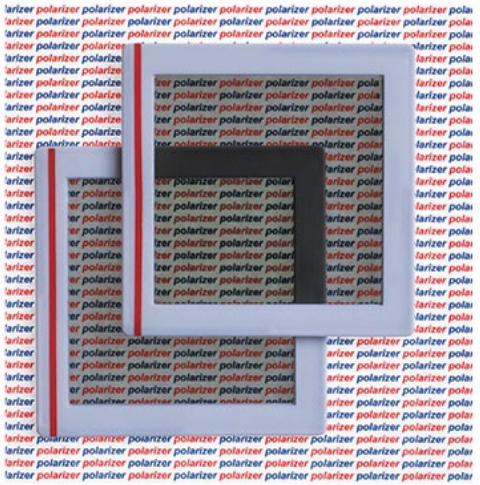
- Malus's law can be demonstrated with two polarizing filters.
- The first, called the *polarizer*, is used to produce polarized light of intensity I_0.



- The second, called the *analyzer*, is rotated by angle θ relative to the polarizer.

Polarizing Filters

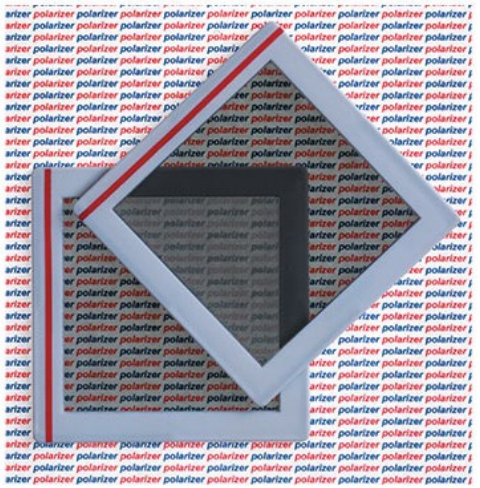
- The transmission of the analyzer is (ideally) 100% when $\theta = 0^{\circ}$, and steadily decreases to zero when $\theta = 90^{\circ}$.
- Two polarizing filters with perpendicular axes, called *crossed polarizers*, block all the light.



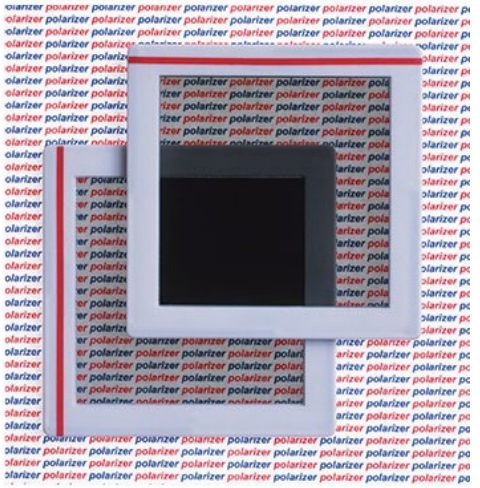
$\theta = 0°$

- If the incident light on a polarizing filter is *unpolarized*, half the intensity is transmitted:

$$I_{\text{transmitted}} = \frac{1}{2} I_0$$



$\theta = 45°$



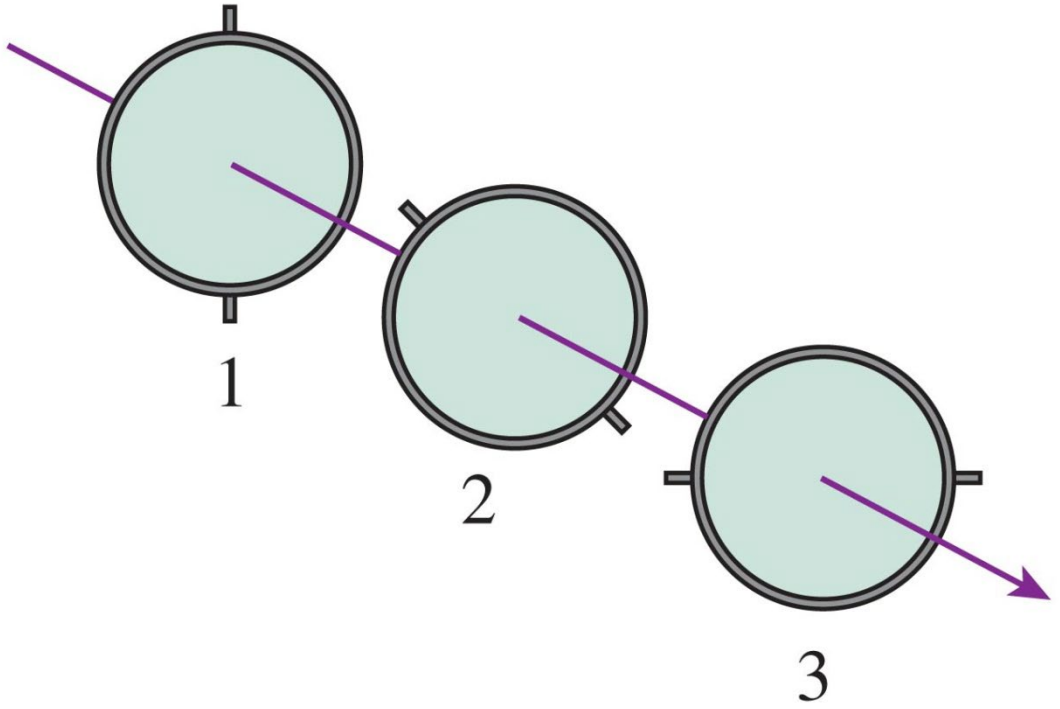
$\theta = 90°$

Q 33.9

Unpolarized light, traveling in the direction shown, is incident on polarizer 1. Does any light emerge from polarizer 3?

A. Yes

B. No.

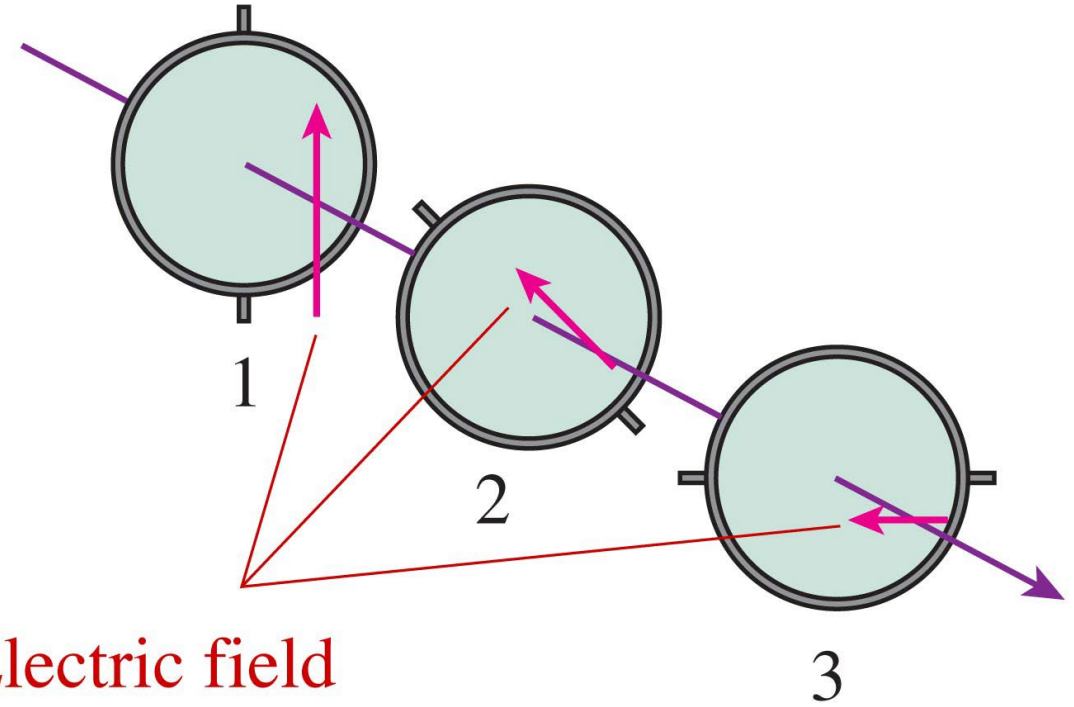


Q&A 33.9

Unpolarized light, traveling in the direction shown, is incident on polarizer 1. Does any light emerge from polarizer 3?

✔ **A. Yes**

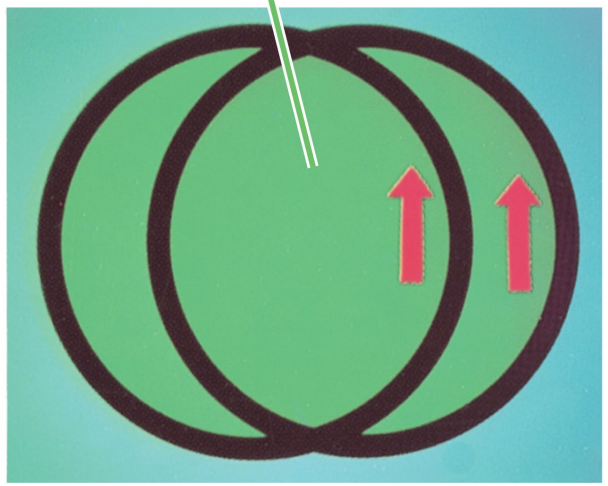
B. No



Polarization by Selective Absorption

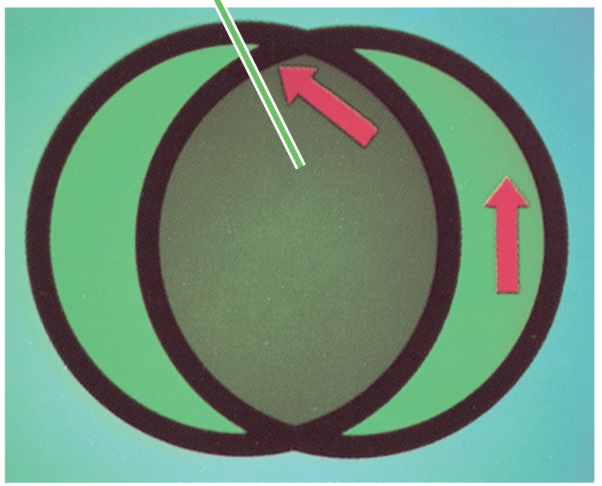
$$I = I_0 \cos^2 \theta$$

The transmitted light has *maximum* intensity when the transmission axes are *aligned* with each other.



a

The transmitted light intensity diminishes when the transmission axes are at an angle of 45° with each other.

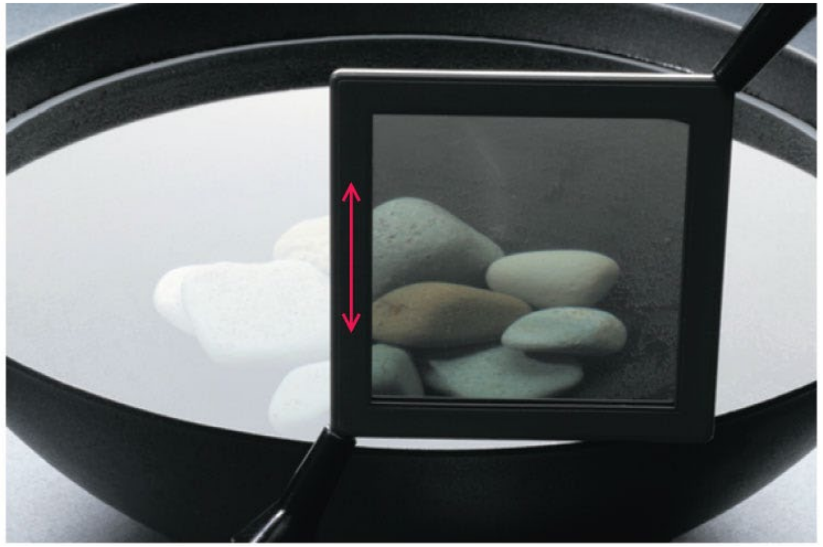


b

Henry Leap and Jim Lehman

Polarizing Sunglasses

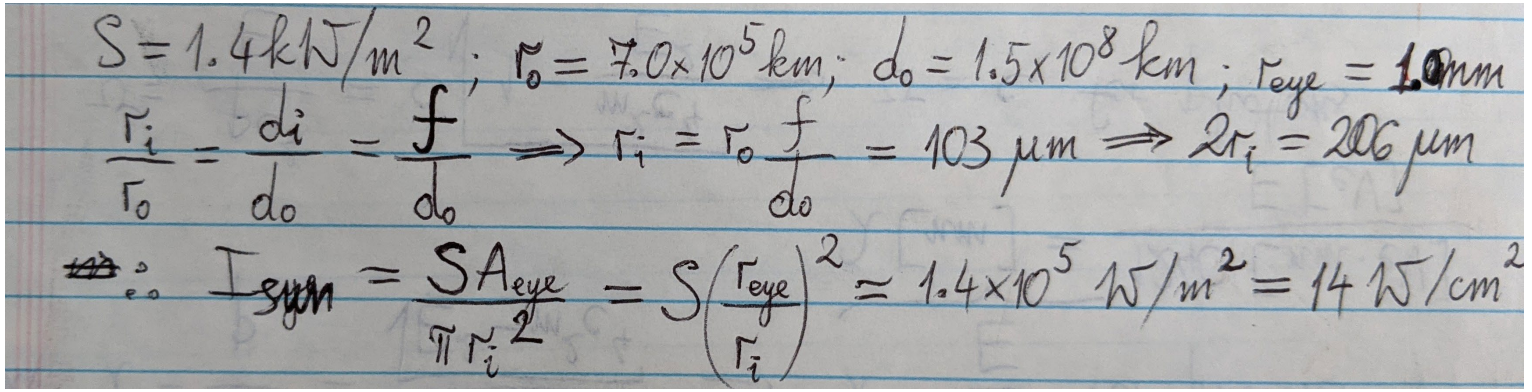
- *Glare*—the reflection of the sun and the skylight from roads and other horizontal surfaces—has a strong horizontal polarization.
- This light is almost completely blocked by a vertical polarizing filter.
- Vertically polarizing sunglasses can “cut glare” without affecting the main scene you wish to see.



Q&A: Sun Focused onto Retina

What is the intensity, in W/cm^2, of the image of the Sun on the retina, as per Problem 2.11(a) in *Peatross & Ware*?

A. 0.0014
B. 0.014
C. 0.14
D. 1.4
✔ E. 14
F. 140
G. 1400
H. 14000



Q: Laser Pointer Focused onto Retina

What is the intensity, in W/cm^2, of the image of the 1 mW HeNe laser, as per Problem 2.11(b) in *Peatross & Ware*?

A. 0.0039
B. 0.039
C. 0.39
D. 3.9
E. 39
F. 390
G. 3900
H. 39000

Example 11.4

Write the beam waist w_0 in terms of the *f-number*, defined to be the ratio of z to the beam diameter $2w(z)$ far from the beam waist.

Solution: Far away from the beam waist (i.e. $z \gg z_0$) the laser beam expands along a cone. That is, its diameter increases in proportion to distance.

$$w(z) = w_0\sqrt{1 + z^2/z_0^2} \to w_0 z/z_0$$

The cone angle is parameterized by the f-number, the ratio of the cone height to its base:

$$f^{\#} \equiv \lim_{z\to\pm\infty} \frac{z}{2w(z)} = \frac{z}{2w_0 z/z_0} = \frac{z_0}{2w_0}$$

Substitution of (11.44) into this expression yields

$$w_0 = \frac{2\lambda f^{\#}}{\pi} \qquad (11.46)$$

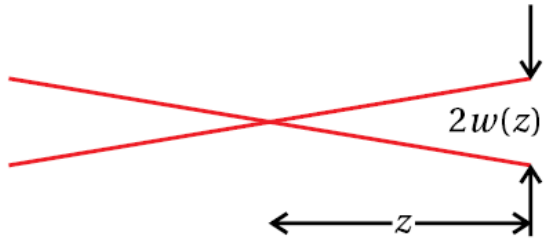


Figure 11.18

Q&A: Laser Pointer Focused onto Retina

What is the intensity, in W/cm^2, of the image of the 1 mW HeNe laser, as per Problem 2.11(b) in *Peatross & Ware*?

A. 0.0039
B. 0.039
C. 0.39
D. 3.9
E. 39
F. 390
G. 3900
H. 39000

λ (nm)	$2w_0$ (mm)	D (mm)	f (mm)	s (m)	z (m)
632.8	1.00	2.0	22.0	2.00	2.00

$2w_0'$ (μm)	
17.726	Term 2 only
vs.	
9.421	Terms 1 & 2
Term 1:	32335
Term 2:	12731
Check:	9.421

$$w_{02} \cong \frac{f\lambda}{\pi w_{01}} = f(\theta_{FF})_1$$

<--- OK w/o Term 1?

$$\frac{1}{w_{02}^2} = \frac{1}{w_{01}^2}\left(1 - \frac{Z_1}{f}\right)^2 + \frac{1}{f^2}\left(\frac{\pi w_{01}}{\lambda}\right)^2$$

Term 1 Term 2

Z_2 (mm)
22.176

$$Z_2 = f + \frac{f^2(Z_1 - f)}{(Z_1 - f)^2 + (\pi w_{01}^2/\lambda)^2}$$

$2w(z = f - Z_2)$ (mm)
17.726

$$w(z) = w_0\sqrt{1 + \left(\frac{\lambda z}{\pi w_0^2}\right)^2}$$

$$I_{\text{HeNe "past retina"}} = \frac{1.0 \times 10^{-3}\ \text{W}}{\pi\left(\frac{9.42 \times 10^{-4}\ \text{cm}}{2}\right)^2} = 1435\ {}^{\text{W}}/_{\text{cm}^2}$$

Q&A: Laser Pointer Focused onto Retina

What is the intensity, in W/cm^2, of the image of the 1 mW HeNe laser, as per Problem 2.11(b) in *Peatross & Ware*?

A. 0.0039
B. 0.039
C. 0.39
D. 3.9
E. 39
F. 390
G. 3900
H. 39000

λ (nm)	$2w_0$ (mm)	D (mm)	f (mm)	s (m)	z (m)
632.8	1.00	2.0	22.0	2.00	2.00

$2w_0'$ (μm)	
17.726	Term 2 only
vs.	
9.421	Terms 1 & 2
Term 1:	32335
Term 2:	12731
Check:	9.421

$$w_{02} \cong \frac{f\lambda}{\pi w_{01}} = f(\theta_{FF})_1$$ <--- OK w/o Term 1?

$$\frac{1}{w_{02}^2} = \frac{1}{w_{01}^2}\left(1 - \frac{Z_1}{f}\right)^2 + \frac{1}{f^2}\left(\frac{\pi w_{01}}{\lambda}\right)^2$$

Term 1 Term 2

Z_2 (mm)
22.176

$$Z_2 = f + \frac{f^2(Z_1 - f)}{(Z_1 - f)^2 + (\pi w_{01}^2/\lambda)^2}$$

$2w(z = f - Z_2)$ (mm)
17.726

$$w(z) = w_0\sqrt{1 + \left(\frac{\lambda z}{\pi w_0^2}\right)^2}$$

$$I_{\text{HeNe "past retina"}} = \frac{1.0 \times 10^{-3}\ \text{W}}{\pi\left(\frac{9.42 \times 10^{-4}\ \text{cm}}{2}\right)^2} = 1435\ {}^{\text{W}}/_{\text{cm}^2}$$

Q&A 1.1

What is the wavelength of a 10-eV electron?

A. 0.39 m
B. 0.39 mm
C. 0.39 μm
✔ **D. 0.39 nm**
E. 0.39 pm

$$\lambda_e = \frac{hc}{pc} = \frac{hc}{\sqrt{E^2 - m_e^2 c^4}}$$

$$hc = 1240\ [\mathrm{nm} \cdot \mathrm{eV}]$$

$$E = K + m_e c^2 = 10\ [\mathrm{eV}] + 511 \cdot 10^3\ [\mathrm{eV}]$$

Q&A 1.1(b)

What is the wavelength of a 10-keV electron?

A. 0.0012 nm

✔ **B. 0.012 nm**

C. 0.12 nm

D. 1.2 nm

E. 12 nm

$$\lambda_e = \frac{hc}{pc} = \frac{hc}{\sqrt{E^2 - m_e^2 c^4}}$$

$$hc = 1240\ [\mathrm{nm} \cdot \mathrm{eV}]$$

$$E = K + m_e c^2 = 10 \cdot 10^3 [\mathrm{eV}] + 511 \cdot 10^3\ [\mathrm{eV}]$$

The Electron Volt (eV)

- The work done to accelerate the proton across a potential difference of 1 V could also be written as

$$W = (1\ e)(1\ \text{V}) = 1\ \text{eV}$$

- Thus eV, pronounced “electron volt,” is also a unit of energy. It is related to the SI (*Système International*) unit joule by the 2 previous equations.

$$1\ \text{eV} = 1.602 \times 10^{-19}\ \text{J}$$

Total Energy and Rest Energy

We rewrite Equation (2.58) in the form

$$\gamma mc^2 = \frac{mc^2}{\sqrt{1-u^2/c^2}} = K + mc^2 \qquad (2.63)$$

The term mc^2 is called the rest energy and is denoted by E_0.

$$E_0 = mc^2 \qquad (2.64)$$

This leaves the sum of the kinetic energy and rest energy to be interpreted as the total energy of the particle. The total energy is denoted by E and is given by

$$E = \gamma mc^2 = \frac{mc^2}{\sqrt{1-u^2/c^2}} = \frac{E_0}{\sqrt{1-u^2/c^2}} = K + E_0 \qquad (2.65)$$

Relationship of Energy and Momentum (1 of 2)

$$p = \gamma mu = \frac{mu}{\sqrt{1 - u^2/c^2}}$$

We square this result, multiply by c^2, and rearrange the result.

$$p^2c^2 = \gamma^2 m^2 u^2 c^2$$

$$= \gamma^2 m^2 c^4 \left(\frac{u^2}{c^2} \right) = \gamma^2 m^2 c^4 \beta^2$$

$$p^2c^2 = \gamma^2 m^2 c^4 \left(1 - \frac{1}{\gamma^2} \right)$$

$$= \gamma^2 m^2 c^4 - m^2 c^4$$

Relationship of Energy and Momentum (2 of 2)

The first term on the right-hand side is just E^2, and the second term is E_0^2. The last equation becomes

$$p^2c^2 = E^2 - E_0^2$$

We rearrange this last equation to find the result we are seeking, a relation between energy and momentum.

$$E^2 = p^2c^2 + E_0^2 \qquad (2.70)$$

or

$$E^2 = p^2c^2 + m^2c^4 \qquad (2.71)$$

Equation (2.70) is a useful result to relate the total energy of a particle with its momentum. The quantities ($E^2 - p^2c^2$) and m are invariant quantities. Note that when a particle's velocity is zero and it has no momentum, Equation (2.70) correctly gives E_0 as the particle's total energy.